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FOREWORD

It is with great pleasure that I welcome you all to the 3rd Biennial International Conference on Hydrocarbon Science and Technology organized by the Petroleum Training Institute in collaboration with African Petroleum Producers' Organization (APPO), Petroleum Technology Development Fund (PTDF), NNPC Academy and NNPC Research Training Innovation (RTI) holding at the PTDF Towers, Abuja on this 23rd -24th October, 2023. The Petroleum Training Institute was established in 1972 by the Federal Government of Nigeria as a prerequisite for membership of the Organization of Petroleum Exporting Countries (OPEC), to train indigenous middle level manpower to meet the labour workforce demands of the Oil and Gas industry in Nigeria and the West Africa sub-region. Other core mandate of the Institute includes Research and Consultancy, organizing Seminars, Symposia and Conferences.

train indigenous
middle level
manpower to
meet the labour
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demands



The International Conference on Hydrocarbon Science and Technology is geared towards bringing the academia, community/opinion leaders, Oil and Gas industry players to critically discuss topical issues in the industry as it relates to policy formulation, investments safety, environment and security.

The theme for this year ICHST-2023 is "THE FUTURE OF OIL AND GAS INDUSTRY: OPPORTUNITIES, CHALLENGES AND DEVELOPMENT". I believe this conference through the keynote speeches, panel sessions will address pertinent issues on Nigeria oil and gas future opportunities,



, challenges and development. Over 30 technical and research papers in various aspects of oil and gas technology are presented by authors within and outside the country. Some of which are published in this book of proceedings after been peer reviewed

I have no doubt that this collection of scientific research materials will meet various research needs on Petroleum-Gas Engineering and emerging technologies. It will be a useful tool for the industry, as well as researchers in the academia. I wish all participants an exciting ICHST-2023.

I wish to sincerely appreciate our authors, reviewers, and the editorial board members Dr. Fredrick B. Owoyemi, Dr. Engr. A. A. Ominike, Engr. D. Atadious, Mr. P. Akusu and Engr. Chukwuka Chinemelu for their efforts at both ensuring quality and accuracy. Our partners and sponsors, the Director of Research and Development Dr. (Mrs.) Tina O. Isichei, the chairman, Conference organizing Committee Mr. E. D. Onakughotor, members of his committee, especially those who worked assiduously to make the conference a huge success, thank you all very much. God bless you

Dr. Henry A. Adimula
Principal & Chief Executive
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ACKNOWLEDGEMENT

The ICHST 2023 Conference Organizing Committee (COC) wishes to acknowledge the enormous contributions and support received from our Principal & Chief Executive Dr. Henry A. Adimula, the Director of Research and Development Dr. (Mrs.) Tina O. Isichei, the entire top management team and staff of the Petroleum Training Institute. We are also very thankful to our Partners / Sponsors: NNPC, PTDF, NNPC-RTI, NNPC-ACADEMY, APPO, NUPRC, NCDMB, NBTC, JOGISCO and E-EMTY.

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Microstructural Characterization and Corrosion Resistance of Mechanically-Deformed Mild Steel in Marine Environment

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Abstract: The influence of mechanical deformation by cold rolling process on the corrosion resistance of mild steel in chloride-containing environment at room temperature was investigated using potentiodynamic polarization (Tafel curve) method and the microstructures were examined via scanning electron microscopic (SEM). The mild steels were initially cold rolled with varying thickness reduction of 5%, 10%, 15% and 20% respectively, and the corrosion resistance of both the unrolled and rolled samples were then determined. The Tafel curve shows that all the rolled samples had a much lower corrosion current density, I_{corr} than the unrolled sample and it also illustrated that corrosion current density decreases with increase in the thickness reduction ratio. Moreover, the results of the curve revealed that the entire cold rolled MS samples (5%, 10%, 15%, and 20%) possessed corrosion potentials, E_{corr} of -0.66V, -0.62V, -0.60V and -0.56V respectively. This tends toward more positive E_{corr} signifying observable effects on the anodic reaction and is further attributed to the formation of more protective passive layers. The SEM reveals that the 20% cold rolled thickness reduction compared to the unrolled sample has higher smooth surface. Therefore, increasing the surface smoothness decreases the corrosion current density and lowers the pitting tendency, implying that increased surface roughness reduces mild steel corrosion resistance. The deformation and corrosion processes of the above study are linked to grain refinement and surface roughness.

Keywords: mild steel; cold rolling; corrosion resistance; scanning electron microscopy; corrosion potential

I. INTRODUCTION

Mild steel has certain unique qualities that make it a promising material for the majority of the building, construction, and automotive industries. These qualities include low costs machinability, good toughness, and ductility (Fayomi et al., 2018). However, mild steel is susceptible to corrosion, especially in hostile environments (Malatji et al., 2017). Metallic materials exposed continuously become increasingly vulnerable to corrosion attack, decreasing their corrosion characteristics (Olugbade and Lu 2019a). Due to mild steel's versatility as an engineering material, efforts have been made to overcome this primary drawback of low corrosion resistance by utilizing several techniques, such as electrodeposition and surface treatment or mechanical deformation processes like drawing, extrusion, or rolling (Nasiri-Abarbekoh et al. 2012; Wronski et al. 2013).

The great dimensional precision and productivity of rolling, however, make it more popular than other techniques (Lee et al. 2009). Interestingly, cold rolling has drawn a lot of attention because of its exceptional qualities. Cold rolling is less costly and easier to handle than other types of rolling, and it provides the benefits of close tolerance, enhanced machinability, and improved dimensional accuracy and surface polish (Wang et al., 2016). The process of refining grains during rolling is connected to the rolling process as well as the operational factors including the roller diameter, roller breadth, temperature, speed, and rolling reduction. As the dislocation density rises, mechanical distortion from rolling process exacerbates the plastic deformation process that activates dislocation. The dislocation of the grain boundaries, which further subdivides the coarse grains into the nanometer regime that brought about the decrease of grain size, is the result of this process.

Grain refining in metallic materials is thought to promote surface reactivity, which may improve passivation or dissolution. This feature, however, changes according to the nature and type of the metallic material. Passive metals, such as titanium (Balyanov et al., 2004) and aluminum (Song et al., 2009), for examples, experience increases in corrosion resistance with grain refinement, whereas active metals, such as copper (Luo et al., 2007), Ni (Rofagha et al., 1991) and low carbon steel (Olugbade and Lu, 2020) show a decrease in corrosion resistance as a result of an increase in the high energy grain boundary area.

It is on this premise that further studies is required to gain insight into the corrosion mechanism in the light of advanced characterization techniques and corrosion testing at the micro-structural levels. Therefore, this research focuses on the investigation of microstructural evolution and corrosion resistance of mechanically deformed mild steel in 0.6%wt. sodium chloride (NaCl) solution.

II. Experimental

Surface Rolling Process

Commercial mild steel (14 x 14 cm) was initially mechanically worked under very high compressive stress by rolling under the impacts of rollers with thickness reductions of 5%, 10%, 15%, and 20%.

Electrochemical Measurements

The mild steel sheet specimen utilized for electrochemical investigations was cut into 1-cm pieces. The specimens were implanted in epoxy resin with connecting terminals, allowing a working surface of 1 cm². Surface preparation of mechanically abraded specimens was performed using different grades of silicon carbide emery paper (400 to 1200 grit), followed by cleaning with acetone and rinsing with distilled water prior to electrochemical measurement in 1 M NaCl solution, which was prepared by diluting 83 ml of 37% NaCl (Merck) stock into a 1 L standard flask with distilled water, which is used as a corrosive environment.

Potentiodynamic polarization (PDP) curves were generated in a standard three-electrode cell utilizing a computer-controlled potentiostat/galvanostat (Autolab PGSTAT 302N). The counter electrode (CE) was platinum, the reference electrode (RE) was Ag/AgCl in 3 M KCl, and the working electrode (WE) was mild steel. The exposed area of the WE to the media was approximately 1 cm². The electrode was allowed to corrode freely before each potentiodynamic polarization (Tafel) investigation.

Surface Morphology

Scanning electron microscopy (SEM) at a voltage of 15 kV was used to analyze the surface morphology following the deformation process, the effect of grain refinement by rolling on the corrosion characteristics of mild steel, as well as the corrosion products and mechanisms.

1. Results

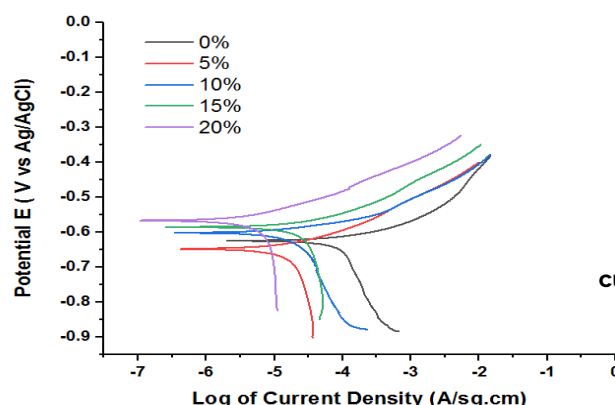


Figure 1: Potentiodynamic polarization curve of as-received and rolled mild steels immersed in 0.6% wt. NaCl solution

Table 1: Composition of the EDS spectra of MS at 0% thickness reduction in 0.6wt. % NaCl solution

Element	Fe	O	Ca
Composit	57.72	33.24	1.32

Table 2: Composition of the EDS spectra of MS at 20% thickness reduction in 0.6wt. % NaCl solution

Element	Fe	O	Ca	Al	Si	N
Composit	77.63	21.05	1.36	1.12	2.15	4.42

Surface Morphology

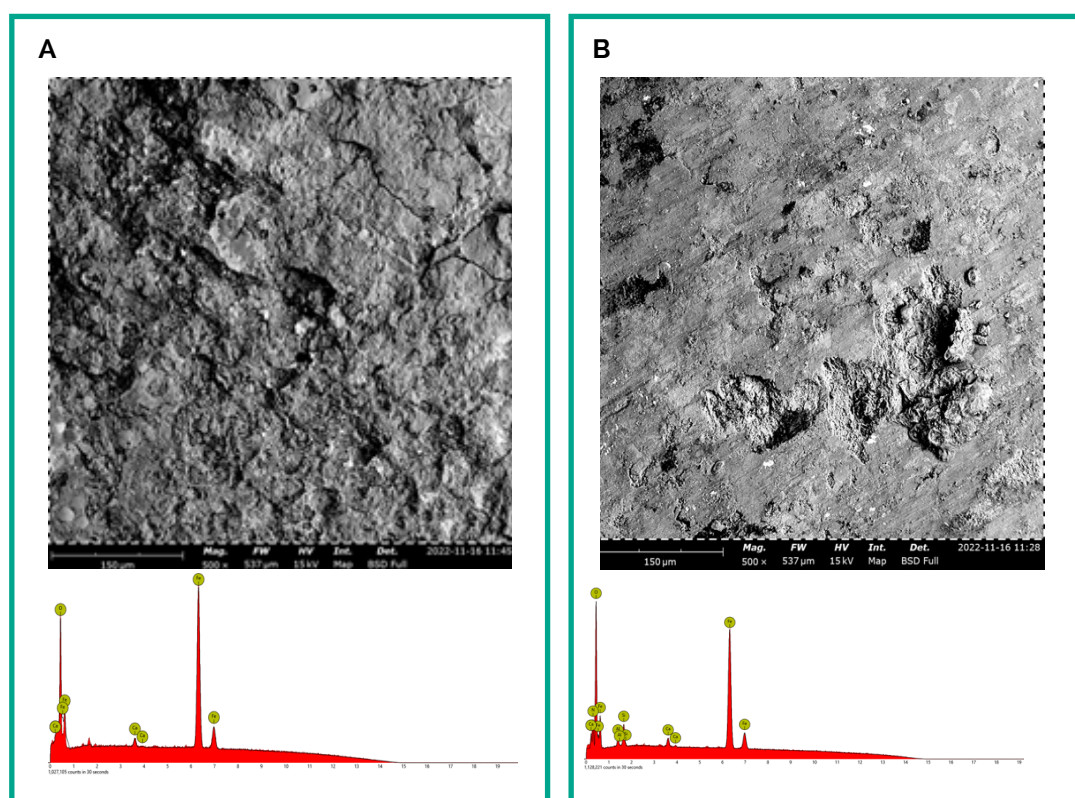


Figure 2: Scanning electron microscopic images of the unrolled and 20% rolled mild steel samples after polarization in saline solution at ambient temperature



III. Discussion

In the present study, the rolled mild steel sample (20%) exhibited a significant reduction in the corrosion current density (Figure 1) and increased corrosion potential compared with the unrolled sample (0%) as shown by the potentiodynamic polarization (Tafel) curve. A decrease in corrosion current density, I_{corr} and increase in corrosion potential, E_{corr} is attributed to the formation of more protective passive layers (since more nucleation sites for passivation activities are formed from the disorientation of the dislocation boundaries); hence increase in corrosion resistance is obtained.

Comparing the rolled samples (5%, 10%, 15%, 20%) with the unrolled, the corrosion resistance improved with the increase in the thickness reduction ratio. Moreover, enhanced corrosion performance with increasing thickness reduction ratio can be attributed to the grain refinement caused by deformation. Mechanical deformation by rolling tends to reduce grain size due to rapid rolling rate, which causes internal compressive residual stress, resulting in additional nucleation sites for passivation activities (Abioye et al., 2020; Aghuy et al., 2015). Figure 2, Tables 1, and 2 present the SEM images of the cross-sectioned samples of the 0% and 20% thickness reduction before and after rolling in 0.6wt. % NaCl solution at room temperature; and the EDS spectra conducted on the sample layers respectively. The SEM images illustrate that the 20% sample compared to the 0%, has higher and significant smooth surface. This is in tandem with the potentiodynamic polarization results. A smooth surface implies an increase in corrosion resistance. Furthermore, EDS spectra were used to determine the components present on the surface of the mild steel (rolled and unrolled) samples. Because of the reduction in grain size due to the rolling process, there is formation of additional nucleation sites for passivation activities. The possible increase in the corrosion resistance of the tested specimens is therefore linked to the above phenomenon (Olugbade et al., 2019; Walter and Kannan, 2011). In conclusion, both grain refinement and surface roughness are found to contribute to the deformation and corrosion process in the cold rolled mild steel.

IV. Conclusion

The corrosion characteristics of mild steel following mechanical deformation by rolling were examined in this work using potentiodynamic polarization (PP) and scanning electron microscopy (SEM). The current findings demonstrated that, when compared to the unrolled mild steel sample, all of the rolled mild steel (5%, 10%, 15%, 20%) displayed a lower corrosion current density, showing a minor gain in corrosion resistance.

The corrosion mechanism in a rolled mild steel sample may be related to grain refinement and reduced surface roughness caused by rolling, because a decrease in surface roughness is expected to increase passivation activities on the material surface layer, resulting in a more protective oxide layer and increased corrosion resistance.

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Optimal Sitting Of Gas-Fired Distributed Generation In Electric Power System For Power Loss Minimization

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Abstract

Distributed generation (DG) systems have emerged as a promising solution for improving the reliability and efficiency of electricity delivery to end-users. One of the key technologies used for the DG is gas power plants. In this research, the new voltage stability pointer (NVSP) was used to identify the soft spot for the installed location of the gas-fired DG plants and FACTS devices in the power network for optimal distribution of electricity. The effectiveness of the proposed technique was implemented in the MATLAB R2018a environment using the IEEE 3-bus test system. This study provides valuable insights into the factors that affect the voltage profile and proffers solutions through the optimal placement of the DG for enhancement of the grid performance.

Keywords: Distributed Generation (DG), New Voltage Stability Pointer (NVSP), Distribution Network, Gas-fired Plant, Voltage Stability Index

INTRODUCTION

In recent years, there has been a growing demand for reliable and efficient power distribution systems to meet the ever-increasing energy requirements of urban and industrial areas (Badrudeen et al. 2022a). With rapid technological advancement, distributed generation (DG) systems have emerged as a promising solution to address these challenges. Among various DG options, gas power plants have gained significant attention due to their versatility, high efficiency, and cleaner energy generation (Jian-xiao et al. 2007). Radial distribution networks are widely adopted in power distribution systems due to their simplicity and cost-effectiveness. However, these networks often face challenges such as high power losses and voltage deviations, especially during peak load conditions. Introducing gas power plants as DG units within these networks presents a potential solution to mitigate these issues. In addition, loss minimization plays a crucial role in enhancing the efficiency and economic viability of power systems. This can be achieved by strategically locating the DG within the transmission or distribution network. In the same vein, voltage profile improvement is another critical factor in maintaining the reliability and quality of power supply (Badrudeen et al. 2023). Gas power plants can contribute to voltage regulation by injecting reactive power into the distribution network. This capability becomes particularly important when dealing with voltage drops and fluctuations caused by high-demand scenarios or line constraints.

Elattar and Elsayed (2020) presented a modified moth flame optimization to improve on the existing optimization techniques in allocating DG placement in radial distribution system. With a glare consideration on operational cost reduction, this approach finds the best location and sizing of the DG for voltage profile improvement and active power loss minimization. The results from the analysis show that the proposed method is relatively better optimization technique for renewable energy sources. Similarly, Gümüş et al. (2023) proposed a novel approach to solve associated problems with sitting and placement of DG in radial distribution system for maximum benefits. The approach was employed to obtain the Thevenin impedance of the considered network at different operational conditions of DG sizing and placement. The best location for sitting and sizing the DG was deduced via modified Genetic Algorithm under different power factors. The suitability of the proposed method was verified with the Grey Wolf optimization algorithm.



In addition, Tolba et al. (2017) presented a hybrid optimization approach of particle swarm optimization and genetic algorithm for detection of optimal sizing and location of DGs. The candidate buses for the DG sitting were determined using loss sensitivity factor. The efficiency of the presented technique was validated with IEEE 69 and 33-bus data. The results from the simulation were compared with other notable optimization techniques. Prabha and Jayabarathi, (2018) presented a new technique for the identification of suitable locations of several DGs in the distribution system. The multi-objectives approach considers loss sensitivity factor for the placement of DGs and invasive weed optimization for the sizing of the DG. The applicability of this technique was demonstrated on IEEE 69 and 33 bus data.

Furthermore, Prakash and Lakshminarayana (2016), considered a particle swarm optimization technique for the enhancement of the voltage profile and quantitative power via optimal sizing and placement of DG. The research investigated the effectiveness of the presented technique with the IEEE radial distribution systems. Ameri et al. (2015) utilized the genetic algorithm to estimate the sizing and best area of situating the multiple DGs in power network. This is with a view to reducing the active power loss while managing the voltage profile constraints. The simulation results from the analysis proved that the technique is efficient in active power loss reduction. Kaur et al. (2016) presented a technique to evaluate the most efficient location and sizing of DG in a radial distribution network for better results in terms of total active power loss and voltage. Pham et al. (2022) proposed a modified mixed-integer quadratic programming algorithm via a DistFlow model. In addition, the study employed CPLEX software for linearization of power grid parameters. The efficacy of the proposed technique was demonstrated on standard radial distribution system.

Research by Ramavat et al. (2018) presented a technique through the particle swarm optimization to determine the capacity and location of the DG. Emphases were placed on the cost reduction through the cost-benefit concept. The present value factor analysis was carried out to estimate the investment cost, maintenance cost and operating cost and the effect of accurate DG placement in reducing these financial obligations. Hemeida et al. (2021) presented a Manta ray

foraging optimization algorithm to solve problem of power losses and drop in available energy in radial distribution network via accurate sizing and placement of DG. The effectiveness of the proposed technique was verified with different standard distribution systems. Prakash and Sujatha (2016) proposed bat optimization technique for finding the effective position and sizing of DG in order to reduce the real power losses in radial distribution network. The research considers cases of a unit DG and multiple DG units' installation and evaluates their performance on the stability of the power network. The results from the analysis proved to superior when compared with the particle swarm optimization technique.

Similarly, Devi et al. (2021) presented a stud krill herd algorithm for locating and sizing of DG units in radial distribution systems. The research explored different objective formulation framework with a view to minimizing the power loss and improving voltage profile. The effectiveness of the proposed technique was tested with IEEE standard systems.

Yuvaraj et al. (2020) presented an approach to size the DG and DSTATCOM simultaneously in radial distribution network using whale optimization algorithm. The method employed the loss sensitivity factor for the placement of DG and DSTATCOM.

The objective functions of the research were set at reducing the operating cost and active power loss. The results showed an improved voltage profile. Banumalar and Manikandan (2022) proposed an hybrid optimization of whale algorithm and genetic algorithm for accurate placement and sizing of DG sources in a radial distribution system. The results from the simulation were applied to standard radial distribution data. Kumar and Kansal (2022) proposed particle swarm optimization technique to estimate the capacity of the DG at each bus of the distribution network via a sample loss formula. Thereafter, the loss sensitivity factor is employed for the sitting of the DG in distribution network. Mandefro and Bantiyrga (2020) proposed PSO for power loss minimization and enhancement of voltage profile in radial distribution network. The results fulfilled the multi objectives target in the research.

This research aims to investigate the effectiveness of optimal placement of DG and Flexible Alternating Current Transmission System (FACTS) in power networks for loss minimization and voltage profile improvement. Power system analysis tools- MATLAB- will be utilized to model the power system and simulate its performance under various scenarios. The NVSP will serve as the optimization algorithm and decision support system to identify the optimal placement of DG units based on predefined objectives and constraints. The outcomes of this research will provide valuable insights into the optimal placement of DG units in power systems.

METHODOLOGY

This research employed the NVSP proposed by Badrudeen et al. (2022b) to determine the best location for sitting the gas-power plant. The voltage stability index, NVSP is derived from a reduced one-line diagram as shown in Figure 1.

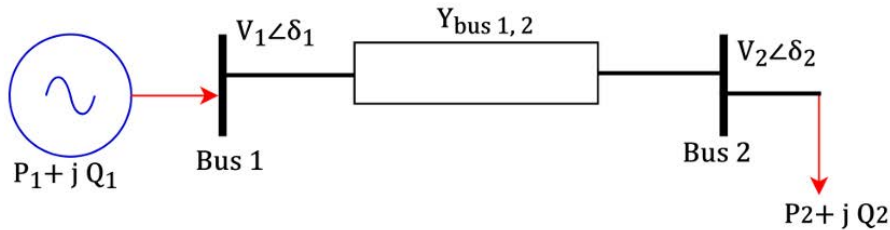


Figure 1. A reduced one-line diagram (Badrudeen et al. 2022a)

The line current (I) from the generator bus is expressed as:

$$I = (V_1 - V_2) \cdot Y_{bus} \quad (1)$$

The current at the load bus can also be calculated as:

$$I = \left(\frac{S_2}{V_2} \right) = \frac{P_2 - jQ_2}{V_2 \angle -\delta_2} \quad (2)$$

Assuming the line loss due to the load current is neglected, then Equation (1) will equate to Equation (2):

$$P_2 - jQ_2 = (V_1 - V_2) \cdot Y_{bus} \cdot V_2 \angle -\delta_2 \quad (3)$$

$$P_2 - jQ_2 = |V_1 V_2 Y_{bus}| \angle(\theta - \delta_2) - |V_2|^2 \cdot |Y_{bus}| \angle \theta \quad (4)$$

Dividing Equation (4) by $|Y_{bus}| \angle \theta$ yields:

$$\frac{P_2 - jQ_2}{|Y_{bus}| \angle \theta} = |V_1 V_2| \angle -\delta_2 - |V_2|^2 \quad (5)$$

Equation (5) can be rewritten as:

$$|V_2|^2 - |V_1 V_2| \angle -\delta_2 + \frac{P_2 - jQ_2}{|Y_{bus}| \angle \theta} = 0 \quad (6)$$

Therefore, NVSP is given as

$$NVSP = \frac{4Q_2 |Z|}{|V_1|^2} \leq 1 \quad (7)$$

where V_1 is the voltage at the sending end bus, Q_2 is the reactive power at the load bus and Z is the line impedance.

The test system is given in Figure 2 which is the IEEE 3-bus system.

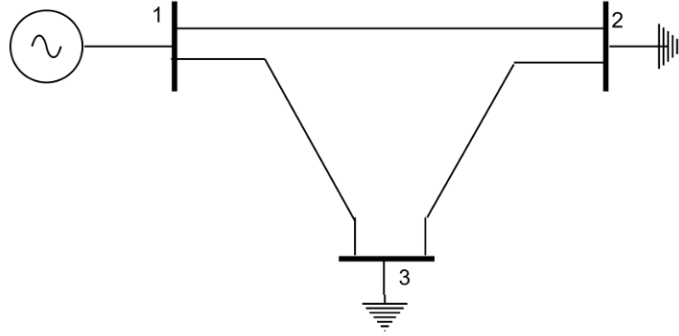


Figure 2: Single line diagram of IEEE 3-bus system

RESULTS AND DISCUSSIONS

The NVSP is employed in this research to accurately site the best location for Distributed generation (DG) and Flexible Alternating Current Transmission System (FACTS) devices in power network. The novel NVSP was developed from a reduced 2-bus one line diagram to generate factor values which represent the stability status of each candidate bus in an interconnected power network. In this case, placement and sizing of DG and FACTS devices in a power network come with several merits including improving the voltage profile, reduction of power loss and operating cost, if adequately optimized. Figures 3 and 4 depict the voltage profile of the IEEE 3-bus network at base case and identification of sitting of DG and FACTS devices on IEEE 3-bus, respectively. The Tables 1 and 2 present the base case using NVSP and identification of optimal sitting of the distributed generation and FACTS devices, respectively.

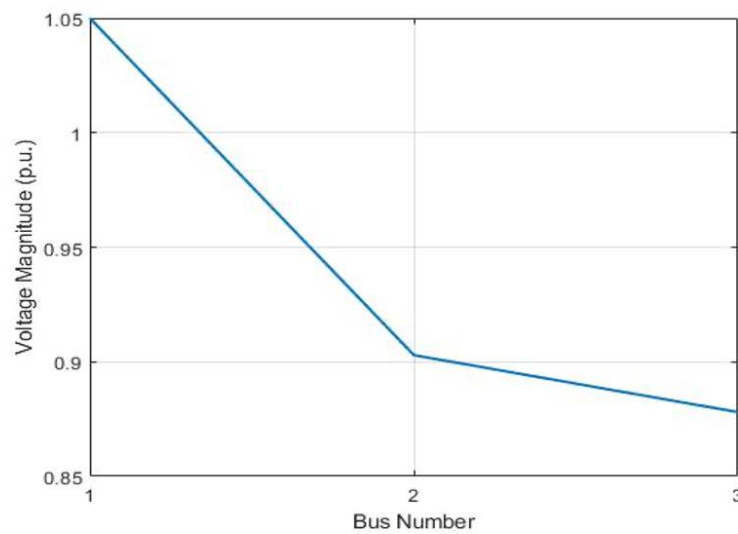


Figure 3: Base case voltage profile of the IEEE 3-bus system

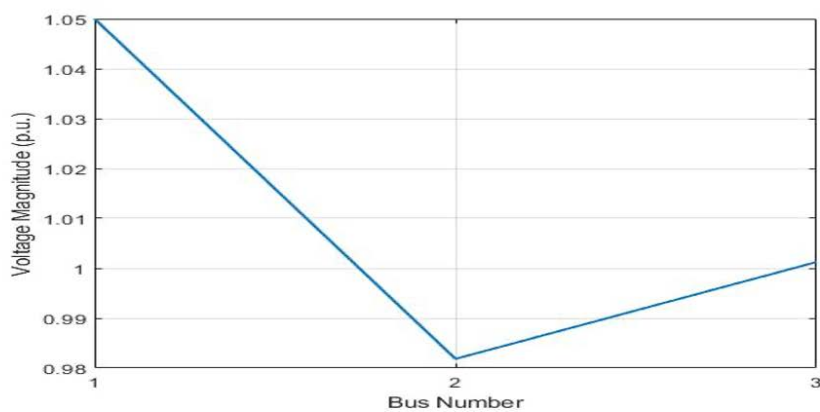


Table 1: Line Stability Assessment of IEEE 3-bus base using NVSP

From	To	NVSP
1	2	0.1788
1	3	0.0519
3	2	0.1229



Table 2: Assessment of Optimal Sitting of DG at IEEE 3-bus using NVSP

Bus No	From	To	Q (Mvar)	NVSP
2	1	2	200.2	0.3248
	1	3		0.0519
	3	2		0.02289
	1	2	400	0.6493
	1	3		0.0519
	3	2		0.4874
	1	2	600.2	0.9739
	1	3		0.0519
	3	2		0.7899
3	1	2	650	0.1785
	1	3		0.7460
	3	2		0.1714

The bus 2 of the IEEE 3-bus is the optimal location for siting the DG and FACTS devices for an improved voltage profile and total power line reduction.

CONCLUSION

This research has presented a new method of identifying the best location for sitting the DG and FACTS devices in power network. The results from the analysis was demonstrated on the IEEE 3 bus system which revealed that bus 2 is the best location for sitting the DG and FACTS for optimal performance and efficiency.

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Assessment Of Corporate Social Responsibility And Oil Companies In Niger Delta: A Case Study Of Platform Petroleum Limited's Community Development Projects In Ukwuani Land

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Abstract:

This study contains the assessment of Corporate Social Responsibility (CSR) and oil companies in Niger Delta: A case study of Platform Petroleum Limited's community development project in Ukwuani Land. The design for this study was survey research design. Based on this, a sample size of 50 management staff and 25 each from the four host communities were purposively selected giving a total sample of 150 respondents. The research instrument for this study was questionnaire which made up of two sections; the first section is the bio data which consisted of items such as status (staff or host). The other section considered factors that dealt with CSR development of the host communities. The instrument was validated with face and content validity. A reliability of the instrument was 0.69 which indicate that the instrument is reliable. Four research questions were raised and six hypotheses were formulated to guide the study. Descriptive statistics and measures of central tendencies were used to answer the research questions while Pearson product moment correlation analysis and independent t-test analysis were employed to test the hypotheses at 0.05 level of significance. The findings revealed that the management staff of PPL and host community indigenes are well aware of the concepts of CSR. Nevertheless, no significant impact of PPL CSR policy and projects were found on their host communities. It further revealed that PPL as a company had stated policies on CSR but not followed to the later. This study concluded based on the findings that the practice of CSR by PPL in host communities in Ukwuani land is very low and the people are yet to feel any serious positive impacts of their oil exploration activities in their community. The researcher recommended that the company should quickly review the implementation of their CSR policies on the interest of their host. Also, as the indigenes request that the PPL should be socially responsible to them through infrastructural development and government should enact laws that compels all business organisations to be socially responsible to their host communities.

INTRODUCTION

Nigeria is blessed with numerous extractive resources such as crude oil, natural gas, gold, limestone, granite, and coal, among others. Hydrocarbon resources are some of the most valuable due to their high demand globally (Borha, 2023). Nigeria is currently the second-largest crude oil producer and one of the world's top gas producers, possessing one of the Africa's biggest gas reserves (Akpan, 2021). Nigeria is set to benefit from the global demands for gas as a result of the ongoing conflict between Russia and Ukraine, as it is expected to benefit from the gaps in the global demands for gas and other supply chain caused by the ongoing war, hence the need to further overhaul the law (Olujobi, 2022). Apart from the degradation of the environment through oil spillage and gas flaring, there are other issues that have affected the relationship of the oil companies and their host communities. Indeed, the relationship between the oil companies and some of their host communities in the Niger-Delta, has, to say the least not been cordial in recent years owing to different perceptions of the role that the oil companies are expected to play in the development process of their host communities.



On the one hand, the host communities claim that oil companies are not doing enough considering the amount of oil wealth taken away from their lands. On the other hand, oil companies feel that they are doing enough and, have, in fact gone beyond the realm of normal corporate social responsibility. In the face of conflicting perceptions of the issues in contention, crises have often erupted in the oil-producing region. What then are the responses of the oil industry to these crises?

Under the Nigeria Companies and Allied Matters Act, 1990 (CAC, 1990) a registered company is statutory required to perform some duties which include the duties to pay tax, rates and levies (Ayorinde, 2007). These duties do not include building of schools, health centres, roads, water supply etc. to its immediate environment. However, a company is under an obligation to have positive impact on its immediate environment. Thus, it becomes a compelling but not legally enforceable duty on a company to be socially responsible with its immediate environment so as to have a peaceful co-existence with them.

The concept of corporate social responsibilities as a new concept in the business world is the ability of a corporation to relate its operations and policies to the needs of the environment and the work force to the benefit of the organisation and the host communities. Onwuchekwa (2000) further gave his definition thus; "business social responsibility" is operationally defined as the managerial intellectual concern and practical involvement on social issues relating to business interdependence with the society of location. Werther (Jr) and Chandler (2011) use the phrases shown below to describe corporate social responsibility: (1) Corporate Responsibility, (2) Corporate Community Engagement, (3) Community Relations, (4) Corporate Stewardship, (5) Social Responsibility, and (6) Strategic Philanthropy.

Post et al., (1999); Benn and Bolton (2013), believe that corporate social responsibility means that a corporation should be held accountable for any of its actions that affect people, their communities and environment. It therefore follows that negative business impacts on people and society should be viewed seriously and corrected. Emphasis on profit maximization should be relaxed to enable businesses consider other stakeholders. They consider these terms by synonymous with corporate social responsibility: (1) Corporate philanthropy, (2) Corporate awards, and (3) Corporate charity.

Corporate Social Responsibility (CSR) therefore is an obligation on companies to consider the interests of the communities by contributing in the provision of social infrastructure such as schools, hospitals, roads, water supply in their area of operations. These duties have been universally acknowledged and accepted as a convention and practice that companies must have policies on Corporate Social Responsibility. It is against this backdrop therefore that the present study hopes to unravel the level of corporate social responsibility of oil companies operating in Ukwuani land with a special focus on Plat-Form Petroleum Company.

Statement of the Problem: Presently, it is obvious that oil companies are not living up to expectations of their duties in the host communities. This contributed immensely to environmental degradation of the region, the air, water and land pollution, destruction of natural habitations and waste disposal system that are harmful and destructive to the host community. Lack of corporation to tackle the problem and reduce the effects of drilling and production activities on the environment and contribute in raising the standard of living of the people in the host communities became a source of concern to the youth of the region. Secondly, lack of infrastructures and social amenities in the host communities of Ukwuani region by the federal government and oil companies operating in the region. Thus, the researcher is intending to find out the cause of these problems in the region.

Objectives of the Study: The aims of the study include the following:

- (1) To determine the opinion of Plat-form Petroleum Limited management on their corporate social responsibility towards the host communities where they operate in Ukwuani?
- (2) To ascertain the extent to which Plat-form Petroleum Limited is living up to their corporate social responsibility? (3) To determine available projects undertaken by Plat-form Petroleum Limited in Host communities that are in Ukwuani?
- (4) To ascertain the level of impact of PPL corporate social responsibilities activities if any has on individuals in their host communities?

Research Questions: To guide this study, the following research questions were formulated to enable the researcher carry out the study: (1) What is the Opinion of Plat-form Petroleum Limited on corporate social responsibility to host communities in Ukwuani? (2) Is Plat-form Petroleum Limited living up to their corporate social responsibility? (3) Are there visible community development projects undertaken by Plat-form Petroleum Limited in Host communities in Ukwuani? (4) Do the operations of Plat-form Petroleum Limited have any significant impact on the lives of individuals in their host communities?

Significance of the Study: The significance of this study lies in its ability to unravel the level of CSR activities carried out by oil multinationals operating in Ukwuani Land. The study apart from contributing to existing literatures is significant to the federal government of Nigeria as findings from this study when completed will be of relevance on how best oil multinationals can be compelled to living out their corporate social responsibilities in their host communities. One great aspect of relevance of this study is the fact that searches in literature reveals a great dearth of research in the assessment of corporate social responsibilities of oil multinationals in Ukwuani land which this study hopes to fill the existing gap in literature.

Scope of the Study: The scope of the study focused on the role of corporate social responsibilities of multinational companies in the community development projects in Ukwuani land. Specifically, the study hopes to assess the perception of PPL management of their corporate social responsibilities to their host communities. It will also assess the extent to which community development projects have been carried out by PPL in their host communities. The study however is delimited to four communities in Ukwuani region which comprises of Umutu, Obi-Iloh (Umu-Osele)/Umu-Ezhiogoli, Adonishaka/Ogbe-Uzu and Akoku- Uno communities where the company currently operates.

Research Methods: Here, we dealt with the method and procedures adopted in carrying out this research. It is discussed under the following sub-headings; research design, the population of the study, the sampling techniques and sample size of the study, validity of the research instrument and reliability of the research instrument, instrument for data collection, the method of data collection as well as the statistical tools used in the analysis of data.

Research Approach: This research work explores the level of corporate social responsibility of oil companies operating in Ukwuani land with a special focus on plat-form Petroleum Limited Company. The research approach adopted for this work is the survey research design. Survey research design is defined as the systematic gathering of information from respondents for the purpose of the study. The justification for this method is that it helps to gather definite answers to specific question related to the area of study.

The Population of the Study: Population in research is the totality of the objects or elements being studied and to which conclusion or generalization of the result will apply. The population of the study therefore,



Sampling Techniques and Sample Size of the Study: Priscilla (2001), defined sampling as a process of selecting a population for judgment. To get the true representation of the sample size, a purposive and random sampling technique was employed. In using the purposive sampling technique only management staff who are mostly at the helm of decision making in the company were contacted because they are the ones that determine policy matters as it relates to implementation of corporate social responsibility while the indigenes were selected purposively by contacting community and opinion leaders in the communities. Based on this therefore, a sample size of 50 management staff and 25 each from the four host communities were purposively selected giving a total sample size of 150 respondents.

Instrument for Data Collection: To guide this research, an instrument named corporate social responsibility questionnaire (CSRQ) was carefully designed to elicit information from the subjects of the study. The questionnaire was made up of two sections; the first section is the bio data which consisted of items such as Status (Staff or Host), gender, Location. The other section considered factors that dealt with CSR development of the host community. Participants of the study were free to agree or disagree with any statement on the questionnaire on a four-point continuum scale ranging from strongly agree to strongly disagree. Strongly Agree (4-Points); Agree (3-Points); Disagree (2-Points) and Strongly Disagree (1-Point).

Validity of Research Instrument: For the instrument to be made valid, it was given to the research supervisor who effected correction to make sure the instrument reflects face and content validity. After due corrections the final instrument was thus validated.

Reliability of the Research Instrument: Reliability is the extent or degree of consistency between two measures. It is the stability of sources obtained in the different measures (Lee and Lings, 2008). To determine the reliability of instrument, a pilot study using the instrument was done sampling the opinions of 15 staff Seplat Nigerian Plc field office Sapele and 15 indigenes of Sapele that is not part of the area of sampling. Their responses were coded and the Split half method was used to determine their reliability. A reliability of 0.69 was yielded which indicate that the instrument is reliable.

Method of Data Collection: The researcher visited the Plat-form Field Offices in the various communities under survey and administers the questionnaire to the management staff which included Head of field operations, Head of community relations, Head of Human Resource and Head of Welfare. Also part of the study were senior field officers in the community under survey. Also, questionnaires were administered to community leaders which included the community PRO on behalf of the Okpala-Uku in council, the Community Youth Executives, Woman and Men Leaders and representatives of the various quarters in the Community Development Unions. Before administering the questionnaire, the instructions for completing the questionnaire was carefully and thoroughly explained to the respondents by the researcher and were allowed to respond to the questions independently. There after the researcher collected the questionnaires from the respondents. Strongly agreed (SA), agreed (A), disagreed (D), and strongly disagreed (SD) were used for their response. At the end of the field work, their response was analysed and generalisations were made.

Method of Data Analysis: The responses of the respondents were worked out employing descriptive statistics and measures of central tendencies. Based on this, the frequencies so expressed guided the mean analysis of each of the items raised before the cumulative mean analysis section by section. In the analysis of the individual questionnaire items, a benchmark of 2.50 was set to accept as true any of the statements made in the questionnaire. This was done because of the mean summary of the 4, 3, 2 1 scale of the instrument used. Data collected was summarized in tables and hypothesis tested with the Pearson product moment correlation analysis and the independent t-test analysis using the statistical

package of the social sciences version 17 (SPSS 17). The choice of the Pearson correlation analysis was made because the correlation analysis establishes relationships and the extent to which an independent variable influences a dependent variable. While the t-test compares differences in mean of two independent samples.

Presentation of Result, Data Analysis and Discussion

Corporate Social Responsibility Questionnaire

Table 1: Demographic Profile of Respondents

Variable	Frequency	Percentage
Gender		
Male	118	79
Female	32	21
Total	150	100
Educational Qualification		
SSCE	26	17
ND/NCE	21	14
HND/BSc	76	51
Postgraduate Degree	27	18
Total	150	100
Experience of PPL Staff		
1-5 Years	10	20
6-10 Years	24	48
Above 10 Years	16	32
Total	50	100

The data presented in table 1 revealed that 118 males comprising of 79% of respondents participate in the study. Likewise, 32 females comprising of 21 % respondents participated. On the highest level of education attained by respondents, 26 comprising 17% had SSCE as their highest qualification, 21 comprising 14% had ND/NCE while 76 comprising of 51% had HND/BSc as their highest qualification. 27 persons which made up 18% of the respondents had bagged various post graduate degrees. Relative to the experience level of the PPL staff, 10 making up 20% had an experience between 1-5 years while 24 of the respondents making up 48% had an experience of 6-10 years and 16 comprising of 32% had an experience above 10 years

Research Question

Research Question One: What is the Opinion of Plat-form Petroleum Limited and Host Communities on corporate social responsibility to host communities in Ukwuani?



Table 2: Perceptions of PPL Staff and Host Community Indigenes on Corporate Social Responsibility

S/N		PPL Staff					Host Community Indigenes				
		N=50					N=100				
		SA	A	D	SD	Mean	SA	A	D	SD	Mean
1	I understand the concept of corporate social responsibility	36	12	2	0	3.68	28	15	18	39	2.32
2	Corporate social responsibility means taking care of the environment we work	23	18	4	5	3.18	12	18	22	48	1.94
3	Corporate social responsibility covers taking the needs of its staff and host as important	32	18	0	0	3.64	65	28	7	0	3.58
4	Corporate social responsibility involves sponsoring projects within the companies host communities.	28	12	8	2	3.32	78	22	0	0	3.78
5	Corporate social responsibility involves taking care of social-welfare needs of the host communities	26	24	0	0	3.52	81	19	0	0	3.81
Grand Mean		PPL Staff				3.47	Host Community Indigenes				3.09

Research Question Two: Is Plat-form Petroleum Limited living up to their corporate social responsibility?

Table 3: Responses on PPL CSR Policy

		SA	A	D	SD	Mean
6.	The concept of social responsibilities of PPL is effective to their host communities.	18	24	6	2	3.16
7.	PPL have a laid down policy for social responsibility.	14	16	15	5	2.78
8	There has been implementation of Corporate social responsibility policies in PPL over the years.	21	11	8	10	2.86
9	At the moment PPL have a social obligation to the communities in which they operate	12	14	16	8	2.60
10.	PPL have been able to handle environmental pollution as part of its social responsibility.	14	11	7	18	2.42
Grand Mean						2.76

Table 3 reveals responses on PPL CSR policy. All issues raised had acceptable mean scores except for item 10 which revealed the inability of PPL to handle environmental pollution squarely as part of its social responsibility. Responses to Item 6 however revealed that PPL staff claimed that the concept of social responsibility is effective to their hosts while item 7 revealed that PPL has laid down policy for social responsibility of the company. Responses to item 8 and 9 further revealed that there has been implementation of CSR policies over the years and that PPL have a social obligation to the communities in which they operate.

Research Question Three: Are there visible community development projects undertaken by Plat-form Petroleum Limited in Host communities in Ukwuani?



Table 4: Responses on Visible Development Projects Undertaken by PPL

		PPL STAFF					Host Community Indigenes				
		SA	A	D	SD	Mean	SA	A	D	SD	Mean
11	PPL have undertaken several infrastructural development projects in host communities	22	10	11	7	2.94	16	22	10	52	2.02
12.	PPL have undertaken road maintenance projects in host communities	16	6	4	26	2.32	12	32	12	44	2.44
13.	PPL over time provides electricity to their host communities	10	4	2	34	1.80	2	21	32	45	2.40
14.	PPL have built markets, town halls and Health centres for host communities	13	2	18	17	2.22	14	16	11	59	1.85
15.	PPL regularly supports the education of indigenes in host communities through scholarship	22	18	4	6	3.12	21	25	10	44	2.33
16	PPL provides education aids to schools in their host communities	10	4	2	34	1.76	12	10	21	57	1.77
Grand Mean		PPL STAFF				2.36	Host Community Indigenes				2.14

Table 5: Impact of CSR on Host communities

		SA	A	D	SD	Mean
17.	I am satisfied with the level of development projects undertaken by PPL	18	17	12	53	2.0
18.	PPL activities have impacted the growth of the host communities to a very great extent	12	15	10	63	1.76
19	Our children's education has been affected positively through scholarships from PPL	16	14	12	58	1.88
20	We have received better health care through PPLs' CSR activities	8	4	6	82	1.38
21	We have experienced clean-up of our environments through adequate CSR by PPL	22	14	8	56	2.02
22	I doubt if we have been positively affected by PPLs' CSR activities.	56	16	20	12	3.24
Grand Mean						2.05

Table 5 reveals the level of impact CSR of PPL has on host communities. The mean scores of all the statements made above fell below the acceptable benchmark revealing a very low level of impact. This is clearly revealed by the grand mean score of 2.05 which is also below the acceptable mean benchmark.

Table 6: Test of significant Correlations between CSR activities and Impact of CSR on the Lives of Host Community Indigenes

		CSR Activities	Impact on Host Communities
CSR Activities	Pearson Correlation	1	-.078
	Sig. (2-tailed)		.883
	N	100	100
Impact on Host Communities	Pearson Correlation	-.078	1
	Sig. (2-tailed)	.883	
	N	100	100



The r calculated shows that there is no significant relationship/impact between CSR activities of PPL and Impact of CSR on the lives of host community indigenes at $p > 0.05$ ($p = 0.883$). Based on this therefore, the null hypothesis that states that the operations of Plat-form Petroleum Limited have no significant impact on the lives of individuals in their host communities is accepted while the alternate hypothesis that the operations of Plat-form Petroleum limited have a significant impact on the lives of individuals in their host communities is rejected.

Discussion of Findings: Findings from this study reveal that the perception of PPL staff and host community indigenes on corporate social responsibility are in a high level of agreement and showed no significant difference. From their submissions the core of CSR comprises care for the environmental in which companies operate. It was also agreed that sponsoring of community projects, taking care of social-welfare needs of the host communities and been responsible for staff and host community needs were all part of corporate social responsibility. These submissions are in line with those of Freeman, (2010) who noted that CSR is an approach which aims at helping organizations implement voluntary obligation to support and solve social issues and to fulfil commitment. Also supporting this submission is the view of Hohen, (2007) who noted that CSR is characterized by the decisions to improve the environmental degradation, social and economic activities of its existence.

Results presented in Table 3 reveals a very clear policy statement on CSR by PPL. As part of the policy, it was submitted that PPL has a policy to handle environmental pollution squarely as part of its social responsibility. Also supporting these arguments are claims that the concept of social responsibility is effective to their hosts while and at the same time there has been implementation of CSR policies over the years and that PPL have a social obligation to the communities in which they operate. Relative to answers to research question three which examined visible developmental projects by PPL in their host communities, responses from the host community indigenes and the administrative staff revealed a positive impact only in the award of scholarship to host community indigenes. These observations are against the claims of PPL that they have undertaken infrastructural development projects in the communities. A further test of hypothesis on this revealed no variance in the opinions of indigenes and those of the PPL staff. It is pertinent therefore to note that this observed difference is not in line with the laid down principles of corporate social responsibility of companies to their host communities. According to Okoye and Nwaigwe (2015), a company that implements properly its corporate social responsibility policy helps in fostering greater peace and promoting maximization of profits. The study further revealed a non-significant impact of the CSR activities of PPL on the host communities. This observation does not go in line with earlier outlined benefits of CSR which include provision of welfare packages for indigenes and attraction of infrastructural development of host communities. The observation thus is in line with the findings of Ejumudo et al., (2011) who came up with the findings that oil companies' activities in the Niger Delta region have had destructive effect on the environment and concluded that oil companies operating in the region has done little or nothing in minimizing the difficulties cost by them.

Conclusions: Based on the outcome of data analysis, this study wishes to conclude that the practice of corporate social responsibility by Plat-form Petroleum Limited in host communities in Ukwuani land is very low and the people are yet to feel any serious positive impacts of their oil exploration activities in their community.

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Integrated System Framework For Monitoring Oil And Gas Pipelines In The Niger Delta Region: Ultrasonic Fault Detection Technology Based

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Abstract:

Oil and Gas spillage continues to be a menace in the Niger Delta region. Many lives and properties have been destroyed including flora and fauna, with air polluted and waters contamination. The Government of Nigeria budget financing suffers too because of loss of revenue to oil and gas leakages, especially due to theft by bunkering. Several solutions have been provided. But all tend to focus attention on either external (i.e. outside the pipe) or internal (i.e. within the hollow of the pipe). These methods frequently miss fundamental parameters - the exact location and time of incidence as the pipeline installation terrain varies. This has often made litigation for compensation between affected community and the operating companies more difficult. Again, in most cases with the existing approaches, spilling of oil and gas would have progressed significantly before detection. Therefore, we intend to harness the potential of ultrasonic flaw detection mechanism to spot out flaws such as corruptions and cuts in the pipe metal to determine interference into the pipeline, using integrated software that can process the signals. This will report the interference incidence precisely before the first drop of fluid. The current flaw detection concept, using ultrasonic, is the method applied in this study. We have created a design framework for the hardware constituents. This can be useful for correct implementation and further studies.

Keywords: ultrasonic flaw detection, oil and gas leakage, pipeline surveillance, spillage location

INTRODUCTION

In most cases in Nigeria, Oil and gas must flow through the pipe to be available for commercial use. As such, loses (including theft through bunkering) of the oil and gas must mostly be through the pipe. To detect and curb the menace, the pipe transporting the oil and gas resources should be the object of focus. In fact when it comes to oil and gas theft, nothing could be more attractive to the bunker than the pipe. The fluid can leak naturally into the soil or water due to damages, and breaks, caused by aging materials and equipments; or be stolen using small containers (like gallons and jerry cans) and bigger ones (like tankers and ships). All these cause huge wastes and damages to government and the host communities.

The flow of oil and gas fluid diverted illegitimately from the pipe creates damages, first to the host communities, by destroying flora and fauna. This is common in the Niger Delta region. Affected communities suffer immeasurable hardship including health hazards. Government of Nigeria on her part has lost much revenue that should have accrued from crude oil, due to the theft. "Between January and July, Nigeria, Africa's biggest oil producer, lost an average of 437,000 barrels of oil a day to criminals" (see figure 1 - Yusuf (2022)). This has often make it difficult to meet national budget financial demand. The overall consequences of this menace is twofold hardship - on the host community and the Federal Government.

Nigeria lost an estimated \$10 billion to large-scale crude theft in seven months

Month	Crude Loss (b/d)	OPEC Quota (b/d)	Estimated revenue loss (\$)
January	270,000	1,683,000	713,458,800
February	328,000	1,701,000	862,836,800
March	378,000	1,718,000	1,329,758,640
April	413,000	1,735,000	1,308,879,600
May	520,000	1,753,000	1,835,584,400
June	534,000	1,772,000	1,885,874,400
July	616,000	1,799,000	2,072,870,800

PREMIUM
Times

Sources: OPEC, PT Data

Total
\$10bn

Fig. 1: Crude oil theft over seven months in Nigeria

This often attract two main reactions, one from each party. On the part of affected villages, litigations are frequently sorted for adequate compensation from the company operating in the area. Government, on the other hand, will quickly respond by sending law enforcement agents to the affected area. In most cases, JTF (Joint Task Force) and Navy and sometimes including mobile police are deployed. If the vandals were not caught on the spot, Government would start investigation immediately. Thus, Government have always be reactive rather than proactive in this regard.

To this end, several approaches have been made and alternative solutions do exist including traditional human physical monitoring. Typical example is local vigilante that helps to secure company facilities, Achilike & MacDonald (2017). Only last year, a contract have been awarded for human physical pipeline surveillance, amounting to NGN48 billion per year, (see Agency Report (2022)). But human engagements in such tasks are hardly reliable. There could be compromise, and life expectancy could affect continuity of the project. Also, change of government may lead to exchanging awardees of the contract which in turn may affect effectiveness of the project.

Now, technological methods also abound. Unfortunately, most of these methods seem to have become obsolete, Achilike & MacDonald (2017). They largely range in the embedding medium of pipeline installment: buried in the ground, immersed in water, and suspended in air. None of these methods have much effect these days. Others include multi-sensor solution, Supervisory control and data acquisition (SCADA) approach, Geiger et al. (2017) and recently, IoT multi-sensors network method, Ayeni and Ayogu (2020). All these have one limitation or the other.

While these methods may be effective in some ways on the oil and gas pipe leakage monitoring, the loss of oil and gas resource, either by natural damage or increasingly by theft, continues to be recorded. Huge amount of moneys keeps on being wasted for the menace and communities terrain damaged due to oil leakages, leading to loss of lives and properties. Some of these effects are actually permanent. Many researchers in this field could easily find motivations into undertaking to provide solution; so do we too.

Perhaps, combined tactics could be a better way. Ultrasonic flow detection technology appears to be silent in all these efforts. Ultrasonic flow detection is a technology based on scientific field of Wave. Successful implementation of this science and technology would be highly sellable and thus introduces us into entrepreneurship opportunities.

In this paper Section 1 is the introduction, Section 2 anchors the study on some very relevant literature, Section 3 presents the material used and the method applied, Section 4 discussed the outcome and conclude the paper.

2. Related Works

Pipeline damages leading to leakages of oil or gas are caused mainly by "external interference, corrosion, construction defects, material failure and ground movement", Murvay and Silea (2012). Studies in the oil and gas leakage concerns are vast, because it encompasses technological, legal and security perspectives, especially in the Niger Delta Area. Technological perspective asks such questions as 'which technology will stop or prevent oil and gas leakages'. Legal concerns would figure out 'who is at fault?'; 'who bears consequences of the leakage?'; 'what is the appropriate compensation for which exact magnitude of damage caused?'; 'who gets the compensation?'. These are all facets of litigation that can emanate from oil and gas loss, in the Niger Delta Area. Security wise asks such questions as who constitutes the most effective security checks against oil and gas, theft in particular? Is local vigilante better or JTF/Navy? Our interest in this study is in the technology aspect. If the right technologies are deployed, the other two (which can be referred to as non-technical, Murvay and Silea (2012)) may be significantly reduced. In fact, the other two are heavily susceptible to corruption possibilities and other human factors and need to be minimized.

It is known in the literature that technological perspective can be classified into hardware and software, Scott and Barrufet (2003). But Murvay and Silea (2012) added the non-technical methods. The hardware cannot function well without software and the software in turn relies on the hardware for data capturing. But Turkowski and Bratek (2007) grouped all these leak detection systems into two, direct (external) and indirect (internal) detection systems. Brauers (2013) gave another categorization to the leak detection systems as continuous and non-continuous systems and thereafter regroup the continuous systems into external and internal detection systems. The external approach relies on observation outside the pipeline. Methods employed include: acoustic which uses sensors to detect noises generated by leaks, cable sensors made of continuity circuit and/or the alarm circuit, concentric cable equipped with a hydrocarbons permeable insulation, and optical wave-guide situated along a pipeline, Turkowski and Bratek (2007), Adegboye, Fung and Karnik (2019). Others include: "soil monitoring, ultrasonic flow meters and vapour sampling", Eze et al (2018). The non-continuous approach also has both external and internal detections. For the external interference, example of the "non-continuous methods of [detection] include UAV, Helicopter, Patrol Team and GIS / GPS methods". The continuous counterpart "include Satellite, WSN, Seismic-Sensor, Video monitoring and acoustic methods", Turkowski and Bratek (2007). The indirect (internal) approach utilizes properties of the flowing fluid in the pipeline, such as pressure, flow rate, volume, temperature etc.

Whether hardware, software, or non-technical approaches, there are key parameters for measurement that are of interest.

All parties to oil and gas leakage surveillance (technological, legal and security) focus on: when, where and how much of oil and gas spill. So, any surveillance system should be able to detect " the location of the leak, its size", and of course, the start time of the leak. Unfortunately, most of the approaches, largely, cannot satisfy all three combined parameters accurately, particularly the exact location and time of the incidence".

Another limitation with these methods is due to different terrain of pipeline routes. Pipeline routes in the Niger Delta region pass through three major terrains: completely buried in the ground, suspended and exposed to atmosphere, and deep water pipeline. (see figures 2 and 3). Some solution may work well with certain terrain pipeline installation but not in others. For example, Ofualagba and Ejofodomi (2020) developed amazing device known as Sensing Unit (SU) to detect pipeline invasion. The device is built of "geophones, an Arduino Mega Board, a GSM/GPRS/GPS shield, a Lora Communication Shield and a Lithium Battery Pack enclosed in a sealed anti-corrosion case" Ofualagba and Ejofodomi (2020). SU is able to sense any moving object including rain droplets falling on the ground. With all kinds of vibrations caused by object in motion, an embedded algorithm in the device is able to distinguish actual drilling or digging of the ground (to vandalize pipeline) from any other vibrations. The device uses geophones as ground vibration receptor, compute response data and send them to mobile phone. The response data include location and time of the vandalism incidence. This could be a ground breaking study.



(a)



(b)

Fig.2: (a) Buried Pipeline, (b) Exposed Pipeline, Achilike & MacDonald (2017)



Fig. 3: Underwater Pipeline, SAIPEM (2019)

However, there is a big gap in the Ofualagba and Ejofodomi (2020) study. First, geophones cannot detect a swimming fish in water neither can they sense a flying bird that can touch the pipe. This automatically cut off two out of the three pipeline route terrains from their study - the atmosphere exposed and the underwater pipeline routes (see figure 2b and 3). Secondly, normal leakages caused by corruptions and faults in the pipe cannot be detected by their solution, since no vibration occurs. By extension, any skillful excavation that avoid significant vibration may also escape SU detection.

Many of the other technical approaches are targeted at detecting vandalism and so the second limitation of Ofualagba and Ejofodomi (2020) study also apply to them. For example, state-of-the-art solution proposed by Eze et al (2018), entails "hardware components that could constitute an integrated real-time pipeline surveillance system. These include WMSN mote, base station, motion sensor, IP Camera, GIS database, GPS receivers, UAV (drone) and Flow Meters (Sensors)". Every component discussed by the reference Eze et al (2018) centers only on external interference. The study by Georgewill, Igulu and Japheth (2016) took advantage of Google Earth tool, internet cloud, Earth Geo-stationary satellite to "intercepts area images and signal" and then route to devices in Google Earth stations. This is still for external. So in all these, leakages due to corrosion and fault of the pipe are not covered.

Interestingly, "ultrasonic", Eze et al (2018) and acoustic, Turkowski and Bratek (2007) are mentioned as techniques for detecting oil and gas leakage. However, these are with respect to flow of fluids. Meanwhile, it is known from study results "that the viscosity of the liquids decreases upon exposure to ultrasound". The liquids here are petroleum products: "paraffin oil, synthetic oil and kerosene", Hamidi et al (2014) and Liu et al (2021). This effect might be suggesting alternative methods. Fortunately, ultrasonic wave can also be propagated through other mediums, including the pipe metal itself. Therefore, to state the subject matter clearly, we intend to harness the potential of ultrasonic flaw detection mechanism to spot out flaws such as corruptions and cuts in the pipe metal to determine interference into the pipeline, using integrated software that can process the signals.

The aim is to bypass the limitation existing in both the external and internal pipeline surveillance. Ability to take care of all incidence stances, whether from normal corrosion and default or from vandals, is in the potential of ultrasonic wave. In other words, any deformation in the pipe created by anything can easily be detected in wave reflection. The study is sellable. We want to see how this wave science, applied in metal flaw detection technology, transforms into entrepreneurship. Already, part of this work is being domiciled in Feasibility Giant Company Ltd.

Objectives are: to tap from the potential of ultrasonic flaw detection; to create software that utilizes data from the ultrasonic wave reflection; to determine real-time response using IoT to meet the needs of the computed results from the waves. At this stage of the research, we are representing the solution in the form of framework.

2. Materials and Methods

Of course, ultrasonic wave has been used for flaw detection in different materials. It is widely used in industries to detect "hidden cracks, voids, porosity and other internal discontinuity in metals, composites, plastics and ceramics", Nelligan (n. d. para. 1). So, we will adapt already existing ultrasonic flaw detection devices (see figure 4). The discussion that follows is largely adapted from Nelligan (n.d).

Figure 4 is a resemblance of part of the hardware component of the integrated system. Although figure 4 is experiment designed to test effect of embedding material, such as sand, on how propagation of ultrasonic wave dies off in the pipe medium, the principles applied are the same with our vision methods. It uses elementary principles of Physics in wave propagation to detect flaws in materials. The more solid and homogeneous a material is the more efficient sound waves are transmitted.

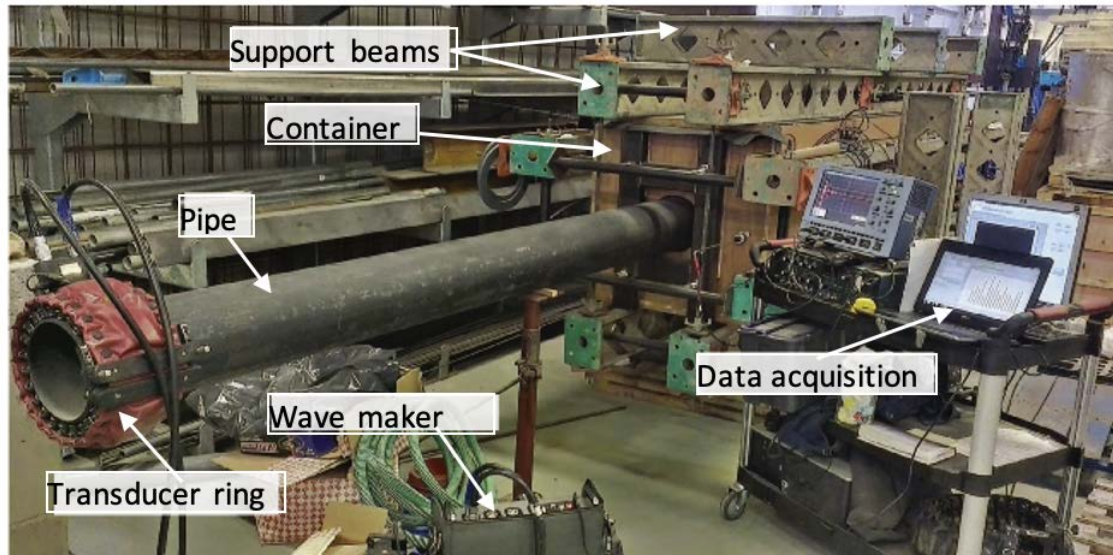


Fig. 4: Ultrasonic wave propagation in metal pipe (adapted from Leinov, Lowe and Cawley (2015))

The basic theory that underlies ultrasonic flaw detection is that travelling wave goes through medium at a specified velocity in very predictable direction. When the wave hits a boundary between two mediums, it is either reflected or transmitted based on simple rules. Three common properties of wave applied in ultrasonic flaw detection are: frequency, velocity and wavelength.

Ultrasonic waves are sound waves but not audible. Frequency is 20 KHz and above. Ultrasonic waves for flaw detection applications propagate with frequencies between 500 KHz to 10 MHz (i.e. 500,000 to 10,000,000 cycles per second). Frequencies that range in the megahertz travels very well through pipelines and other engineering materials.

Velocity of a sound wave varies according to the medium of propagation, which is affected by the elastic and density properties of the material. Different types of sound waves will travel at different velocities.

Wavelength is the distance between any two consecutive corresponding points in the wave cycle as it travels through a medium. Interestingly, wavelength relates frequency and velocity by the following simple equation:

$$\lambda = \frac{c}{f} \quad (1)$$

where λ = wavelength, c = sound velocity, f = frequency.

It is clear from this equation that the two crucial parameters required for pipeline surveillance are embedded in the equation - the exact location and time of leakage incidence. Note that wavelength limits the amount of information derivable from the behavior of a wave. The generally accepted lower limit of detection for a small flaw is one-half wavelength for the ultrasonic flaw detection.

Again, ultrasonic flaw detection utilizes another property of wave which is reflection at a boundary between two mediums. This is crucial to our study. Recall that ultrasonic wave propagation is more effective in solid and homogeneous materials. That means any flaw such as corrosion, defect, or outright cut on the pipe, using hack saw, would create two mediums. In the case of cut into the pipe, the flaw would be the homogeneous pipe and air gap in between (see figure 5). Now, as ultrasonic wave signals travels to the cut sport, there is bound to be reflection, right at the beginning of the of the cutting attempt.

Reflections at Boundary occur when sound wave energy moving through a medium encounters a boundary with another medium, part of the energy reflects back while the other is transmitted through. The quantum of energy reflected (i.e. reflection coefficient) has relationship with the relative acoustic impedance of the two mediums. Acoustic impedance is defined as density times the velocity of sound in a given medium. So, for any two mediums, the reflection coefficient may be calculated using the equation

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (2)$$

where R = reflection coefficient (energy reflection percentage), Z_1 = acoustic impedance of first medium, Z_2 = acoustic impedance of second medium.

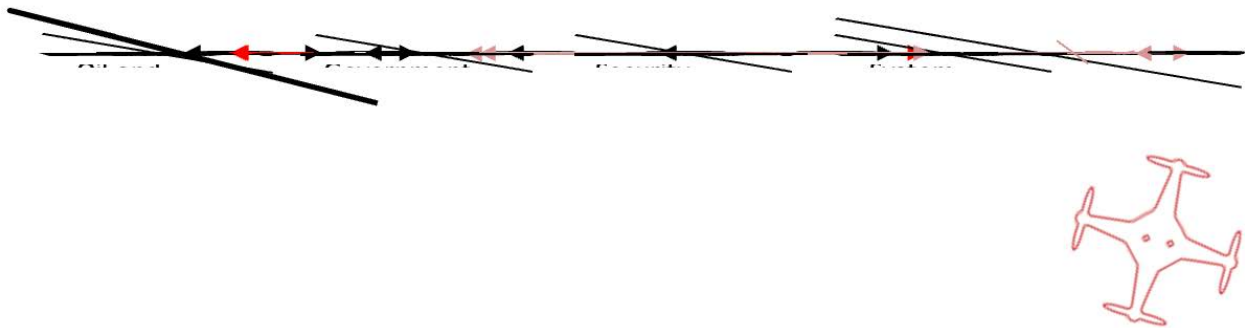


Fig.5: Proposed ultrasonic based surveillance system

Key:

1. Pipe under surveillance
2. Oil or Gas flowing through pipe being monitored against leakages
3. Transducer/wave maker constantly creating ultrasonic wave and listening to any reflection
4. propagated wave without notable reflection
5. Ultrasonic wave signal with significant reflection due to deformation of pipe
6. A mark appearing on the pipe due to cutting attempt by vandals using hack saw
7. Actual ultrasonic wave reflection from the boundary between the pipe medium and the air gap in the opening
8. A base station intercepting wave signal, forming them and transmitting them to surveillance server
9. The server process the data determines the exact location and time of leakage and quickly transmit these to workstations that would trigger alarm and activate drone movement to the location
10. Client machines for all stakeholders which include government, security, and the oil and gas company
11. Surveillance drone that moves at high speed to the incidence location



The next vital aspect of this technique is the angle of reflection and refraction because amount of wave energy that can be captured depends on the shape of the boundary between the two mediums.

Since ultrasonic beams applied for flaws detection are well defined, if the beams are reflected off a boundary, then the angle of reflection and that of incidence are equals. A wave beam hitting material surface perpendicularly reflects straight back. A wave beam hitting material surface at an angle reflects forward at that same angle.

Sound energy transmission from one medium to another will bend following Snell's Law of refraction. Just like the reflect, a beam that travels in a straight direction continues in a straight path, but a beam striking boundary at an angle get bent in accordance with the equation:

$$\frac{\sin \phi_1}{\sin \phi_2} = \frac{V_1}{V_2} \quad (3)$$

where ϕ_1 = incident angle in first medium, ϕ_2 = refracted angle in second medium, V_1 = sound velocity in first medium, V_2 = sound velocity in second medium

This relationship is an important factor in angle beam testing.

The procedure for ultrasonic flaw detection is comparative technique. Using a right reference standards together with understanding of sound wave transmission an operator can identify specific echo patterns that matches echo responses from good parts and from representative flaws. The echo pattern results may then be compared to the patterns from these calibration standards to determine its condition. This is the core of our study and research is ongoing.

4. Discussion and Conclusion

This paper discussed menace of oil and gas spillage caused either by natural means, such as corrosion and pipeline defect, or by outright sabotage in the Niger Delta area. The disaster from leakages has lead to loss of several lives and properties and very large area of land and waters have been damaged. Government of Nigeria has also lost huge revenue due to oil and gas spillage. They have spent much money to curb the incessant occurrence and indeed different approached have been used. Yet the ugly trend fails to stop.

We now studied a well known physical science discipline phenomenon that is now widely applied in technology, to propose a solution to the oil and gas leakage catastrophe. The actual window to the leakage incidence is through the pipe. Therefore, a holistic solution to the problem would be targeted at the pipe. Most of the existing solution are terrain specific (external) or resource targeted (internal) and hence fails to provide generic solution.

However, our proposed solution targets the medium of oil and gas transportation - in this case the pipe. Physics concept (ultrasonic wave) popularly used in technology for flaw detection in physical material is what we are proposing, with full hardware constituent framework provided. Since in ultrasonic flaw detection, the metal/air boundary reflection coefficient is close to 100% Nelligan (n.d), we are confident that before any interference, including corrosion, pipeline defect and cuts by vandals, gets to the point of oil and gas spillage, it would have been reported with full details. These details include exact location and precise time of vandalism or any damage. This is quite significant, because it can lead to prevention by proactive action, before the spill, and ease of litigation cases between communities and companies.

It is obvious from the foregoing that Science (Physics) meets Technology. But in combination can they meet entrepreneurship or opportunities? Intuitively, any research in Oil and Gas should be sellable. The study is part of a more robust research currently ongoing with Feasibility Giant Company and currently under incubation by the Nigerian Content Development & Monitoring Board, Yenagoa. What could be more opportunities oriented. Clearly, this is more fit for the current study.

The study is ongoing and this is the hardware constituent of the work. We are yet to look at software that puts life into it. This is subject for future insight.

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Harnessing the Potential of Biogas from Selected Grasses by Inoculation and Pretreatment Methods: A Systematic Review

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Abstract: The numerous negative environmental issues associated with the use of conventional fossil fuels for energy production, has prompted the need to engage in research studies to design more sustainable sources of fuel, to meet our ever increasing energy needs. This paper reviews, harnessing the potential of biogas from selected grasses through inoculation and pre-treatment methods, as vital source of energy in both developed and developing countries. Biogas can be produced from lignocellulose biomass, agricultural residues, municipal solid wastes, and animal wastes. Second generation lignocellulose biomass such as; elephant grass, industrial hemp, switch grass, lemon grass, and sliver grass, are often used as feedstock's for anaerobic digestion process, and as such, do not compete with food supply. The major challenge hindering the effective use of lignocellulose biomass as feedstock is the recalcitrant nature of the biomass; which hinders microbial access to the cellulose, lignin, and hemicellulose components of the plant cell structures. To increase biogas yield of the feedstock therefore, pre-treatment process need to be carried out prior to the anaerobic digestion of the biomass. Similarly, animal wastes like; cow dung, poultry droppings, swine manure, etc., can also be added to the anaerobic digestion process to serves as buffer substrate, in order to increase the rate of microbial activities; thereby, increasing biogas yields. In Nigeria, the biomass potential estimate is approximately 144 million tones/yr, plus, 13 million hectares of fuel wood, and also, potential to generate 6.8 million m3 of biogas daily from 227,500 tons of animal wastes produced per day. Nigeria reportedly has more than enough lignocellulose biomasses and animal wastes to contribute to the nation's energy demands, if properly and adequately harnessed.

Keywords: Biogas, lignocellulose biomass, renewable energy, sustainable energy.



1.1 Introduction

Biogas production is one of the emerging technologies that can contribute to the national energy demands, and also significantly reduce the amount of greenhouse emissions associated with the generation of energy through the use of conventional fossil fuels (Yoo & Pan, 2015). In this context, biogas is one of the sustainable energy sources that can be produced from many types of feedstock, including; municipal waste, agricultural/ crop residues, biomass, and animal wastes. In recent years, the use of conventional fossil fuels for human activities and the resultant effects on the environment have been a cause of great concern, especially, to policy makers. In view of this, the recent eco-technological trend in the world has resulted in the gradual shift from the conventional fossil fuels to the use of clean sources of energy. A wide range of alternative / renewable and sustainable energy sources, such as, solar, wind, geothermal, hydropower, and biomass, are currently being used to address the growing environmental issues associated with the use of fossil fuels (Wei, 2016). Extensive research studies are also constantly being carried out on how to improve these alternative energy sources to meet human energy needs.

Biomass is a renewable and sustainable biogenic organic matter that originated from natural or anthropogenic sources, obtained from plant photosynthesis, and in the presence of sunlight produce carbohydrates, the major component of the biomass (Warnars & Hivos, 2014). The term biomass, refers to crop residues, energy crops, animal wastes, wood finishes, industrial effluents, sewage, municipal solid wastes, and agricultural residues, while lignocellulose biomass comprises mainly of cellulose and hemicelluloses held together by a network of interconnected component called lignin, which physically hinders or protects the utilisable cellulose and hemicelluloses part of the cell structure from the enzymatic degradation (Warnars & Hivos, 2014). Biomass can be categorised into, first generation, second generation, and third generation biomass (Aftab et al., 2019). Energy crops like Napier grass, switch grass, miscanthus, lemon grass, are classified as second-generation feed stocks, and lignocellulose biomass, specifically grown for energy (bioenergy) production. They are widely used because they have fast growing abilities and they require less cultivation and maintenance. The perennial grasses are a perfect example of energy crops, due to their all-year round availability, less cultivation cost, less moisture, rapid growth, and ability to withstand some extreme weather conditions (Aftab et al., 2019).

The lignocellulose biomass are recalcitrant to anaerobic degradation, due to the chemical structures and properties, being one of the major challenges hindering the commercial use of the biomass (Aftab et al., 2019; Nanda et al., 2018). Thus, the application of pre-treatment procedures, which help improve the accessibility of microorganism to the cell structures; and thereby, improving biogas-yield (Nanda et al., 2018). In addition, the anaerobic digestion process can also be improved by the use of microbial inoculum (such as, animal manure which is obtained from cow dung, poultry droppings, swine dung, and rabbit dung), to facilitate the fermentative phase of the anaerobic digestion process. The addition of an inoculum is vital for the successful anaerobic digestion process, by providing a consortium of microorganisms required for the bio-conversion of the feedstock to biogas (Martinez et al., 2019; Yarberry et al., 2019).

Biogas technology if properly harnessed, offers a suitable platform to utilise the available biomass resources and animal wastes, as feed stocks for bioenergy production. The use of this technology can therefore, be a vital measure of maintaining energy supply in both developed and developing countries (Warnars & Hivos, 2014). In view of the above facts on the use of lignocellulose biomass for biogas production, the prime objectives of this paper are;

- To review the biogas potentials of selected tropical and sub-tropical grasses (elephant grass, industrial hemp plant, sliver grass, switch grass, and lemon grass).
- To review the various pre-treatment methods and inoculum used to enhance biogas yield of lignocellulose biomass.
- To assess the potentials of utilising lignocellulose biomass and livestock manure for bioenergy (biogas) generation in Nigeria.

2.1 Biogas Potentials of Selected Tropical and Sub-tropical Grasses

The biogas yields of five different tropical and sub-tropical grasses, namely; elephant grass, sliver grass, switch grass, industrial hemp, and lemon grass are reviewed in the next session.

2.2 Biogas Production from Elephant Grass (*Pennisetum purpureum*)

Elephant grass, otherwise known as Napier grass, is found in Eastern, Western and Central Africa. It is a tall plant with leaves measuring approximately 30-90 cm long and 3cm wide. Studies have reported that sixty days after cultivation the fresh biomass, the plant can yield up to (70-100) tons of biomass per hectare (Haryanto et al., 2018). It is a high-energy lignocellulose biomass with high-yield matter, and also used primarily, as animal feed, mulch to improve soil quality, and erosion control (Minmunina et al., 2015; Rusdy, 2016). Elephant grass is a fast-growing perennial energy plant, requiring low crop management and propagation practices. It is also tolerant in acidic soil, highly-adaptive, and requires little or no water for cultivation (Haryanto et al., 2018). The proximate composition of elephant grass are highlighted in table 2.1;

Table 2.1: Proximate Analysis of Elephant Grass (*Pennisetum purpureum*)

Parameter	Dry weight
Moisture (%)	68.4
Total solids (%)	31.6
Volatile solids (%)	29.6
Ash content (%)	2.0
Moisture (%)	87.3
Total carbon content (%)	10.0
Nitrogen Content (%)	2.0
pH	5.2

Source: Mbachu et al., (2019)



Studies revealed that elephant grass has a good potential as a co-substrate for biogas production, due to its high organic substrate (Mbachu et al., 2019). A higher yield of methane yield is obtained when it is co-digested with cow dung, pig manure, and poultry manure (Haryanto et al., 2018).

The use of mechanical pre-treatment of elephant grass with native potash, and subsequent inoculation with cow-dung from an already running bioreactor, resulted in production of biogas with a methane content of approximately 64.3% (Mbachu et al., 2019). However, anaerobic digestion of elephant grass and native potash, with elephant grass and cow-dung as inoculum, and raw / untreated biogas, did not result in the production of biogas, rather, only CO₂ was produced when a flame test was carried out (Mbachu et al., 2019). Furthermore, when Haryanto et al. (2018), carried out a study on the co- digestion of elephant grass and cow dung, using a batch digester, a low methane yield of 31.37% was obtained.

The study thus, recommended that the reduction in particle size (mechanical pre-treatment) of the feedstock has a significant role to play in the increase in biogas yield. This can be explained that, smaller particle sizes enable microorganisms to gain access into the cell wall structures (hemicellulose, lignin, and cellulose); thus, increasing biogas yield.

A study carried out by Sinbuathong et al. (2018) revealed that the cutting age of elephant grass has an effect on methane yield, as grass cut at 30 days after planting produced the highest biogas yield, achieving 197 (L/ kg TVS added); thus, the higher the cutting/harvesting interval, the lower the methane yield obtained. The older the age of the grass however, the higher the fiber and lignin contents, though, these are more difficult to breakdown. Pre-treatment is however, required prior to anaerobic digestion, due to the high lignin content, and thereby, creating access of microbes to cellulose and hemicellulose.

2.3 Biogas Production from Industrial Hemp Plant (*Cannabis sativa*)

Industrial hemp plant (*Cannabis sativa*), is cultivated in Africa, North America, Europe, South America, and Asia. It is an annual plant historically grown predominantly for fiber production (Kreuger et al., 2011; Prade et al., 2011). It can also be used in the manufacture of cars (car seats, dashboards and door panels) household products, clothing, bio-plastics, and industrial ropes.

It is a lignocellulose biomass with very high yields, and can withstand pests and microbial attacks i.e., it can suppress weed attack and some soil borne diseases (Kreuger et al., 2011). The composition of hemp plant is stated in table 2.2.

Biogas can be obtained from dry or ensiled hemp biomass (Rheay et al., 2020), and several studies have shown that the type of pre-treatment (especially, steam pre-treatment); the time of harvest, and plant species/variety, have an influence on the biogas yield and methane contents (Prade et al., 2011). The use of hemp biomass as a feedstock for biogas production, resulted in an optimal yield of biogas 296 GJ ha⁻¹ year⁻¹ of energy, which was obtained when harvesting was carried out in autumn (September – October). This was based on the fact that, the dry matter yield is highest during the specified period (Kreuger et al., 2011). Similarly, the anaerobic digestion of hemp hurds and cheese whey was also carried out by Papirio et al. (2020), and this produced a bio-methane yield of 242ml CH₄:g VS⁻¹ and 446 CH₄:g VS⁻¹ respectively.

Table 2.2: Proximate Analysis of Industrial Hemp Plant

Parameter	Dry Weight
Moisture (%)	7
Total Ash (%)	1.2
Lignin (%)	23
Hemicellulose (%)	25
Alpha cellulose (%)	44
Extractives (oil, proteins, amino acids, pectin) (%)	4.0

Source: Gandolfi et al. (2013)

The co-digestion of the hemp hurds and cheese whey in a ratio of 70:30 however, increased the bio-methane yield by 10.70%, as compared to the previous calculated values. Extensive use of industrial hemp plant for biogas yield is restricted, due to the legal restrictions on the cultivation of industrial hemp in many countries of the world. The other limitation is the perception barriers, as most people mistake it for marijuana, which is a close relative of the plant, and higher tetrahydrocannabinol (THC) content, than the industrial hemp plant (Kreuger et al., 2011).

2.4 Biogas Production from Sliver grass (*Miscanthus sinensis*)

Sliver grass (*Miscanthus sinensis*) is a lignocellulose biomass that can be found in Southeast Asia, Africa, Pacific Islands, and parts of Europe. It is a perennial and sustainable energy food crop that requires little rainfall, little or no fertiliser and pesticide for growth (low agrochemical inputs), and does not compete with food crops, since it requires low nutrients, and low water consumption requirements (Kiesel & Lewandowski, 2017). The composition of the plant is summarised in table 2.3.



Table 2.3: Proximate Analysis of Sliver Grass

Parameter	Dry weight
Sulphur (%)	0.1
Moisture (%)	2.9
Carbon (%)	42.5
Nitrogen (%)	1.21
Volatile (%)	72.5
Fixed carbon (%)	21.22
Ash (%)	3.38
Char (%)	27.5

Source: Osman *et al.* (2017)

Sliver grass biomass has a high potential for biogas utilisation, and it can be planted alongside food crops during crop rotation, so as, to ensure all the arable lands are not used for cultivation of energy crops. According to Kiesel. & Lewandowski (2017), the highest yield of methane gas on average 6153 m³/ha was obtained when the sliver grass was harvested in October. Maximum biogas yield of 0.0420 Nm³/kg was obtained, from anaerobic digestion of *Miscanthus sinensis* with cow dung, after 15 days Hydraulic Retention Time (Dinca *et al.*, 2017). Sliver grass is second generation lignocellulose biomass, hence pre-treatment is required prior to anaerobic digestion, due to high lignin contents; thus, paving way for microbial access to cellulose and hemicellulose contents of the cell wall (Kiesel & Lewandowski, 2017; Ben-Iwo *et al.*, 2016; Kulichkova *et al.*, 2020 & Rodriguez *et al.*, 2017). According to Menardo *et al.* (2013), the pre-treatment of *Miscanthus* with steam explosion resulted in a methane yield of 374 kgVS⁻¹, and this is three times more than the methane yield of untreated *Miscanthus sinensis*.

2.5 Biogas Production from Switch Grass (*Panicum virgatum*)

Switch grass (*Panicum virgatum*) is a tall perennial grass commonly found in USA, Canada, Mexico, and selected parts (Northern) of Nigeria. Its lignocellulose feedstock is suitable for biogas production; though, it has low fertilisation, and low nutrient requirements; but it can be harvested early, unlike the *Miscanthus*, in which optimum biogas yield is obtained during autumn harvest (Kiesel & Lewandowski, 2011; Zheng *et al.*, 2015). The proximate and ultimate composition of switch grass is presented in table 2.4.

Table 2.4: Proximate Analysis of Switch Grass

Parameter	Dry Weight
Proximate Analysis	
Moisture (%)	6.01
Ash (%)	4.01
Volatile's (%)	73.32
Fixed Carbon (%)	16.66
Ultimate Composition	
Carbon (%)	42.08
Hydrogen (%)	5.44
Nitrogen (%)	0.36
Sulphur (%)	0.05
Oxygen (%)	41.38
Hydrogen/ Carbon ratio	1.52
Oxygen/Carbon content ratio	1.45
Higher heating value	17.36

Source: Tumuluru, (2015)

The mono digestion of switch grass resulted in a biogas yield of 268.80 NmLgVS-1 and a biomethane yield of 135.31 NmLgVS-1, with an average methane content of 50.34% (Niu et al., 2015). More biogas yield was however obtained when the plant was inoculated, and co-digested with animal manure. In a study carried out by Zheng et al. (2016), the mono digestion of switch grass resulted in accumulation of volatile fatty acids; however, co-digestion of switch grass with diary manure (as inoculum) in a ratio of 2: 2, resulted in the highest methane yield of 158.6 mL/g VS-1. This was 39.0% increase in methane yield, as compared to the methane yield obtained from mono-digestion of the switch grass. Switch grass has a lower biogas yield than Miscanthus, a low germination rate, low competitiveness of seedlings, and limited availability of herbicides (Tumuluru, 2015). Switch grass is a lignocellulose biomass; thus, requiring strict and costly pre-treatment, to improve hydrolysis, minimise carbohydrate losses, and avoid formation of by-product inhibitor called pseudo lignin. The use of switch grass as a single feedstock for biogas production can result in accumulation of volatile fatty acids and digester operational problems but, the pre-treatment and co-digestion of switch grass and animal manure can improve methane production and biogas yields (Zheng et al., 2015).

2.6 Biogas Production from Lemon Grass (*Cymbopogon citratus*)

Lemon grass (*Cymbopogon citratus*) is a rich lignocellulose biomass commonly found in temperate and tropical regions of the world, and it is a perennial crop mainly cultivated for the production of essential oil, as well as, for its citral content (Bekele et al., 2017; Alfa et al., 2012). It can be planted alongside food crops during crop rotation programme, so as, to ensure that all the available arable lands are not used for cultivation of energy crops. The proximate and ultimate composition of crop is presented in table 2.5.



In a study by Alfa et al. (2012) on the production of biogas from lemon grass, it was revealed that, $6.95 \times 10^{-4} \text{ m}^3/\text{kg}/\text{day}$ of biogas was obtained before scrubbing and 0.0897 m^3 ($4.98 \times 10^{-4} \text{ m}^3/\text{kg}/\text{day}$) of bio-methane was produced after scrubbing respectively. Cooking test was conducted with the generated gas after scrubbing, and was found to have higher cooking rates for both water and rice than the un-scrubbed gas. Another study revealed that 0.125 m^3 , 0.191 m^3 and 0.211 m^3 volume of biogas were produced from lemon grass, cow dung, and poultry droppings respectively; however, lemon grass produced less volume of biogas, than the other two biomass wastes (Alfa et al., 2014), although, it produced better quality gas than poultry droppings and cow dung. The biomass also had the highest cooking rate when the cooking test was conducted, while the mono-digestion of lemon grass had the highest gas production, with a pressure of 41.8 kPa (Olaoluwa et al., 2018). In addition, digestion with only 100% chicken droppings resulted in slightly high biogas production, with a pressure of about 11.5 kPa; however, when the lemon grass was co-digested with poultry, the biogas pressure was raised to 26.5 kPa. These findings were in agreement with studies carried out by Bekele et al. (2017).

Table 2.5: Proximate Analysis of Lemon Grass

Parameters	Dry Weight
Hemicellulose (%)	28.8
Cellulose (%)	30.7
Lignin (%)	38.4
Ratio hemicellulose: cellulose: lignin	0.024
Ash (%)	2.1
pH	6.5
Total solids (%)	3.0
Volatile solids (%)	1.2
C/N ratio (%)	76.4

Source: ~~Owamah~~ *Owamah et al. (2014) & Holder et al. (2018)*

The drawback for the use of lemon grass in biogas production is that pre-treatment is required, due to its chemical composition. Lemon grass is a common medicinal plant in Nigeria, and it is widely used for citral content; thus, the cultivation for biogas production maybe faced with strict opposition from the environmentalists. On the contrary, after the plant is used for its medicinal purposes, the left-over leaves after boiling (for medicinal purpose) can serve as feedstock for biogas production. Alfa et al. (2012) suggested that, boiling of the leaves for medicinal purpose can be a form of pre-fermentation process for optimum gas production, implying that, the lemon grass leaves, which would have been disposed as wastes (after medicinal use) can be subsequently adapted for biogas production. Comparative studies of the selected lignocellulose biomass reviewed in this research are presented in table 2.6;

Table 2.6: Comparative studies of the biogas and methane potentials of the reviewed biomass

S/N	Type of lignocellulose Biomass	Biogas Volume	Methane Volume	Methane (%)	Methane Volume (inoculum)	Methane% (With inoculum)	Source
1	Elephant grass	-	-	-	-	64%	Mbachu <i>et al.</i> , (2019)
		-	-	-	-	31.37%	Haryonto <i>et al.</i> , (2017)
		197 (L/kgTVS added)	-	-	-	-	Sinabuathong <i>et al.</i> , (2018)
2	Industrial Hemp Plant	242 CH ₄ - gVS ⁻¹	-	-	-	-	Papiro <i>et al.</i> , (2020)
		296GJha ⁻¹	-	-	-	10.70% increase	Padre <i>et al.</i> , (2011)
3	Sliver grass	-	6153m ³	-	-	-	Lewandowski & Kiesel, (2017)
		-	-	-	0.0420 Nm ³ /kg	-	Dinca <i>et al.</i> , (2017)
4	Switch grass	268.80 NmLgVS ⁻¹	135.31 NmLgVS ⁻¹	50.34%	-	-	Niu <i>et al.</i> , (2015).
		-	-	-	158.6 mL/g VS ⁻¹	39% increase	Zheng <i>et al.</i> , (2015)
5	Lemon grass	6.95x10 ⁻⁴ m ³ /kg/day	0.0897m ³ (4.98x10 ⁻⁴ m ³ /kg/day)	-	-	-	Alfa <i>et al.</i> , (2012)
		-	0.125 m ³	-	-	-	Alfa <i>et al.</i> , (2014)
		41.8 Kpa	-	-	26.5 kPa	-	Olaoluwa <i>et al.</i> , (2018)

3.1 Pre-treatment of lignocellulose biomass

Majority of the microorganisms involved in the anaerobic digestion processes have slow growth rates, and thus, require long residence time, to improve the digestion process, but the best reported technique was to disrupt the chemical bonds present in the feedstock, prior to hydrolysis (Karupiah & Azariah, 2019). The feedstocks used for biogas fermentation is very abundant, in which, lignocellulosic materials, for example, crop stalks, livestock manure, domestic wastes, and some kinds of industrial wastes are the most common and easily-accessible. Due to the recalcitrant nature of lignocellulosic material however, the biogas yield usually was not high (Wei, 2016). So, in order to increase biogas production from lignocellulosic materials, some procedures like pre-treatment and/or bio-augmentation should be adopted. Since pre-treatment alters the physical, biological, and chemical structures of the feedstock, it is therefore, very important to consider the type of pre-treatment that will be used (Aftab *et al.*, 2019). Pre-treatment is an expensive process, particularly, at industrial level, and result of pre-treatment also differs, due to the difference in the ratio of cell wall constituents of the different varieties or species of plants.



3.2 Physical Pre-treatment

The different types of physical pre-treatments include; milling, ultrasonic, microwave, mechanical extrusion. Milling involves reduction in the size of biomass, which increases the surface area of the biomass; thus, allowing microbes to have access to the plant cell structures, for depolymerisation to take place (Harmsen et al., 2010; Penn State College of Health and Mineral Resources, 2020). This process is otherwise known as comminution, and this is usually performed before any processing step. Reduction in particle size can be done by milling, grinding, and chipping, which help to reduce particle sizes and organic residues, without any negative effect (Penn State College of Health and Mineral Resources, 2020). In other words, the use of mechanical/physical pre-treatment is known to increase biogas yield without the formation of products that may have inhibitory effects (Salihu & Alam, 2016). Tools, such as, hammer mills, ball mills, knife mills, and electric power blender, can be used to reduce the particle sizes; however, the efficiency of the process is based on whether the sample is dry or wet. In the milling process, factors like capital and operating costs, as well as, energy use, should be considered.

Microwave pre-treatment is another example of physical pre-treatment, and it was first used in 1984 in Japan. The benefits of using this method includes; fast processing time, less inhibitors generation, and ease of the pre-treatment (does not involve a lot of steps/processes). This method is carried out under high pressures and temperature environment which are carried out by using a closed reactor, and the temperature set between 1500C to 2500C (Aftab et al., 2019). In a study by Baruah et al. (2018), microwave pre-treatment of *Hyacinthus* spp. resulted in a 38.3% higher methane yields than the one pre-treated with water heater. In microwave pre-treatment, temperatures between 1600C and 2500C, and high steam injection are applied to the biomass; but, for better efficiency, and to avoid temperature gradients, the biomass should be reduced to smaller sizes, and afterwards, the microwave uniformly distributes the heat, to avoid formation of intermediates. Heating with microwave has a higher heating efficiency than using the water bath heating or other convention heating methods (Zhang et al., 2020). The use of microwave pre-treatment on switch grass resulted in a 53.0% total sugar, than that pre-treated using a water bath /conventional heating methods. Meanwhile, for optimum results, the microwave pre-treatment should be combined with alkaline pre-treatment methods.

Ultrasonification is a common technique for sludge treatment, and it is used in waste-water treatment facilities, as it is effective in pre-treating organic wastes, prior to anaerobic digestion (Harmsen et al., 2010). The ultrasonification process is carried out by the release nutrients via hydrolysis, caused by the disruption of bio-solid flocs in the bacterial cells, which assist in the solubilisation of the nutrient, and thereby, enhancing the anaerobic digestion process. Unlike other physical processes, ultrasonic process involves a series of combined reactions like, shearing, radicals, pyrolysis, and combustion. In a study by Salihu and Alam (2016), the ultrasonic pre-treatment of organic wastes resulted in more soluble chemical oxygen demand (Scod); thus, resulting in the 10-60% increase on biogas production as compared to the untreated sample. The study further stated that ultrasonic pre-treatment is the most efficient method in pre-treating sludges for anaerobic digestion process, and the effectiveness of the process is highly dependent on the characteristics of the sludge.

3.3 Chemical Pre-treatment

Chemical pre-treatment process involves the use of chemical reactions to break the chemical structure of the lignocellulose biomass. Processes under this method includes; acid pre-treatment, alkaline pre-treatment, organosolv, etc. Acid pre-treatment is an effective pre-treatment method that uses weak acid or strong acids (Harmsen et al., 2010) to pre-treat the biomass; however, the generation of inhibitors like furfural, and hydroxymethyl furfural make the process less attractive. The acid pre-treatment can either be a short duration and high temperature treatment (1-5 mins. temperature above 180°C; with a 5-10% wt% substrate concentration) or long duration and low treatment (30-90 mins, temperature of less than 120°C with a 10-40% substrate loading) (Aftab et al., 2019)

Chemical treatments involving dilute acids are the most common and efficient type of pre-treatment, and in this process, dilute sulphuric acid for example, is sprayed into the biomass and kept within a temperature of 160 – 220°C, for a short time /few minutes (Harmsen et al., 2010) Subsequently, the hydrolysis of the hemicellulose leads to removal of hemicellulose; thereby, increasing porosity, and allowing access of enzymes to cell structure. Inorganic acids like fumaric acid and maleic acid can be used for dilute acid pre-treatment; however, as stated earlier, one of the disadvantages of this method is the formation of inhibitors that can slow down the anaerobic digestion process. Another drawback is that, the acids can corrode the reactor vessel; thus, leading to formation of solid residues. This process is suitable for substrates/ feedstock with low lignin contents, as the process does not remove the lignin content of the feedstock (Harmsen et al., 2010). The major disadvantage is that, to low concentration (pH) of the acids, they are highly corrosive in nature, and there would be need to recycle the acids, so as, to lower operation/production costs; thus, making the process feasible (Harmsen et al., 2010; Kurmar & Sharma, 2017).

Alkali pre-treatment is usually conducted at ambient pressure and temperature, and for experiments conducted at higher temperatures, longer duration of pre-treatment is required Kurmar & Sharma, (2017). Examples of reagents used for alkali pre-treatment are KOH, NaOH, and ammonium salts, with NaOH being the most efficient in modification of chemical structure of the cell wall. In alkali pre-treatment, the solubility of hemicellulose and cellulose is reduced, unlike the in acid pre-treatment. The process involves the use of alkali reagents, such as, KOH, to break down the side chains of the esters and glycosides; thus, resulting in modification of lignin structure, swelling of the cellulose, decrystallisation of cellulose, and subsequently, dissolution of hemicellulose. In a study conducted by Kurmar & Sharma (2017), NaOH pre-treatment of different feedstocks like, hardwood, switch grass, soft-wood, was effective on feedstock containing less than 26.0% lignin contents, but no effect was obtained when dilute NaOH was used on soft-wood with a lignin content higher than 26.0%. There was a 37.0% increase in biogas production when corn stover was treated with NaOH, compared to the untreated corn stover. This treatment seemed not expensive; but quite arduous, as it involves the use of large amount of water for removal of salts from the lignocellulose (Kurmar & Sharma, 2017).

Ozonolysis pre-treatment method has been considered highly suitable for the removal of the lignin contents from the lignocellulose biomass; although, it has a negligible effect on the hemicellulose and cellulose contents of the biomass. Studies have also shown that ozonolysis of poplar sawdust resulted in reduction of its lignin content to 8.0%, and a 57.0% increase in sugar yields (Kurmar & Sharma, 2017).



Ozonolysis technique is undertaken at an optimum pressure and temperature, which does not involve any production of inhibitors, compared to other types of chemical pre-treatment. In addition, ozonolysis does not have an effect on the processes (enzymatic hydrolysis), after the treatment of feedstock; whereas, the moisture content of the feedstock has an effect on the ozonolysis, and the higher the moisture content of the feedstock, the lower the amount of lignin to be oxidised. The drawback of this method is that, it is very expensive, and requires large amount of ozone; thus, this technique is not recommended for industrial application (Kurmar & Sharma, 2017).

Organosolv pre-treatment involves the pre-treatment of the lignocellulose, with combination of organic solvents and water, which breaks down lignin, prior to the enzymatic hydrolysis of the cellulose part of the cell wall structure. The hemicellulose is also removed and this causes a further breakdown of the cellulose structure (Harmsen et al., 2010). The different solvents used for this process are; ethanol, ethylene glycol, and methanol, and the reaction requires high temperatures of up to 200°C; though, lower temperatures can be used, but this depends on the type of lignocellulose, as well as, the catalyst used (Harmsen et al., 2010). Organic and inorganic acids can serve as catalysts for the process, as the catalyst helps to reduce the temperature of the pre-treatment, and also helps to hasten up the delignification process. One of the drawback of this process is that, the solvent can inhibit the hydrolysis and fermentation step; and therefore, must be removed before the commencement of the fermentation process. To reduce the cost of the process, the solvent should be recovered, so as to improve the feasibility of the process, and also, to reduce the environmental impacts of the process, since it is a highly volatile, flammable, and high energy-consuming pre-treatment technique (Aftab et al., 2019). It is also a very expensive method; so, suitable considerations have to be made, prior to reduction of the operating costs.

3.4 Thermal Pre-treatment

Thermal pre-treatment method helps in hydrolysis of complex organic molecules of organic wastes products, such as, animal manure; thereby, enhancing anaerobic digestion. Thermal treatment results in highly soluble organic waste components, leading to transformation of hydrolysed components in the absence of oxygen, to biogas, via the production of volatile organic acid. The optimum temperature for thermal pre-treatment is between 60 °C and 180°C, and if the temperatures are higher, inhibitory compounds may be formed (Salihu & Alam, 2016). Studies have further shown that, thermal pre-treatment of swine manure, cow dung, and municipal sewage, at 170 °C for one hour resulted in 18.3%, 7.8%, and 67.8% increase in biogas productions respectively, as against the control (Salihu & Alam, 2016).

3.5 Combined Pre-treatments

Combined pre-treatments are the combination of more than one pre-treatment method, in order to enhance solubilisation, and increase biogas yield. The combinations may involve combination of physical (thermal, mechanical) with chemical (alkali and acid) pre-treatment. Steam explosion pre-treatment occurs when the lignocellulose biomass is injected with high pressure (0.7-4.8MPa) at temperatures ranging between 160 and 260°C, in a batch reactor for few minutes, and subsequently, the pressure is reduced suddenly to atmospheric pressure (Salihu & Alam, 2016). This explosive decompression results in partial hydrolysis, with degradation of hemicellulose, and distortion of lignin structure (Harmsen et al., 2010 & Salihu & Alam, 2016). The efficiency of this process depends on the size of the particles, temperature, residence time, and moisture content. The downside of this process is that, it can lead to the formation of inhibitors that hinder the fermentation process at high temperature, and also, lead to incomplete break-down of lignin-carbohydrate matrix (Kurmar & Sharma, 2017).

Mechanical/alkaline pre-treatment involves the combination of mechanical such as, milling, and alkali pre-treatment method. This process results in the creation of two parts of cell structure, one consisting of the soluble part, for example; lignin, hemicellulose and inorganic compounds, and the other part consisting of cellulose-enriched part of the biomass (Harmsen et al., 2010). The mechanical/alkaline pre-treatment process helps in degradation of lignin, while the cellulose and hemicellulose components are still present in the solid fraction. Enzymatic degradation occurs in the solid part to produce C6 and C5 sugars, and this can be combined with other fermentation routes to convert the sugars to other by-products. The combination of mechanical and alkali pre-treatment increases the effectiveness of the pre-treatment process; however, the use of chemicals are mandatory for this process, but expensive to purchase; thus, the recycling chemicals is encouraged, to reduce operational costs (Harmsen et al., 2010).

3.6 Biological Pre-treatment

Biological pre-treatment process, microorganisms, such as, white, soft-rot and brown fungi break down hemicellulose, lignin, and a little quantity of cellulose (Kurmar & Sharma, 2017). The pre-treatment requires little energy, and can take place under normal operation conditions; but the drawback back of the process is that, it has a very low rate of biological hydrolysis, and very time consuming.

3.7 Assessment of the Effectiveness of Pre-treatment Process

It is worthy to note that for any pre-treatment method to be effective, there are several factors that need to be considered. These factors include; the use of less expensive materials, less energy consumption, reduction in the use of chemicals, cost effectiveness, sustainability, reduction in size of feedstock, protection of the cellulose, and prevention of the hemicelluloses from losing their natural state/ structure (Aftab et al., 2019). Furthermore, the effectiveness of the pre-treatment process can be determined/evaluated, based on the biogas or methane yield, energy used for pre-treatment process, and the cost of the pre-treatment procedure (Li et al., 2019).

4.1 Use Inoculum in Biogas Production

The use of pre-treatment process to delignify cell component can be quite expensive and energy consumptive. To improve the digestion process, a buffering substrate can be added to the digestion process, thereby, increasing the stability of the process (Zheng et al., 2019). The alternative is, the use of inoculum rich in microorganisms, to facilitate the fermentative phase of the anaerobic digestion process. Examples of this inoculum is animal manure, like, cow dung, poultry droppings, swine dung, and rabbit dung, which contains an array of microorganisms capable of degrading the lignocellulose biomass (Martinez et al., 2019). The addition of an inoculum is vital for the successful anaerobic digestion process, by providing a consortium of bacteria required for the conversion of the feedstock to biogas (Yarberry et al., 2019).

The livestock manures have high microbial activities; rich in micronutrients and good buffering capacity; therefore, the use of livestock manure as an inoculum for anaerobic digestion is of numerous advantages. Majority of the livestock manure produced are not properly disposed and this can pose serious threat to land, water, and air, and can similarly cause serious health issues in both humans and animals, through indigestion, skin contact, and inhalation of the livestock manure (Zheng et al., 2019 & Oyedepo et al., 2019). Proper use of livestock manure will not only lead to significant reduction of these environmental and health concerns, but can foster sustainable energy production. The wastes from animals like cattle, pigs, poultry, goats, donkeys, and sheep can yield daily production of 227,500 tons, and when converted to biogas, will yield approximately 6.8 million m³ of biogas daily (Oyedepo et al., 2019). An average biogas production rates from animal wastes are summarised in table 4.1.



Table 4.1: Animal population, bio-methane potentials, and biogas production

Resources	Livestock population ('000)	Dry dung output (kg/head/day)	Animal dung output (ton)	Energy value (GJ/t)	Total <u>energy potential</u> (PJ)	Gas production (m ³ /Kg)	Methane content (%)	Retention time (days)
Poultry	184,500	0.06	4,040.55	11.0	44.45	0.31-0.56	58-50	9-30
Pigs	7,184.35	0.80	2,097,830.2	11.0	23.08	0.49-0.75	58-61	10-15
Goats	55,145.40	0.40	8,051,228.4	14.0	112.72	-	-	-
Sheep	34,687.30	0.40	5,064,345.80	14.0	70.90	0.37-0.61	64	20
Cattle	16,400	1.80	10,774,800	18.5	199.33	0.20-0.33	-	-

Source: Oyedepo *et al.* (2019)

Several studies have been carried out on the use of livestock manure as an inoculum in the anaerobic digestion process. Zheng *et al.* (2019) reported that, the addition of a 20.0% wet-mass ratio of switch grass to livestock manure resulted in 86.0% rise in methane production, as against the mono-digestion of switch grass alone. In a similar experiment, on production of biogas from co-digestion of cassava and wood ash; the use of pig manure as inoculum resulted in an increase in methane production, compared to the co-digested mixture without pig manure (Sokan-Adeaga, & Ana, 2015).

4.2 Potential Capacity of Biomass for Biogas Production in Nigeria

The generation of energy from biological sources like wood, biological wastes, animal wastes, agricultural residues, energy crops, etc., is referred to as biomass energy (Okonkwo *et al.*, 2021). The feedstock (organic matter) can be transformed through anaerobic digestion to produce biogas, for heat or electricity consumption. Nigeria has an estimated biomass and bioenergy potentials of approximately 144 million tones/yr., and 13 million hectares of wood fuel respectively. The renewable energy resources estimate of Nigeria is summarised in table 4.2.

Table 4.2: Renewable Energy Resources of Nigeria

Energy Type	Resources Estimate
Hydropower	23,400 MW
Small Hydropower	3500 MW
Solar Radiation	3.5-7.0 kWh/m ² -day
Wind Energy	2.0 m/s at 10m height (mainland)
Nuclear	Not yet quantified
Biomass	144 million <u>tones/yr.</u> plus 13 million hectares of fuel wood
Wave and Tidal Energy	Not yet quantified
Geo thermal	Not yet quantified

Source: Energy Commission of Nigeria (2021).

Nigeria has an estimated energy content of 6.0×10^9 MJ, and currently uses 80 million m³ (4791.7kg) of wood fuel yearly, of the total energy content for cooking and other domestic purposes. Approximately 200 million tons of dry lignocellulose biomass can be obtained from grasses, shrubs, and forage, which contributes to a total of 2.28×10^6 MJ of energy in Nigeria, (Okonkwo et al., 2021).

According to Ngumah et al. (2018), Nigeria has one of the highest rates of deforestation in the world, with approximately 55.7% of the forest reserves destroyed between 2000 and 2005. Research data indicated that Nigeria is the 8th largest emitter of methane gas, with approximately 20 billion m³ accounting for 13.0% of the world's total emissions. Gas flaring, from our abundant oil and gas sources make up 69.0%, and the other 28.8% originates from untreated organic wastes. Moreover, the country's wood fuel and charcoal consumption rates between 1989 and 2000 were estimated at 32.0-40.0%; and as at the year 2000, the nation's energy demand for both wood fuel and charcoal, was reported to be 39 million tons. Nevertheless, the substitution of the use of fossil fuels like kerosene to biogas, for cooking, will help reduce amount of green-house gas emissions. The biogas research studies in Nigeria begun in 1982, and since then, research activity has been on-going; however, not sufficient enough to make the technology attain commercial status in the country (Okonkwo et al., 2021). The status of biogas technology is currently at institutional research level and pilot studies; thus, the commercialisation of this technology, and connection to national energy grid are faced with challenges, such as, lack of funding, poor coordination between research units, lack of awareness, government policies, cost recovery, and no clear business model. There is therefore, urgent need to diversify the nation's energy mix to incorporate all sustainable sources of energy, and this will be a right step in achieving the United Nations Sustainable Development Goals 1(No Poverty), 2 (Zero Hunger), 6 (Clean water and sanitation), 7 (Affordable and Clean Energy) and 13 (Climate Action).



5.1 Conclusion

In Africa, the technological advancement on the use of biogas energy on a commercial scale has been really slow, in spite of the continent's huge biomass resources. In a country like Nigeria, the biomass resources estimate is approximately 144 million tones/yr. plus /and 13 million hectares of fuel wood. Furthermore, the country has a potential to generate 6.8 million m³ of biogas daily from the 227,500 tons of animal wastes produced per day. Nonetheless, the continent's biogas potential on an industrial level has not been fully harnessed due to a number of limiting factors. Biogas technology is therefore, a sustainable source of energy, which can even, be obtained from the anaerobic digestion of various very cheap, and easily-accessible feedstock, for production of heat and electricity. The feedstock used for biogas fermentation is very abundant, in which, lignocellulose materials like, crop residues, livestock manure, domestic wastes, and some types of industrial wastes are the most-common, and easily available. The structural and chemical properties of lignocellulosic biomass make them resistant to anaerobic degradation, and this is one of the factors is also responsible for their limited commercial usage in sustainable energy productions.

Due to the recalcitrant nature of lignocellulose biomass however, suitable pre-treatment methods need to be carried out to increase microbial access to the cell wall structures, during the hydrolysis phase of the anaerobic digestion process. This increases the biogas yields, and reduces the hydraulic retention time of the anaerobic digestion process. Depending on the type of process, the pre-treatment of lignocellulose biomass may be viewed as energy consumptive, time consuming, and expensive. But, there are certain factors that need to be considered in the choice of a suitable pre-treatment method, to ensure optimum biogas yields. The biogas yields of a lignocellulose biomass can also be improved by the addition of an inoculum to the anaerobic digestion process. Animal wastes like poultry droppings, cow dung, swine manure etc., can be added as inoculum to the anaerobic digestion of lignocellulose biomass, and this can serve as a buffer substrate to increase microbial activities; hence, facilitating optimum biogas yields. In this regard, to improve and develop the biogas technology, there needs to be collaborative efforts between the government, research institutions, and private sectors, to identify the gaps in the possibility of large scale commercialisation of biogas technology, for lasting supplementary solution to the continent's looming energy crisis.

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Domestic Market Prospects For White Petroleum Products In Nigeria – An Emprical Analysis Of The Price-Effect

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Abstract: The ancestry of the Nigerian petroleum industry is traced o 1956 when oil was struck in commercial quantities in the country. The first refinery came into existence in 1965, giving rise to the expansion of the oil industry vocabulary to include white products. The study focused on the trends and prospects for PMS, Kerosene and LPG in the domestic market. A 22-year time series data was collected from reliable data bases on the prices, production and consumption of the white products. Treated econometrically, trend equations were estimated which showed that, both consumption and prices were upward in the period under review for PMS, consumption was going downwards while prices were increasing for kerosene, and consumption was increasing for LPG even with higher prices. Generally, the domestic market prospects were bright for all the three white products investigated. The raw data were then transformed into their natural logarithm equivalents to estimate price elasticities for PMS and kerosene, and to estimate the cross elasticity between the rival products kerosene and LPG. The own-price elasticity of demand for petrol came up as $LPMS_C = 1.816 + 0.02605 LPMS_P$, as $LKero_C = 2.110 - 0.3542 LKero_P$ for kerosene, and $LGLPG_C = -0.608 + 0.497 LKero_P$ for the case of cross elasticity. These interpreted to an unusual inelastic demand for PMS, an expected highly elastic demand for kerosene, and a theoretically correct positive cross elasticity between kerosene and LPG. These provided basis for recommendations that urgent steps be taken to find substitutes for PMS to eliminate the captive market situation, and that the promotional efforts at LPG consumption be intensified, among others.

Keywords: Consumption, prices, price-effect, substitution-effect, elasticity.

Introduction

The ancestry of the world's petroleum industry is formally traced to 1859, in Pennsylvania, USA. According to Nnadi (1997), it was Edwin Drake that first struck oil following earlier studies conducted by Benjamin Siliman. In Nigeria, the era of oil began effectively in 1956 when oil was discovered in Oloibiri, in present Bayelsa State.

According to the Department of Petroleum Resources, Nigeria currently has five operational refineries (DPR). The Nigerian National Petroleum Corporation (NNPC), which controls four of these facilities, and Niger Delta Petroleum Resources, which controls the fifth refinery, are both owned and run by the Federal Government of Nigeria (NDPR). Two of the refining facilities, KRPC and WRPC, include petrochemical complexes that utilise their refinery intermediates to produce petrochemical precursors, which are employed in the production of petrochemical intermediates, in addition to the one fully-fledged petrochemical plant.

The Port Harcourt Refining Company (PHRC) Limited, a division of the Nigerian National Petroleum Corporation, runs the 210,000 barrels per day complicated conversion unit in the Port Harcourt refinery (NNPC). Two refineries in Alesa-Elеме, Rivers State, make up the PHRC. The new plant's nameplate capacity is 150,000 barrels per day, whereas the old refinery's nameplate capacity is 60,000 barrels per day. The old refinery was first put into operation in 1965. Liquefied petroleum gas (LPG), premium motor spirit (PMS), dual purpose kerosene (DPK), automotive gas oil (AGO), low pour fuel oil (LPFO), and high pour fuel oil are all produced at the plant using bonny light crude oil (HPFO).

With an estimated capacity of 100,000 barrels per stream day when it was built in 1978, the Warri refinery was debottlenecked to 125,000 barrels per stream day in 1987. The Warri Refining & Petrochemicals Company (WRPC) Limited, a NNPC subsidiary, runs the refinery, which is situated at Ekan, Warri, DeltaState.



The Kaduna refinery, which is situated in Kaduna, Kaduna state, has an estimated refining capacity of 110,000 barrels per day. The facility, which has a complicated conversion design, is administered by Kaduna Refining and Petrochemicals (KRPC) Limited, a division of the Nigerian National Petroleum Corporation (NNPC). The KRPC has a fuels plant that was put into operation in 1983 and a 30,000 MT per year petrochemical plant that was built in 1988.

The 1,000 barrels per day NDPR refinery is situated in Ogbelle, Rivers State. The facility is designed to produce diesel for internal use, and any extra is sold to the neighborhood. The Niger Delta Exploration and Production (NDEP) Company, an upstream affiliate, runs a flow station within its marginal field from which the plant receives crude oil.

As the Core Investors, Indorama purchased the Eleme Petrochemicals Company Limited (EPCL), formerly known as the Indorama Eleme Petrochemicals Company Limited (IEPL), from the NNPC in 2006 as part of the privatization program. The complex, which is in Eleme, Rivers State, processes natural gas liquids (NGL) from the Nigerian Agip Oil Company (NAOC) and propylene-rich feed from the Port Harcourt refinery to produce a variety of polyethylene and polypropylene products.

The operational activities of these refineries have given rise to an enlargement of the industry vocabulary to include what are now called white and black petroleum products. Wartsila (2023) lists black oils to include crude oil, furnace oil, fuel oil, tar and asphalt while the white span kerosene, petrol and benzene.

The Nigerian domestic market for oil is dominated by the white products, including cooking gas or, more technically, liquefied petroleum gas. The paper is concerned with the trends in this market for these white products, namely petrol, kerosene and cooking gas. The objective is to establish whether the market for each is expanding or contracting as a basis for extrapolating into the future; to pinpoint the role of open market prices in these trends; and to evaluate the alleged substitutability between kerosene and cooking gas on account of the trends affecting them.

Literature review

Traditionally, white oil products in Nigeria have exhibited price volatilities. It is very apposite to begin with a conceptualization of the term volatilities as applied in economic characterization of commodity markets. Lamy (2011) defines volatility as a measure of price variation from period $t - 1$ to time period t . It is a measure of price movement between two selected points in time. The vocabulary of price management recognises a period of sustained price increases as high volatility and the obverse of sustained decrease in prices as low volatility.

Among those who have theorized on commodity price fluctuations are Chambers and Bailey (1996). The authors studied price fluctuations of seven storable commodities tradeable in the open market and subject to random shocks both to supply and demand. Before then, Haldane (1934) had simply subscribed to the theory of business cycles as the major explanatory factor for commodity price variations. Speculative consumer response to price oscillations have also been implicated in the causation of further instability of prices. Kilian (2010) is confident that unlike shocks to the flow demand or flow supply, "speculative demand shocks can cause large immediate effects on the real price of oil, for example in response to geopolitical events". The final consequences can adversely affect the market for the products involved. It is against this gloomy background that Yepez-Garcia and Dana (2012) emphasise the relationship between demand elasticities and price volatilities.

The literature so far has not addressed the specific Nigerian market consequences for white petroleum products. This is the gap to be filled by the present undertaking.

Economic theorising

Economics is the theoretical ancestor of the market as a concept. A pedestrian understanding of the market is that it is a place where buying and selling takes, the transactions being mediated by the agreed prices. Ordinarily, buyers, representing demand, prefer lower prices while sellers, representing supply, prefer higher prices (Nnadi and Falodun, 2003). These contradictory tendencies create a constant tension in the market around the prevailing prices. It follows that higher prices create an economic incentive for buyers to reduce their consumption and foment a rivalrous economic incentive for sellers to increase their production.

This brings up what the economist has characterised as the price-effect. The price effect is the outcome of a change in the ratio of prices between goods in the market place, which manifests in evident variations in their consumption. The changes in consumption are a direct function of the fact that the price effect bifurcates into the substitution effect and the income effect. The substitution effect occurs when more of a product, say LPG, is consumed as a direct result of a fall in its price as more of it is substituted for another product, say kerosene. According to McKenzie (2022), "reducing the relative price of an item makes consumers more likely to purchase it when compared to other options". On the other hand, the income effect occurs when the price ratio so changes as to make a consumer end up with more money to spend, as if his income increased. As explained by McKenzie (2022), "a consumer can now buy more of the item with the same amount of money". The consumer could, equivalently, buy the old quantity and still go home with some change, called the consumer surplus, simply because there is a fall in the price of the product.

However, the ease with which the substitution and income effects operate depends on the price elasticities of the goods involved and the cross elasticities between them, bordering on whether they relate in complementarity or in substitutionality.

The foregoing scenarios will shortly become more comprehensible as empirical Nigerian white product market data are presented and analysed.

Methodology

To begin with, the study follows an ex post facto research design, a quasi-experimental procedure in which historical or secondary data are collected, analysed quantitatively and relied upon for interpretations, conclusions, projections and extrapolations.

Numerical data relating to production, consumption, and pricing of white petroleum products were sourced from internationally recognised reliable data bases. The data span a 22-year time series from 2000 to 2021. Specifically, the variables of interest are Premium Motor Spirit (PMS) or petrol; Kerosene and Liquefied Petroleum Gas (LPG) or cooking gas. The sources are the US Energy Information Administration (EIA) and the National Bureau of Statistics.

Analysis of data

The statistics collected were analysed using an econometric approach. At the first stage, the raw data were used to calculate the trends showing price movements and the corresponding consumption of the products in the period under review.

In the next stage, the figures were transformed into their natural logarithm equivalents and subjected to the Ordinary Least Squares (OLS) regression to produce the elasticity coefficients.

All computations were done applying the Minitab Version 18 software. The log transformations were carried out with the IBM SPSS Version 20 software.

The data details are presented in the Appendix 1.

Results

The Trends

First, the trends for the consumption of PMS, Kerosene and LPG are presented both geometrically and algebraically in Figures 1 – 6.

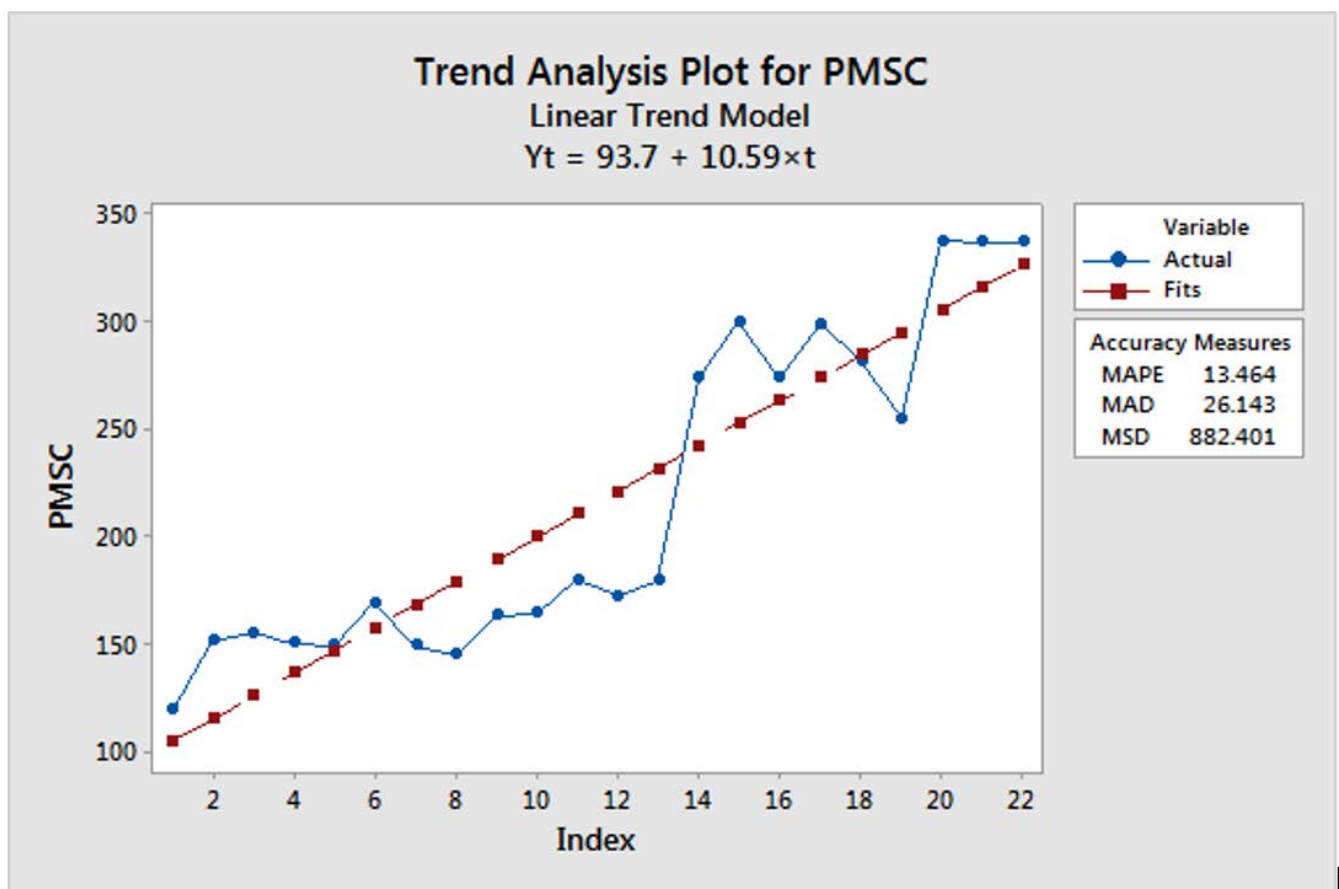


Figure 1: Petrol consumption trend

In the period under review, petrol consumption was rising as represented by the positive coefficient (10.59) in the embedded equation of the trend and demonstrated by the upward slope of the fitted line in Figure 1.

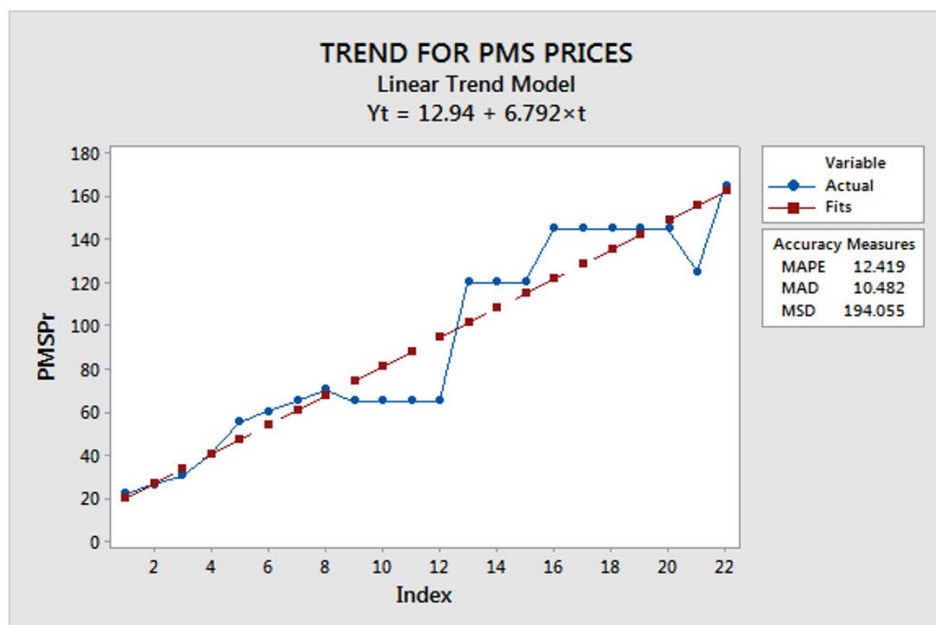


Figure 2: Petrol price trend

In the period under review, petrol prices were rising as represented by the coefficient (6.792) and demonstrated by the upward fitted line in Figure 2.

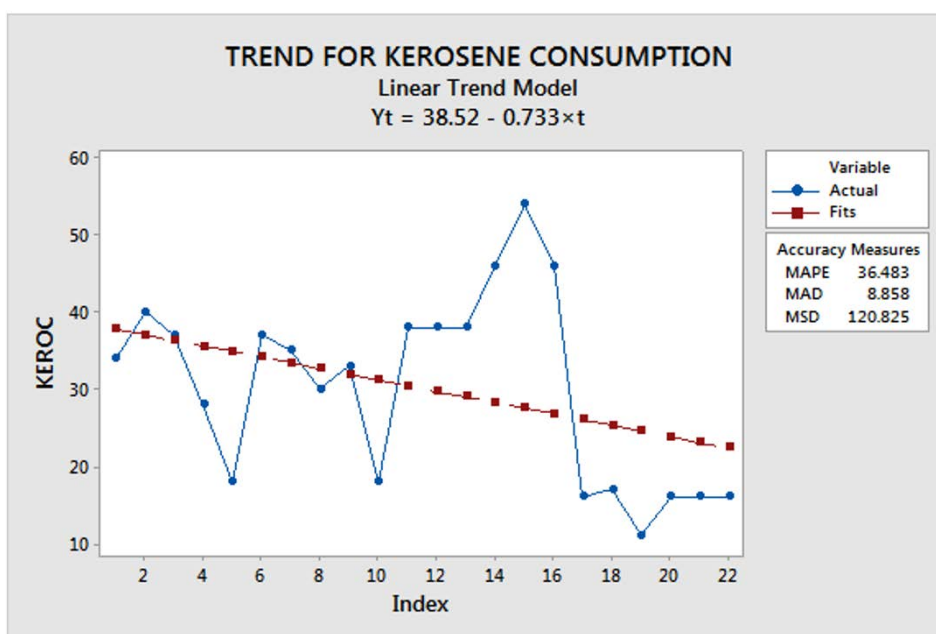


Figure 3: Kerosene consumption trend

In Figure 3, kerosene consumption was decreasing as indicated by the negative coefficient of the trend function (-0.733) and reflected in the downward sloping fitted line.

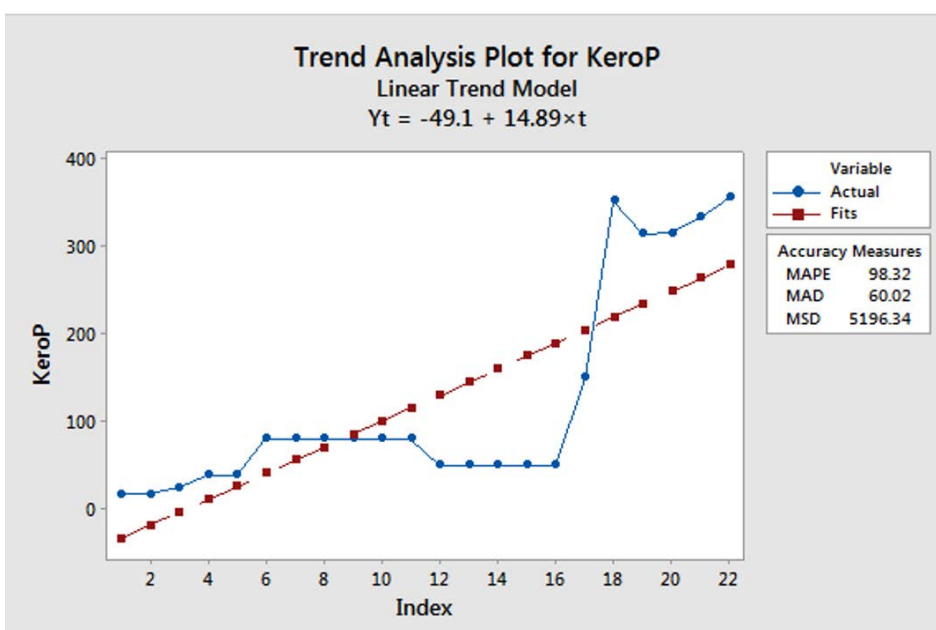


Figure 4: Kerosene price trend

In Figure 4, kerosene prices were increasing as indicated by the positive coefficient of the trend function (14.89) and reflected in the upward sloping fitted line.

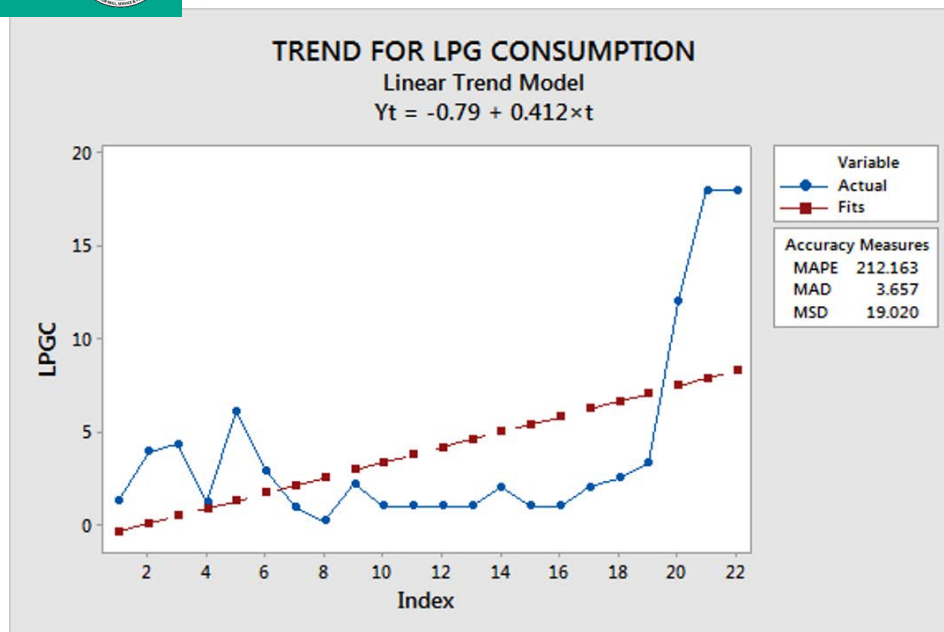


Figure 5: Cooking gas consumption trend

Figure 5 shows that cooking gas consumption was rising in the period under review. Algebraically, this is indicated by the positive parameter estimate of the trend function (0.412) and reflected in the upward slope of the fitted line.

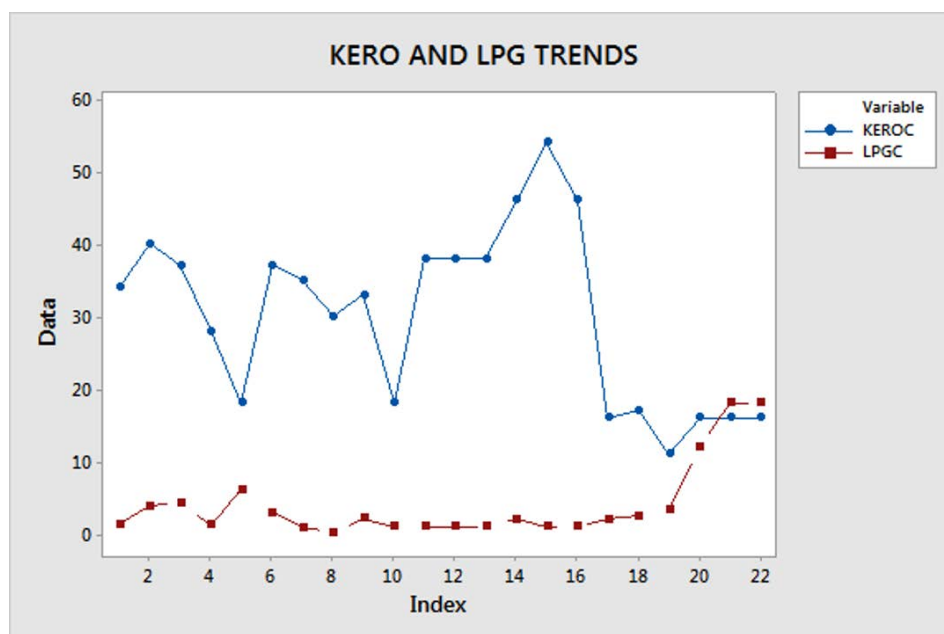


Figure 6: Kerosene and cooking gas consumption trends

Figure 6 features the trends for kerosene and cooking gas consumption together. The graphs show that kerosene consumption has been much higher than that of cooking gas, until after 2020 when the later equalised and outpaced the former.

The Elasticities

The elasticities here, presented as mathematical equations, are of two kinds. The first is the own-price elasticity – which measures the responsiveness of quantity demanded or consumed of a product to small changes in its own price. Of interest here are PMS and kerosene. How do these products respond to changes in their own prices as an index of the market outlook ahead? This is answered by looking at the coefficients of own-price elasticities.

The other elasticity is a cross-price elasticity – which measures the responsiveness of the demand or consumption for one product to changes in the price of another product, usually a rival or substitute. In this case, the attention is focused the responsiveness of the demand for cooking gas to small changes in the prices of kerosene, a well known rival and substitute for cooking purposes.

Details of the estimation procedure and output from computations are shown in the Appendix 2.

According to Li et al (2023), the elasticity of demand is a measure of the response of demand to changes in factors that affect it. Demand elasticity is an economic ratio without units, which can be defined as the ratio of the relative (percentage) change in demand scale to the relative (percentage) change in demand influencing factors.

There are identifiable positive and negative differences in demand elasticity. "A negative value means that demand-influencing factors are negatively correlated with demand, and a positive value means that demand-influencing factors are positively correlated with demand".

Urbanek (2019) introduced the research results of demand price elasticity in the field of public transport and determined the non-price factors of urban public transport service demand changes based on the public transport ticket sales data of a city in Poland.

Regression Equation 1: Own-price elasticity of demand for petrol

LPMSC	=	1.816 + 0.2605 LPMSP
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Regression Equation -2: Own-price elasticity of demand for kerosene

LKeroC	=	2.110 - 0.3542 LkeroP
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Regression Equation 3: Cross-price elasticity of demand between LPG and Kerosene

LGLPGC	=	-0.608 + 0.497 LKeroP
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Findings

The major findings from the results of the analysis so far are now itemised.

First, in the case of PMS reflected in equation 1, both prices and consumption were rising in the period under review. This is at variance with conventional economic wisdom which dictates that rises in price should lead to downfall in demand and consumption. However, the upward trend in both prices and consumption of the product is a direct outcome of own-price inelasticity of demand, which has been satisfactorily analysed by Wu (2020), in relation to social services. That is, demand continues to grow even in the face of rising prices. Why this condition can prevail is the fact that the product of concern, PMS, currently does not have a close substitute in the Nigerian domestic market. Virtually all automobiles in the country run on petrol. Electric cars and their solar driven counterparts are at present, not in vogue in the country. Compressed Natural Gas (CNG) and Liquefied Natural Gas (LPG) have been proposed as alternative to petrol in Nigeria due to the country's significant natural gas reserves. Yet, in the foreseeable future, cars on Nigerian roads will continue to run on petrol. What all of this scenario boils down to is that the domestic market is still very ripe for PMS. Consumers do not respond to price increases by curtailing their consumption in the same proportion.

An important explanatory factor in this respect is that car ownership, in the meantime, is a status symbol in Nigeria. It represents part of the verifiable evidence of arrival at, at least, the middle class of income group. No participant of that class would risk the loss of prestige and status associated with inability to keep driving at full mast on account of fuel costs. Another explanation is that many vehicles on the road are driven for commercial purposes; so that there is a compulsive desire to remain on road to make more money from passengers and commuters. This finding bears close affinity to results from studies in the demand for public transport by Yaman and Offiaeli (2022). It needs to be remarked that PMS is viewed as



a necessary product by the Nigerian consumers of the product. Several studies confirm that necessary products manifest the characteristic of own-price inelasticity of demand as confirmed in out-of-pocket health care expenditures in Mauritius (Jeetoo and Jaunky, 2022).

Second, findings relating to kerosene and based on equation 2 are presented. Computations emerge with a negatively signed parameter estimate for demand in response to price movements. It is hereby reported that the analysis finds that the consumption of kerosene is quite both alert and responsive to own-price shifts. The response takes the form of a decline in consumption following an increase in prevailing prices. The situation is characterised as one of own-price elastic demand.

The situation of kerosene, in which the consumption is reducing as price increases is usually a function of certain factors, among which are the type of product, the availability of substitutes, income levels and the time period. The nature of the product is its existence as a necessity or a luxury. If a necessary good, demand will not respond so noticeably at a slight price increase. On the other hand, if a luxury good, the demand will swiftly and evidently respond to price changes. The ease with which substitutes can be found for the product is another reason it could have an elastic demand and consumption pattern. The easier it is to find substitutes, the higher the speed and magnitude of change in demand sequel to a price change. Income levels are such that at low levels, consumers are touchy in the face of price changes and so easily reduce their consumption as price increases. According to Jeff Bezos of Amazon, “we’ve done price elasticity studies, and the answer is always that we should raise prices. We don’t do that, because we believe – and we have to take this as an article of faith, -- that by keeping our prices very, very low, we earn trust with customers over time, and that that actually does maximize free cash flow over the long term.”

The third scenario here relates to equation 3 – the cross elasticity between cooking gas and kerosene. The results show that as kerosene prices go up, more of cooking gas is consumed. What does this reveal concerning the relationship between kerosene and cooking gas? Is this indicative of complementarity or substitutability? Pathak (2021) defines “complements as goods that individuals prefer to consume with another good, and substitutes as goods individuals prefer to consume instead of another good.” If the price of a complement rises the demand for both goods will fall, if the price of a substitute rises the demand will rise for the rival product. When estimated numerically as in equation 3, cross-price elasticity means that a complement will have a negative cross-price elasticity, since if the change in price is positive, the change in quantity will be negative and vice-versa; while a substitute will have a positive cross-price elasticity, since if the change in price is positive, the change in quantity will be positive and vice-versa.

The case of kerosene and cooking gas under consideration therefore emerges as a classical example of substitutes since the coefficient of cross elasticity came up with a positive sign (+ 0.497).

Conclusions

The market trends and prospects for PMS are still excellent at present. The characteristics of the outlook are one of price inelasticity and a growing trend of consumption. The combination of these two features means that pump prices can be increased without loss of demand and patronage. In fact, historical data shows that PMS customers will keep up their consumption levels and even increase this irrespective of pump price movements. A product enjoying such patronage is experiencing an ongoing boom. The main reason for this state of affairs is that for now, PMS has no close substitute and therefore seems to be benefitting from a captive market in consumers have nowhere else to turn to.

In respect of kerosene as a white product, the reverse is the case. That is, the market prospects are grim and the message could be said to be one of an obituary! Analysis of the database made it clear that there is unusual sensitivity of consumption to price movements. The upward trend in prices gave rise to a downward trend in consumption as patronage experiences a decline in the face of any increase in open

market prices. Any market manifesting this characteristic is exiting and both production and sales will begin to nosedive. The major explanatory factor for this gloomy market outlook for kerosene is that it suffers from very keen competition from the substitute and rival product LPG.

The domestic market prospects for LPG are analogous to that for PMS. It is shown in the foregoing sections that LPG consumption is characterised by an upward trend. This is in spite of recent price increases. The rationale for the brightening prospects of LPG in the domestic market is that as the price of kerosene goes up, demand shifts away from it to LPG. Secondly, money not spent on high-priced kerosene becomes available as extra spending power to LPG users who thus increase their consumption. With this background, the current campaign for LPG consumption should continue.

Recommendations

The data yielded by this study provides adequate basis to recommend that conscious and more sustained effort be made to promote the use of fuels alternative to PMS. It is not cheering that demand remains strong for the product in the face of rising prices simply because there are no substitutes and consumers find themselves in a captive market. Captivity and bondage whether economic , political or spiritual are objectionable.

It has been proposed that PMS consumers consider Compressed Natural Gas (CNG) and Liquefied Natural Gas (LPG) as alternative to petrol. To begin with, Nigeria has substantial natural gas reserves. According to PricewaterhouseCoopers (2020, the country has

the

ninth

(9th)

largest

proven

reserves

globally, with

202

trillion

cubic

feet

(tcf)

of

untapped

proven

gas



reserves, of which the estimated

recoverable

gas

is

139.4

tcf. For another, CNG will yield some consumer surplus to car owners on account of its lower comparative prices per unit energy output. It has been found also that CNG offers improved fuel efficiency and is more environmentally friendly in terms of the total quantum of emissions of atmospheric pollutants.

Also, on the basis of available evidence, the campaign to promote a switch to LPG seems to be yielding results. It is therefore recommended that the promotion be intensified through public enlightenment using the now heavily patronised social media, in addition to the traditional use of the print and electronic media. Although the foregoing time series shows an upward trend in the use of the commodity, consumption is still low in Nigeria compared to some West African countries that do not produce oil.

The advocacy needs to expand to creating sufficient terminals, enhancing current storage facilities and the supply of the pressurised transportation vehicles that can take the commodity from the source to the ultimate consumers.

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Appendix 1

Year	Petroleum consumption (Mb/d)	PMS Consumption	Per capita consumption (litre per day)	GDP (B2015\$PPP)	Dry Natural Gas production (billion cubic feet)	Dry Natural Gas consumption (billion cubic feet)	Kerosene production (million barrels per day)	Kerosene consumption (million barrels per day)	Kerosene Average Price/litre (Naira)	Liquefied Petroleum Gas production (billion cubic feet)	Liquefied Petroleum Gas consumption (million barrels per day)	Average Price/5kg (Naira)
2000	227	119	22	358.6	440	238	13	34	17	1	1.3	
2001	287	151	26	378.8	526	219	31	40	17	4.2	3.9	
2002	292	155	30	438	501	225	29	37	24	4.9	4.3	
2003	280	150	40	470.2	717	301	16	28	38	1.1	1.2	
2004	280	149	55	513.7	770	329	12	18	38	0.7	6.1	



2005	310	169	60	546.8	791	366	25	37	80	2.9	2.9	
2006	250	149	65	579.9	1006	386	15	35	80	0.3	0.9	
2007	217	145	70	618.1	1148	374	6.3	30	80	0.2	0.2	
2008	277	163	65	659.9	1159	433	13	33	80	2.2	2.2	
2009	237	164	65	713	912	147	6.1	18	80	1	1	
2010	292	180	65	781.3	1024	178	12	38	80	2.8	1	
2011	280	172	65	819.9	1459	542	14	38	50	4.3	1	
2012	278	179	120	854.7	1503	547	11	38	50	3.4	1	
2013	429	274	120	900.4	1356	585	14	46	50	6.4	2	
2014	454	300	120	957.1	1548	667	13	54	50	2.4	1	
2015	410	274	145	982.9	1594	665		46	50		1	
2016	425	299	145	967.5	1475	653		16	150		2	2002
2017	426	282	145	975	1571	609		17	352		2.5	2377
2018	452	255	145	993.8	1609	654		11	315		3.3	2276
2019	483	338	145	1015.9	1635	663		16	316		12	1976
2020	493	337	125	997.5	1644	668		16	334		18	1979
2021	494	337	165	1033.5	1568	777		16	356		18	2923
2022												

Source: NBS - PMS Price Watch

US Energy Information Administration (EIA)

a. Statista

b. US Energy Information Administration (EIA)

c. National Bureau of Statistics

Appendix 2

MINITAB OUTPUT

Regression Analysis: LGLPGC versus LPMSP

Method

Rows unused	1
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Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	0.8905	0.8905	4.72	0.042
LPMSP	1	0.8905	0.8905	4.72	0.042
Error	20	3.7725	0.1886		
Lack-of-Fit	9	2.5157	0.2795	2.45	0.082
Pure Error	11	1.2568	0.1143		
Total	21	4.6630			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.434309	19.10%	15.05%	0.00%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-0.608	0.444	-1.37	0.186	
LPMSP	0.497	0.229	2.17	0.042	1.00

Regression Equation

LGLPGC	=	-0.608 + 0.497 LPMSP
--------	---	-------------------------

Fits and Diagnostics for Unusual Observations

Obs	LGLPGC	Fit	Resid	Std Resid	
8	-0.699	0.339	-1.038	-2.45	R

R Large residual

LPGC PMSP LGLPGC

LPMSP



1.3	17	0.11394	1.23045
3.9	17	0.59106	1.23045
4.3	24	0.63347	1.38021
1.2	38	0.07918	1.57978
6.1	38	0.78533	1.57978
2.9	80	0.46240	1.90309
0.9	80	-0.04576	1.90309
0.2	80	-0.69897	1.90309
2.2	80	0.34242	1.90309
1.0	80	0.00000	1.90309
1.0	80	0.00000	1.90309
1.0	50	0.00000	1.69897
1.0	50	0.00000	1.69897
2.0	50	0.30103	1.69897
1.0	50	0.00000	1.69897
1.0	50	0.00000	1.69897
2.0	150	0.30103	2.17609
2.5	352	0.39794	2.54654
3.3	315	0.51851	2.49831
12.0	316	1.07918	2.49969
18.0	334	1.25527	2.52375
18.0	356	1.25527	2.55145
*	*	*	*

SPSS OUTPUT

Variables Entered/Removed ^a			
Model	Variables Entered	Variables Removed	Method

1	Log10PMSP ^b	.	Enter
a. Dependent Variable: Log10LGPC			
b. All requested variables entered.			

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.437 ^a	.191	.151	.43431
a. Predictors: (Constant), Log10PMSP				

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.891	1	.891	4.721	.042 ^b
	Residual	3.772	20	.189		
	Total	4.663	21			
a. Dependent Variable: Log10LGPC						
b. Predictors: (Constant), Log10PMSP						

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Fraction Missing Info.	Relative Increase Variance
		B	Std. Error	Beta				
1	(Constant)	-.608	.444		-1.370	.186		
	Log10PMSP	.497	.229	.437	2.173	.042		
a. Dependent Variable: Log10LGPC								

Regression Analysis: LGLPGC versus LPMSP
Method



Rows unused	1
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Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	0.8905	0.8905	4.72	0.042
LPMSP	1	0.8905	0.8905	4.72	0.042
Error	20	3.7725	0.1886		
Lack-of-Fit	9	2.5157	0.2795	2.45	0.082
Pure Error	11	1.2568	0.1143		
Total	21	4.6630			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.434309	19.10%	15.05%	0.00%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-0.608	0.444	-1.37	0.186	
LPMSP	0.497	0.229	2.17	0.042	1.00

Regression Equation

LGLPGC	=	-0.608 + 0.497 LPMSP
--------	---	-------------------------

Fits and Diagnostics for Unusual Observations

Obs	LGLPGC	Fit	Resid	Std Resid	
8	-0.699	0.339	-1.038	-2.45	R

R Large residual

Regression Analysis: LGLPGC versus LKeroP Method

Rows unused	1
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Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	0.8906	0.8906	4.72	0.042

LKeroP	1	0.8906	0.8906	4.72	0.042
Error	20	3.7724	0.1886		
Lack-of-Fit	7	1.9910	0.2844	2.08	0.121
Pure Error	13	1.7814	0.1370		
Total	21	4.6630			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.434305	19.10%	15.05%	0.00%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-0.608	0.444	-1.37	0.186	
LKeroP	0.497	0.229	2.17	0.042	1.00

Regression Equation

LGLPGC	=	-0.608 + 0.497 LKeroP
--------	---	--------------------------

Fits and Diagnostics for Unusual Observations

Obs	LGLPGC	Fit	Resid	Std Resid	
8	-0.699	0.337	-1.036	-2.44	R

R Large residual

Regression Analysis: LKeroC versus LKeroP

Method

Rows unused	1
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Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	0.4518	0.451790	24.24	0.000
LKeroP	1	0.4518	0.451790	24.24	0.000
Error	20	0.3727	0.018637		
Lack-of-Fit	7	0.2520	0.036003	3.88	0.017
Pure Error	13	0.1207	0.009286		



Total	21	0.8245			
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Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.136517	54.79%	52.53%	46.83%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	2.110	0.139	15.13	0.000	
LKeroP	-0.3542	0.0719	-4.92	0.000	1.00

Regression Equation

LKeroC	=	2.110 - 0.3542 LKeroP
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Fits and Diagnostics for Unusual Observations

Obs	LKeroC	Fit	Resid	Std Resid	
5	1.2600	1.5499	-0.2899	-2.21	R

R Large residual

Cross Correlation Function: KeroP, KeroC

CCF - correlates KeroP(t) and KeroC(t+k)

Cross Correlations

Lag	CCF
-14	-0.050871
-13	-0.006702
-12	-0.070951
-11	-0.004714
-10	0.006743
-9	0.029254
-8	0.087507
-7	0.351461
-6	0.525624
-5	0.332555
-4	0.196643
-3	-

	0.056668
-2	- 0.332288
-1	- 0.622979
0	- 0.724250
1	- 0.549574
2	- 0.358424
3	- 0.189271
4	- 0.062475
5	0.068664
6	0.086249
7	0.071539
8	0.041610
9	0.010917
10	- 0.077786
11	- 0.096188
12	- 0.063263
13	- 0.033425
14	0.044611

Regression Analysis: LKeroP versus LKeroC

The regression equation is

LKeroP = 4.120 - 1.547 LKeroC

Model Summary

S	R-sq	R-sq(adj)
0.285270	54.79%	52.53%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	1.97277	1.97277	24.24	0.000
Error	20	1.62758	0.08138		
Total	21	3.60035			



LPGC	PMSP	LGLPGC	LPMSP	LKeroP	LKeroC	KeroP	LKeroPr	KeroC	
1.3	17	0.11394	1.23045		1.23	1.53	17	1.23	34
3.9	17	0.59106	1.23045		1.23	1.60	17	1.23	40
4.3	24	0.63347	1.38021		1.38	1.57	24	1.38	37
1.2	38	0.07918	1.57978		1.58	1.45	38	1.58	28
6.1	38	0.78533	1.57978		1.58	1.26	38	1.58	18
2.9	80	0.46240	1.90309		1.90	1.57	80	1.90	37
0.9	80	-0.04576	1.90309		1.90	1.54	80	1.90	35
0.2	80	-0.69897	1.90309		1.90	1.48	80	1.90	30
2.2	80	0.34242	1.90309		1.90	1.52	80	1.90	33
1.0	80	0.00000	1.90309		1.90	1.26	80	1.90	18
1.0	80	0.00000	1.90309		1.90	1.58	80	1.90	38
1.0	50	0.00000	1.69897		1.70	1.58	50	1.70	38
1.0	50	0.00000	1.69897		1.70	1.58	50	1.70	38
2.0	50	0.30103	1.69897		1.70	1.66	50	1.70	46
1.0	50	0.00000	1.69897		1.70	1.73	50	1.70	54
1.0	50	0.00000	1.69897		1.70	1.66	50	1.70	46
2.0	150	0.30103	2.17609		2.18	1.20	150	2.18	16
2.5	352	0.39794	2.54654		2.55	1.23	352	2.55	17
3.3	315	0.51851	2.49831		2.50	1.04	315	2.50	11
12.0	316	1.07918	2.49969		2.50	1.20	316	2.50	16
18.0	334	1.25527	2.52375		2.52	1.20	334	2.52	16
18.0	356	1.25527	2.55145		2.55	1.20	356	2.55	16
*	*	*	*	*	*	17	1.23	34	

Effect Of Thermal Treatment On Metakaolinization Of Ikere Kaolin

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ABSTRACT

This research work studied the effects of heat treatment on transformation of kaolinite based clay mineral, obtained from IkereEkiti, Ekiti state, Nigeria, to a more reactive metakaolinite phase. The chemical analyses of the sourced sample showed predominance of Al₂O₃ (54.40 %), SiO₂ (48.76%), and a trace of Fe₂O. The samples were heat-treated at varied degree of temperature (600-800oC) for 30 minutes. X-ray diffractogram (XRD) and scanning electron microscope (SEM) showed a transformation from native crystalline kaolinite structure to amorphous metakaoline phase, as temperature increases until 800oC. Brunner emmett teller (BET) for surface area measurement shows an increase of 70.3% and 932.45% in surface area and pore volume respectively at 800oC.

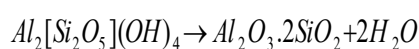
Keywords: metakaolinization, dehydroxilation, Ikere kaolin, silica and alumina



1. INTRODUCTION

Kaolin, also known as china clay (Tsegahun *et al.*, 2021) is a clay mineral that is rich in kaolinite (Saikia *et al.*, 2003) and other non-accessory minerals, otherwise called non-clay minerals, such as carbonates, feldspars, quartz (hydro) oxides of iron and aluminum (Salahudeen *et al.*, 2015). Abundance deposits of kaolin have been reported across Nigeria, including Ikere deposit of Ekiti state (Ogundana *et al.*, 2015). Altogether, Nigeria is said to have a reserve estimate of about 3 billion metric tonnes of kaolin (Alabi and Omojola, 2013). Ikere Ekiti is a local government in Ekiti state Nigeria, flanked in the west by Ise Ekiti having a coverage area of about 2km. the town is located between latitude 7°00', and longitude 5°15'E (Ogundana *et al.*, 2015)

Metakaolin is the reactive phase of silica and alumina (Tironi, *et al.*, 2012) after kaolin has been thermally treated at high temperature. Kaolin is inert in its natural state (Suphada *et al.*, 2018), hence, there is a need to transform it to a reactive phase called metakaolin before it could be used for certain processes. Hong and his team gave an equation (eqn.1) showing metakaolinization of kaolin, where SiO_4 tetrahedral structure is retained after heat-treatment while the AlO_6 octahedral layer is transformed to tetrahedral shape, oxide and water vapour through a process known as dehydroxylation.



The possible transformation phases of kaolin when heat-treated are; metakaolin (500-850°C), spinel (900-950°C) and mulite cristobalite ($\geq 1050^\circ\text{C}$) (Salahudeen, 2018). Due to its very reactive nature, metakaolin is used in the production of catalysts used in fluid catalytic cracking (FCC), aromatic alkylation and adsorbents (Thiago *et al.*, 2012). Metakaolin is also a source of supplementary cementitious materials to reduce the CO_2 emissions and energy consumption originated in cement production (Tironi, 2012). There have been several published research articles on thermal activation of kaolinite from various deposits within and outside Nigeria. For example, these authors have respectively worked on kaolin deposits; (Victor2020) from Kankara (Lawrence *et al.*, 2018); Darazo (Mgbemere *et al.*, 2018); Aoko (Kovo *et al.*, 2009); Ado Odo (Adeoye *et al.*, 2017) and, Elefun (Ayoola *et al.*, 2018).

Therefore, it is important to note that kaolin compositions, crystallinity and mineralogy, are major factors in thermal activation temperature determination of kaolin which vary with location (Bergaya, 2006; Murat *et al.*, 1992). Hence, this work aims at determining the appropriate metakaolinization temperature for Ikere kaolin.

2. Method

Kaolin samples were taken from the site at three different locations of 7 meters away from each other. The samples were washed and sun-dried. The dried samples were ground to particle size below 0.5mm with mortar and pestle 2kg of each

of the three ground samples was sieved with 300 μm sieve. The sieved samples were homogenized and characterized using Electron Microscope with Energy Dispersive X-ray (SEM/EDX), X-ray Diffractometer (XRD), X-ray fluorescence (XRF) and Fourier Transform Infrared (FTIR).

The thermal treatment of the homogenized kaolin was carried out at varied temperatures; 600°C, 700°C and 800°C for 30 minutes in a muffle furnace, after which each of the heat-treated samples was characterized with SEM/EDS, XRD, XRF, FTIR and BET.

3. Result and Discussion

3.1. Chemical and Mineralogical Analysis

Chemical analysis of the sourced kaolin (Table 1) carried out on XRF shows percentages of the major components as SiO_2 (48.76 %), Al_2O_3 (54.4 %), and Fe_2O_3 (2.86 %), while the following oxides occurred in trace amounts; P_2O_5 (0.233 %), V_2O_5 (0.144 %) and MnO (0.044). This indicates a silica-alumina rich raw material. The silica content may have been complimented by the presence of quartz.

The percentage concentration of Fe_2O_3 (2.86) is high which may be responsible for the seemingly red coloration of the kaolin. Hence, beneficiation to reduce the iron content may be necessary for certain applications. The molar ratio of Si?Al is 1.3, comparing favourably with theoretical composition of kaolinite in terms of oxides; 46.5

SiO_2 , 39.5% Al_2O_3 , and 13.96% H_2O (Siaka et al.,2003).

The XRD pattern of the crude kaolin in Figure 1 shows crystalline peaks that correspond to kaolinite, quartz, orthoclase, nacrite, garnet and illite. If comparison is drawn between the crude and the calcined samples, a transformation from crystalline kaolinite phase to metakaolinized amorphous phase (Figures 1-2) is observed as heating temperature rose to 800°C. However, there is an observable quartz mineral phase peak at about 5000 counts in the metakaolinized XRD diffractogram in Figure 2, this indicates the presence of structurally stable quartz mineral, when treated at temperature below 1050°C (Salahudeen, 2018).

Table 1: XRF analysis of Ikere kaolin

Oxide	Fe_2O_3	TiO_2	CaO	K_2O	SiO_2	Al_2O_3	MgO
percentage	2.86	1.10	2.82	1.04	48.76	54.40	0.48

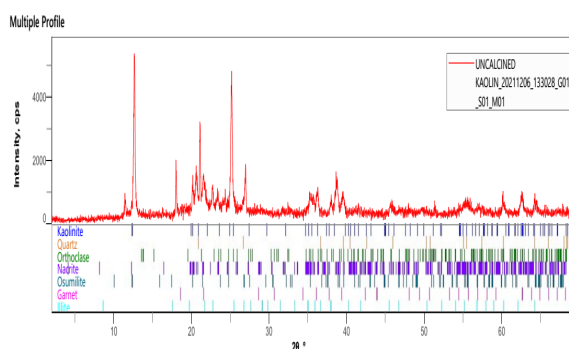


Figure 1: XRD pattern of the raw kaolin

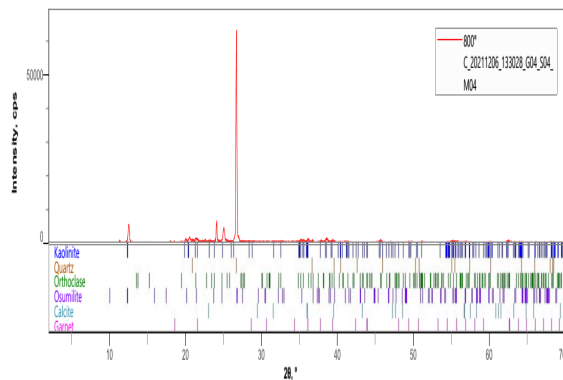


Figure 2: XRD pattern at 800°C

3.2. FTIR analysis

The IR spectroscopy of the raw kaolin in Figure 3 shows the characteristics kaolinite band 3683.3 cm^{-1} high vibration frequency which corresponds to O-H. Similarly, the functional group bands at 1248 cm^{-1} , 1118 cm^{-1} and 997 cm^{-1} are bands of Si-O-Si stretching of kaolinite. However, the low spectra intensity signals poor clay crystallinity. IR spectrogram of metakaolinite in Figure 4 shows disappearance of 3683 cm^{-1} band. This indicates the removal of constituent water and formation of metakaolin.

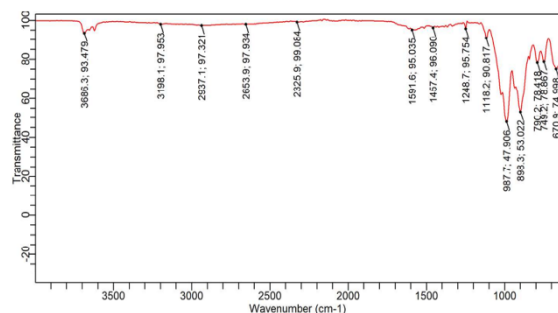


Figure 3: FTIR spectrum of raw kaolin

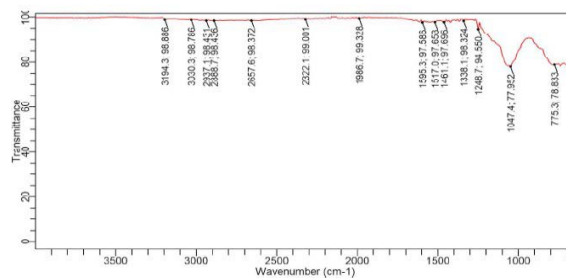


Fig. 4: FTIR spectrum of metakaolin at 800°C

3.3. SEM analysis

Figures 5 and 6 show the SEM images of the raw and dehydroxylated kaolin. The morphology of the raw kaolin shows a platelet structure of kaolinite, while Figure 6 shows distorted amorphous morphology of a metakaolin at 800°C

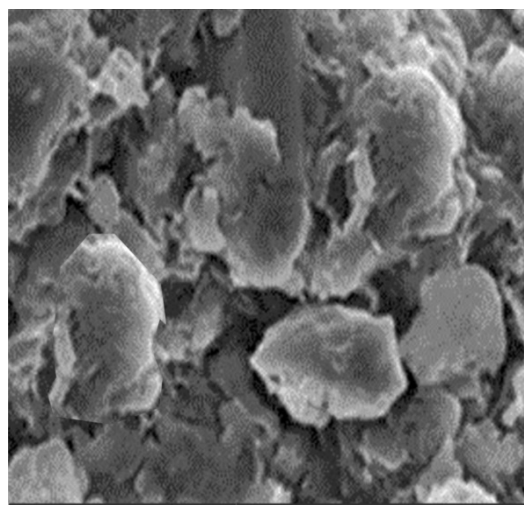


Figure 5: SEM of kaolinite

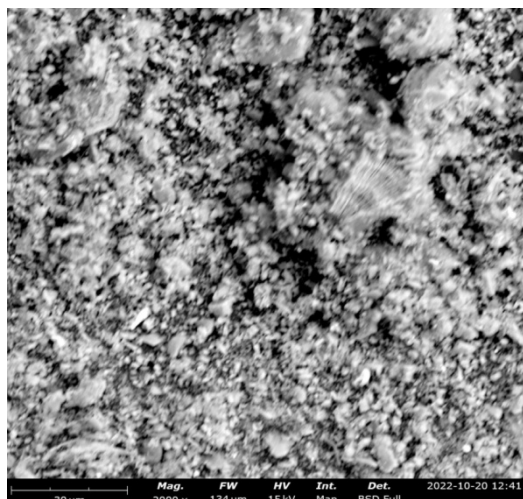


Figure 6: SEM of metakaolin at 800°C

The BET result of Ikere kaolin in Table 2 shows the surface area increased from 10.98 m²/g and pore volume from 0.0000493cm³/g to 18.7m²/g and 0.000509 cm³/g respectively at 800°C. Hence, respective percentages increase of 70.3% and 932.45 cm³/g in surface area and pore volume. The increase in surface area and pore volume might be due to collapse in the crystalline structure of kaolinite (Salaudeen, 2018) at elevated temperature as revealed in the SEM image, the property which is responsible for the high reactivity of metakaolin

3.4. BET Analysis

Table 2: BET analysis of kaolinite and metakaolin

Temperature	Surface area (m ² /g)	Pore volume (cm ³ /g)
Room Temperature	10.98	0.0000493
800°C	18.7	0.000509
% increase	70.3	932.45

4.0. Conclusion

This research work shows phase transformation to metakaolin at temperature of 600-800 as seen in the XRD.

The metakaolin obtained is amorphous as revealed by SEM and FTIR

There is a high increase in the surface area (70.3) and the pore volume (932.45) of the metakaolin as shown by BET, which makes the metakaolin obtained a suitable raw material for catalyst synthesis

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Developing Present and Future Skills for The Oil and Gas Industry in Nigeria: An Assessment Of The Contribution Of Petroleum Training Institute, Effurun To Development Of Manpower

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Abstract: The oil and gas industry is rapidly evolving in response to technological advancement. The industry is fast embracing digital technology in terms of automation and artificial intelligence in its operation in line with the Fourth Industrial Revolution. Today's workforce more than ever before requires advanced skills to be able to perform efficiently in the oil and gas industry due to this shift in technology. Overall, the aim of this paper is to inform policy makers, educators and members of the oil and gas industry of the various roles and responsibilities that human resources play in the oil and gas industry and how these can be fulfilled through the Petroleum Training Institute, Effurun (PTI). This research paper is specifically aimed at evaluating the effectiveness of the Petroleum Training Institute, Effurun in providing high technological training and skills for the oil and gas industry going into the future. To achieve this objective, the study relies on Modern Organizational Theory which describes an organization such as PTI in this instance as an open system that interacts with its environments (internal and external) to develop and grow in the market; including that it encompasses multiple aspects and levels of organizational environments—micro- and macro-environment and that an organization is a dynamic system that responds and adjusts to the changing environment such as the new global dynamics in technological advancement in meeting human and industrial needs. The study adopted the method of empirical survey for data gathering through which sample of 25 participants were drawn for participation in 5 Focus Group Discussions (FGDs) from five (5) selected oil and gas companies in the Niger Delta region. In other words, Focus Group Discussion (FGD) is the data collection technique adopted while content analysis was deployed as method of data analysis and discussion of findings. Essentially, this study evaluated the quality of training and efficacy of skills offered by PTI as being experienced by the selected oil and gas companies in the Niger Delta who engaged graduates of PTI in their various establishments. The results of this research revealed the richness of the goals of PTI and its efficiency in providing human resources support to the development of the oil and gas industry in Nigeria. However, the study recommended on ways to increase the capacity of the Institute to adequately provide competent and efficient human resources support to the development of the oil and gas industry in Nigeria as well as the Sub-Sahara Africa using a comprehensive digital deployment in teaching and learning in order to measure up to global competition in technological skill and knowledge production.

Key Words: Artificial Intelligence, Automation, Manpower, Oil and Gas Industry, Technological Advancement

1. Introduction

The oil and gas industry in Nigeria has undergone significant growth and development. However, there have been challenges in terms of the development of manpower and the acquisition of necessary skills and expertise (Akinwale, 2018). The lack of requisite skills and technical expertise has hindered the participation of small and medium-sized enterprises (SMEs) in the industry (Akinwale, 2018). This has been attributed to a low level of local content implementation (Akinwale, 2018). The introduction of the marginal field development initiative and the local content act by the Federal Government of Nigeria aimed to address these challenges and promote the inclusion of SMEs in the industry (Akinwale, 2018). Institutions such as the Petroleum Training Institute (PTI) in Effurun have played a crucial role in addressing the need for skilled labour in the oil and gas industry (Akinwale, 2018).

PTI is involved in research and development in various areas including petroleum research and development (Abdullahi et al., 2021). This institute has been instrumental in providing training and development opportunities for personnel in the power sector, both in technical and managerial aspects (Abdullahi et al., 2021). The development of the oil and gas industry in Nigeria has been driven by the exploration and exploitation of crude oil, which has become a major source of income for governments and citizens (Okonofua et al., 2020).

However, there is a need to improve the efficiency of resource use in industry (Agri et al., 2018). Technological innovation and the acquisition of technology-using skills and knowledge have been identified as key factors in promoting efficiency in the industry (Agri et al., 2018). As the energy sector transitions towards renewable and sustainable energy sources, it is important to identify the relevant skill sets that will be needed in the future (Okoroafor et al., 2022). A study conducted in Nigeria aimed to identify the skill sets of oil and gas students and professionals that will be relevant to advancing a renewable and sustainable energy future (Okoroafor et al., 2022). This study highlights the need for a coordinated approach from government, industry, and academia to facilitate this transition (Okoroafor et al., 2022).

The Nigerian government has recognized the importance of petroleum engineering education in the development of the oil and gas industry (Amadi et al., 2022). Higher institutions of learning, such as universities and polytechnics, have been established to provide training in petroleum engineering and related courses (Iyeke & Bello, 2019). These institutions aim to develop locally qualified Nigerians to operate in the oil industry (Iyeke and Bello 2019). The Federal Ministry of Education is also involved in curriculum reviews to foster sustainability in the industry (Amadi et al., 2022).

As technology advances and the industry becomes increasingly complex, it is essential for educational institutions to provide high-quality training and skills development programs to meet the evolving needs of the sector. One such institution is the Petroleum Training Institute (PTI) located in Effurun, Nigeria. This evaluation aims to assess the effectiveness of PTI in



equipping students with the necessary technological training and skills required for successful careers in the oil and gas industry.

Founded in 1973, PTI has established itself as a premier institution for technical education in the petroleum sector. The institute offers a wide range of programs and courses, including engineering, geosciences, drilling technology, petroleum marketing, and industrial safety. By focusing exclusively on the oil and gas industry, PTI aims to provide students with specialized knowledge and skills that are directly applicable to the sector's demands.

The effectiveness of PTI's training programs can be gauged by the employability of its graduates in the oil and gas industry. Alumni success stories and the job placement rate serve as crucial indicators of the institute's effectiveness. PTI's reputation for producing skilled and competent professionals has led to its graduates being sought after by leading companies in the sector. Furthermore, feedback from employers regarding the competence and performance of PTI graduates provides valuable insights into the quality of training imparted by the institute.

This is crucial as rising skills shortages have become one of the main problems facing the oil and gas industry in Nigeria. To meet rising demand, oil and gas exploration and production need to increase significantly in the coming years, and this requires more petroleum engineers, production engineers, drilling crews and geoscientists. However the supply of technical staff is by no means sufficient to meet demand. Increasingly there are delays and contracts have to be renegotiated (Financial Times, 2007). Many organisations face growing difficulty in meeting their schedules as they have found out that simply paying higher salaries is not sufficient to secure enough qualified personnel.

It is thus important to assess the ongoing efforts of PTI to adapt to industry advancements and emerging trends. The institute should have mechanisms in place to review and update its programs to incorporate technological advancements, industry best practices, and environmental sustainability considerations. By continuously evolving and aligning its programs with industry requirements, PTI can ensure that its graduates remain competent and relevant in the ever-changing oil and gas sector.

2. Statement of Problem

Nigeria's oil and gas industry is a critical sector that contributes significantly to the country's economy. However, the industry faces challenges in terms of developing and maintaining a skilled workforce with the necessary present and future skills. This hinders the industry's growth, efficiency, and ability to adapt to changing technologies and market demands. One of the key institutions addressing this issue is the Petroleum Training Institute (PTI) in Effurun, Nigeria. PTI plays a crucial role in providing training and development opportunities for personnel in the oil and gas industry.

However, it is essential to assess the effectiveness of PTI's contribution to the development of manpower in the oil and gas industry. The problem statement revolves around the need to evaluate the extent to which PTI training programs align with the present and future skills required in the oil and gas industry. This assessment will help identify gaps or areas of improvement in the training provided by PTI. Additionally, it will determine the effectiveness of PTI in equipping graduates with the skills necessary to meet industry demands.

Furthermore, the problem statement also addresses the need to evaluate the impact of PTI training programs on the overall development of manpower in Nigeria's oil and gas industry. This assessment examines whether PTI graduates are adequately prepared to contribute effectively to the industry and meet the industry's manpower needs. By conducting a comprehensive assessment of PTI's contribution to the development of manpower in the oil and gas industry, policymakers, industry stakeholders, and PTI itself can identify strategies to enhance training programs and bridge any skill gaps.

This evaluation will also provide insights into the future skills required in the industry and help PTI align its training programs. In conclusion, the problem statement focuses on assessing the contribution of the Petroleum Training Institute, Effurun, to the development of present and future skills for the oil and gas industry in Nigeria. By evaluating the effectiveness of PTI training programs and their impact on the industry's manpower development, stakeholders can identify areas for improvement and ensure that the industry has a skilled workforce capable of meeting evolving needs.

3. Literature Review

Technology was defined by Omorogbe (2003) as the systematic knowledge for the manufacture of a product, the application of a process or the rendering of a service, whether that knowledge be reflected in an invention, an industrial design, a utility model, or a new plant variety, or in technical information or skills, or in the services and assistance provided by experts for the design, installation, operation or maintenance of an industrial plant or for the management of an industrial or commercial enterprise or its activities.

For Alleyne (cited in Atsegbua, 2004), petroleum technology is the systematic and formulated knowledge about the industrial art in the field of petroleum which is capable of changing as a result of researches leading to new discoveries. It has two aspects: the tangible and the intangible. The intangible relates to knowledge and skill and the ability to use the knowledge to repair; maintain and create modifications and innovations of the physical assets which are imported while the tangible aspect is the physical embodiment of the technology: the equipment, tools, machinery and plant.

According to Okwelum (2022), technology transfer is first and foremost a transfer of knowledge and not just of a finished product that embodies high technology. A transfer occurs



when a technological application that has been developed and applied by one organization is utilized and effectively applied by another organization. For a transfer to be successful, the personnel of the recipient organization must be capable of applying the knowledge transferred by virtue of adequate education. In other words, education forms the bed-rock upon which the drive for technology transfer can be predicated. It also requires continued education by way of in service training and retraining. Furthermore, technology transfer does not lie solely with the industry but through the process of training and adequate exposure to same. A transfer of only one aspect of technology does not meet the ultimate goal of technology transfer or solve the dependency syndrome.

The transfer of petroleum technology from the international oil companies to Nigeria is one of the most complex issues confronting the nation in the oil and gas sector. Although State participation in the industry has been achieved, the management and technical knowledge of the industry is still controlled by the international oil companies. The issue is whether the nation should rely on the oil companies being the principal experts and custodians of petroleum technology for the conveyance of petroleum technology to her or whether she should make efforts, devise programmes and create arrangements which compel transfer of technology or whether she should steal same.

Technology transfer is ordinarily not in the interest of the owner organisation, thus its quest has the potentials of giving rise to an adverse relationship between the economic power of the multi-nationals and the political sovereignty of developing nations not only because the ultimate goal of technology transfer is the control of all the phases of the industry which no doubt is at cross purposes with the ultimate goal of the multi-national oil companies, but because the owner must have invested in terms of money, research, resources, time and energy to develop a technology which may not be adequately compensated whereas the recipient state or organization may hanker to acquire a technology with ease and as some may advocate, may steal the technology (Okwelum, 2022).

According to Alegimenlen (1999), another reason for monopoly in technological acquisition is traceable to the sophisticated and capital intensive nature of technology which changes rapidly and becomes obsolete quickly because of the fast-growing and changing nature of the industry and the need to find new ideas to deal with the changes. In their work, *Oil and Class Struggle*, Nore and Turner (1980) have canvassed that oil companies resist technology transfer as part of their efforts to avoid displacement in the industry. The companies, according to the authors, strive to block entry into the downstream operations by creating obstacles to transfer. A transfer of petroleum technology is not profitable to the oil companies and it does not further their main concern which is to extract oil and make profit. The national companies are viewed in the light of potential competitors and disrupters of market shares and profits in the industry, thus the multi-nationals are reluctant to transfer technology and

managerial skills and their development policies making the host-nations technological dependent.

In his own view, Atseugba (2004) stated that the clamour for technology transfer is actuated by the desire of the host-state to fulfill the principles of permanent sovereignty and control of the oil industry by the State. However, a nation state that cannot boast of its technology to operate and run its oil industry cannot effectively assert its sovereign rights over its mineral resources. In fact, at the base of the sovereign rights of a state over its mineral resources is technology. It is also fundamental to note that the responsibility for the transfer of technology does not lie on oil companies since it does further their interests. In the light of the fact that the international oil companies would not readily transfer petroleum technology to the host-state which needs same for its economic growth and political autonomy under market economy dictated by the invisible hands of demand and supply, the host-state has to recourse to legal regimes and regulations to compel transfer of technology from the international oil companies to the host-state.

Meanwhile, Imomoh (cited in Atseugba, 2004) affirmed that the best way to bring technology is to bring people into Nigeria to work for a long time. As postulated by Shell, a leading player in the Nigerian oil and gas industry, oil technology is driven by five imperatives: to maximize long term production; access new hydrocarbon reserves; drive down unit costs; reduce negative impact on the society and the environment; and use the most appropriate technologies. In these five directions, the oil industry technology must be actuated to benefit from the 'heavily funded world-wide technology research efforts and the best-practice process sharing to attain world class performances.

Similarly, Nathaniel (2008) concentrated on developing Nigeria's oil and gas industry's technological capabilities. Among other things, the study's goal was to pinpoint and examine the factors that influence the development of technological capabilities in Nigeria's oil and gas sector. The study's approach included a thorough literature search that was augmented by interviews conducted while researchers visited the industry and other organisations. According to the report, creating partnerships and alliances is crucial for the growth of the oil and gas sector, particularly when the necessary funding and local technical expertise are few.

And in their own research work, Mohamed and Barjoyai (2013) looked at the effects of technology transfer in the oil and gas industry of the Libyan economy from 1996 to 2010. To better understand how technology transfer affected the overall economy during this time, both quantitative and qualitative methods were used. The qualitative method was based on focus group discussions and interviews with some of the major stakeholders in the oil and gas business, while the quantitative approach used time series data of several important economic indicators in the oil and gas industry. The NVivo9 software was used to analyse the qualitative data, while the Statistical Package for Social Science (SPSS) software was utilised



to analyse the quantitative data. The study discovered that the oil and gas industry and various other sectors of the national economy were both negatively and favourably impacted by technological transfer.

Lastly, in a study titled “Influence of information literacy skills on Petroleum Training Institute (PTI) students' use of electronic resources: Library overview” by Sambo et al (2022), it was reported that respondents affirmed the access and uses of the following for e-resources at PTI, Internet to sources materials and information, Online Public Catalogue (OPAC) for assignments, e-resources journals provided by the library for my research, online databases for searching information to augment my class note, CDROM for assignment and offline databases provided by the library for assignments and they use e-books provided in the library among others. The result further revealed that the majority of the respondents use the Internet daily and the level of information literacy skills the respondents is very high/high level. The study concluded that the management of the petroleum training institute, Effurun should encourage a higher level of resource utilisation, and the teaching of information literacy skills to PTI students should be pushed and encouraged. In addition, the management of the Effurun Petroleum Training Institute should guarantee that electronic materials are continuously available, with suitable information communication technology tools to ease their use.

Gap in Literature

However, in all the studies and materials explored, no where was an assessment of PTI in terms of efficiency and effectiveness in relation to providing human resources support to the oil and gas industry in Nigeria was carried out. This study provides empirical evidence for the effective and efficient contribution of the Petroleum Technology Institute to the development of manpower in the oil and gas industry in Nigeria, a feat that no known study to this research has previously attained.

4. Theoretical Framework

Human capital theory serves as the theoretical foundation for this study. According to Armstrong (2009), a company's human capital consists of the knowledge, abilities, and skills of its employees. Because it assumes that education or training has the potential to stimulate economic growth, technological advancement, and productivity because it transfers useful knowledge, dexterities, and skills for better lifetime earnings, the human capital theory of Schultz (1975) is the theoretical foundation for the majority of research works in the field of education (Becker, 1964; Schultz, 1975; Robert, 1991; Ladipo, Akhuemonkhan, and Raimi, 2013). From a different angle, Klein and Cook (2006) claim that investing in people is a sort of human capital that stimulates social transformation. Hands-on human capacity can be

improved by education, training, experience, health care, and other factors, just like other types of human capital (Akhuemonkhan, Raimi, and Dada, 2014).

According to this study's application of the aforementioned principle, oil workers must have the necessary expertise to run the business of the industry, especially given that it is the backbone of the Nigerian economy. On the other hand, Bontis, Dragonetti, Jacobson, and Roos (1999), cited in Armstrong (2009), stated that human capital reflects the human aspect of the firm; their knowledge, abilities, and skills provide the company with a distinguishing advantage over its rivals. This suggests that competence is an ability that every organisation needs to fulfil its aims. According to McKinsey and Company (2001), businesses need the appropriate and pertinent skills required by that sector of the economy to survive. Additionally, the workers in this study reflect the human capital, or possibly the human resources, that comprise the Nigerian oil and gas companies. Human capital theory's basic assumption is that individuals who make up oil and gas companies each have a unique set of skills and competencies that must be identified, developed, and maintained.

Sunday (2015) conducted a study on local content policy, human capital development and sustainable business performance in the Nigerian oil and gas industry. The study concluded that the Local Content Policy had achieved significant success in enhancing the development of human capital which in turn positively influenced business performance of indigenous companies in the Oil and Gas Industry in Nigeria.

Kingston and Frank-Igwe (2019) conducted a study on the negative effects of petroleum gaming on Nigeria's institutional and legal regimes. The central goal of this paper is to unmask the fundamental reasons for the ineffectiveness of all the Nigerian laws that are supposed to regulate and control the oil and gas enterprises. The paper presents the game theory effects which weigh the parties' rational choices leading to perpetual losing position of the Nigerian government to the effect the loser in the game is compelled to lower standards and weaken her laws to accommodate the winners' investment preferences. It argues that, the irrational choice of the Nigerian government which creates laxity in the laws to accommodate the oil companies is a race to the bottom with consequential permanent damages to the growth and development of the national economy. Among others, the paper highlights the effect of weak regulatory regime on profit oil, royalty receipts and environmental protections. It recommends that, Nigeria should develop local manpower and task the universities to produce indigenous personnel capable of replacing the foreign corporations in the oil and gas sector.

Agwu and Ogiriki (2014) discussed human resource development (HRD) and organizational performance in the Nigeria liquefied natural gas company limited, Bonny. It views HRD as the integrated use of training and development, organizational development and career development to improve individual, group and organizational effectiveness. It assumes that



HRD practices directly/indirectly enhance business performance through improvement in employees' skills, attitudes and behavior. The research question addressed the extent of the relationship between HRD practices (training and development) and increased employees motivation/commitment and organizational productivity in the Nigeria liquefied natural gas company limited, Bonny. Results from the data analysis indicated that significant relationship exists between HRD practices (training and development) and increased employees motivation/commitment and organizational productivity in the Nigeria liquefied natural gas company limited, bonny and recommends among others: review of current training/development procedure, regular appraisal of employees job skill requirements, regular staff training/development, regular improvement of employees' condition of service and sustenance of current team building efforts/ incentive scheme.

Akinwale, Akinbami and Akarakiri (2018) investigated the factors influencing technology and innovation capabilities in Nigerian indigenous oil and gas firms. Primary data were collected through the use of questionnaire to elicit information from 150 indigenous oil firms. The result of the ordered logit regression showed that the extent of the firm's R&D captured by in-house R&D and R&D fund allocation were significant at 10% level of significance. Also, work experience of staff, qualification of staff and size of technical staff which captured human and knowledge capital of firm were significant in influencing the firms' technology capabilities. Acquisition of advanced machinery, firm's size and age and training were also significant in influencing technology capabilities of indigenous oil firms. The study suggests that both government and industry should ensure improvement in extensive training, R&D and education of the work force to enhance local capabilities in oil and gas field development.

5. Methodology

The study adopted a survey design as key methodology. Survey method involves collection of information from primary sources and Focus Group Discussion (FGD) was adopted in this instance. FGD is an organised informal discussion with the intent of acquiring in-depth data on a particular topic as a means to achieve full participation by respondents and full control of the data gathering process by the researcher. The usefulness of the FGD in this study, as Gibbs (1997) describes it, was to draw upon the respondents' attitudes, feelings, beliefs, experiences and reactions about the matter under investigation.

In determining the participants in the FGD, a non-probability criterion-based purposive sampling was used to select 25 participants in the FGD in order to achieve symbolic representation and diversity. The selection was based on specified characteristics such as being a graduate from any programme at PTI or playing supervisory role to graduates of PTI in any of the selected oil companies in the Niger Delta Region.

Focus Group Discussions were held with 5 groups of 5 respondents each based on the 5 selected oil companies who have engaged graduates of PTI. The aim of the FGD was to elicit information on views of participants on evaluating the effectiveness of the PTI in providing high technological training and skills for the oil and gas industry going into the future. The suggestions derived possibly can create an enriched learning environment that promotes skill acquisition and prepares students for successful careers in the industry at the PTI.

The data collected from the FGD(s) were analysed using content analysis technique known as NVIVO – a computer assisted qualitative data analysis (CAQDAS) package. The software facilitated the construction of a coding index for identifying key words, phrases, themes and patterns that emerged from the discussions.

Table 1: The Profile of Participants in the Focus Group Discussions

Oil Company	FGD Code	Location of FGD	Gender of Participants
Agip Oil	Agip 1	Port Harcourt	M
	Agip 2	Port Harcourt	M
	Agip 3	Port Harcourt	F
	Agip 4	Port Harcourt	F
	Agip 5	Port Harcourt	M
Nigerian Petroleum Development Company (NPDC)	NPDC 1	Benin City	M
	NPDC 2	Benin City	F
	NPDC 3	Benin City	F
	NPDC 4	Benin City	M
	NPDC 5	Benin City	M
Chevron	Chevron1	Warri	M
	Chevron 2	Warri	M
	Chevron 3	Warri	M
	Chevron 4	Warri	F
	Chevron 5	Warri	F



ExxonMobil	Mobil 1	Uyo	F
	Mobil 2	Uyo	M
	Mobil 3	Uyo	M
	Mobil 4	Uyo	M
	Mobil 5	Uyo	M
Shell Petroleum Development Company (SPDC)	SPDC 1	Port Harcourt	F
	SPDC 2	Port Harcourt	F
	SPDC 3	Port Harcourt	M
	SPDC 4	Port Harcourt	M
	SPDC 5	Port Harcourt	M

6. Data Analysis and Discussion of Findings

a. Result

After processing the results arising from the discussions made during the course of each Focus Group, the views presented to the interlocutors, which have synthesized in a statistic for each question. It specifies that any opinion expressed was not circumvented in this study and the findings are consistent with the succession of questions in interview guide.

6.1 *How familiar are you with the PTI?*

To this question, 70.6% answered that they were familiar with the PTI while 29.4% said that they were not familiar with the institution. According to some discussants, PTI is a recognized institution that provides technical and vocational education in various areas, including engineering, petroleum technology, and related fields. Some also noted that while PTI offers programs that are relevant to the oil and gas industry, it is not exclusively focused on training manpower specifically for that sector. During a discussion about PTI, the opinions of the discussants regarding their familiarity with the institution vary. The following responses were given by the participants which showed their familiarity of the institution;

I have heard of the Petroleum Training Institute in Effurun. It is a well-known institution that offers technical and vocational education in the petroleum industry. I am not familiar with the specific programs they offer, but I know they have a good reputation in the field (NPDC 1).

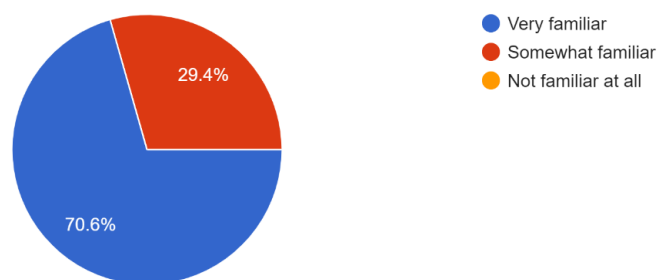
I have some familiarity with the Petroleum Training Institute in Effurun. I know that it is a specialized educational institution focused on petroleum-related disciplines. I have heard positive feedback about the quality of their programs and the practical training they provide to students (Mobil 2).

I am not very familiar with the Petroleum Training Institute in Effurun. I have heard the name before, but I don't know much about the institution or the programs they offer. It seems like it is related to the petroleum industry, but I cannot provide any specific detail (SPDC 1).

I have attended the Petroleum Training Institute in Effurun. It is a prominent institution that offers specialized training in various fields related to the petroleum industry. I have completed a diploma program there and found the education to be valuable in terms of industry knowledge and practical skills (Agip 3).

1. How familiar are you with the Petroleum Training Institute (PTI), Effurun?

17 responses



6.2 Have you attended any training programmes or courses offered by PTI, Effurun

Majority of participants – 58.8% - have declared that yes they have attended some form of training programme or attended a course at the institution. A 41.2% of participants said they have not attended PTI for any course or programme. Some attendees have positive opinions about PTI because it offers specialized training in the petroleum industry, which enhanced their career prospects. They also appreciated the practical knowledge and skills gained during their time at PTI. Some of them asserted:

Yes, I attended a training program at PTI, Effurun, and it was a great experience. The course provided practical knowledge and hands-on training that greatly enhanced my skills in my field of work (Agip 1).

I had the opportunity to attend a course at PTI, Effurun, and I must say it was worth every minute. The trainers were highly knowledgeable and



engaged, and the course content was comprehensive and relevant to my career goals (Chevron 2).

I participated in a training program at PTI, Effurun, and it exceeded my expectations. The instructors were skilled professionals who provided practical insights and shared valuable industry experiences. I left the course feeling confident and equipped with new knowledge (Mobil 4).

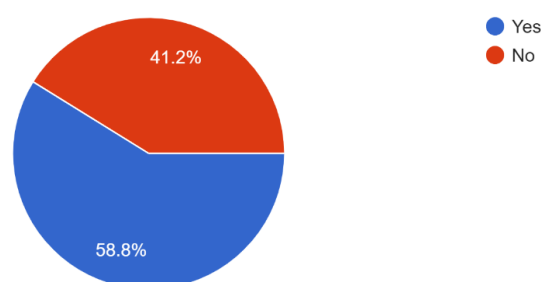
However, some of them who have not attended PTI also remarked in positive light:

Unfortunately, I couldn't attend any training programs at PTI, Effurun, but I've heard positive feedback from colleagues who have. They mentioned that the courses were well-structured and tailored to industry needs, giving participants a competitive edge (Agip 2).

I haven't attended any training programs at PTI, Effurun yet, but I've seen their course offerings, and they seem impressive. The variety of topics covered is extensive, and the positive reviews from previous attendees indicate that the training is of high quality (NPDC 4).

2. Have you attended any training programs or courses offered by PTI, Effurun?

17 responses



6.3 *If you have attended training programme at PTI, please rate the effectiveness of the training programme in providing high technological training and skills for the oil and gas industry*

To assess the effectiveness of a training program, it's important to consider feedback from past students and industry professionals who have undergone training at PTI. They can provide insights into the quality of education, the relevance of the curriculum, the practical training opportunities, and the overall preparation provided for the oil and gas industry. 29.4% of participants agreed that PTI was highly effective in providing high technological training and skills for the oil and gas industry and 58.8% said it was effective. Together, 88.2% of participants agreed the effectiveness of PTI in providing high technological training and skills for the oil and gas industry while 11.8% of participants did not view PTI training as being effective. A discussant asserted that:

I believe PTI has been quite effective in providing training and skills for the oil and gas industry. The institute offers a wide range of specialized courses and programs that cater to various disciplines within the industry.

The curriculum is well-structured and focuses on practical applications, which is crucial for working in the industry (SPDC1).

Other discussants agreed with the position above, as they remarked:

As a recent graduate, I can say that PTI has equipped me with the necessary knowledge and skills to start my career in the oil and gas sector. The faculty members are experienced professionals who have worked in the industry, which adds value to the learning experience. Additionally, the institute has well-equipped labs and facilities that allow students to gain hands-on experience (SPDC 3).

Some of the discussants also highlighted the program's track record of producing skilled professionals who have successfully transitioned into the industry and contributed significantly to their respective roles. In their view, the training program effectively equips individuals with the necessary knowledge and skills to excel in the technology-driven oil and gas sector.

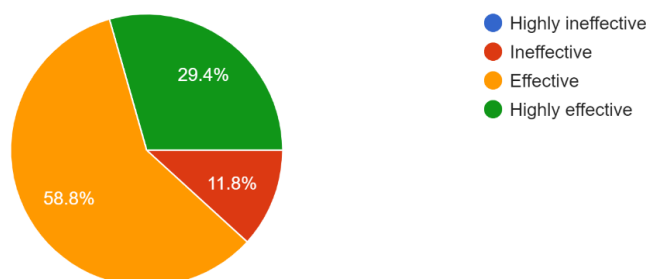
I work in the oil and gas industry. From my experience, PTI has been instrumental in training skilled professionals who are well-versed in the latest technological advancements. Their programs have a good balance of theory and practical training (NPDC 5).

I work as a project engineer in an oil and gas company. From my experience, I believe PTI has been effective in providing high technological training and skills. Many of my colleagues and I have attended their programs and found them to be comprehensive and relevant to our industry needs (SPDC 5).

I believe PTI has done a good job in aligning their courses with industry needs. The curriculum covers essential areas such as drilling, production, refining, and safety, which are vital for the oil and gas industry (Chevron 4).

3. If you have attended training programs at PTI, please rate the effectiveness of the training in providing high technological training and skills for the oil and gas industry.

17 responses



6.4 How will you rate the level of technological competence of PTI graduates at your establishment?

PTI aims to equip its graduates with the necessary knowledge and skills to excel in their chosen fields. In attempting to investigate the technological competence of PTI graduates at



their work places, 94.1% of participants were of the opinion that the level of technological competence of PTI graduates was high, only 5.3% disagreed. Some of the participants have positive ratings about the technological competence of PTI graduates. According some of the discussants:

I would rate the level of technological competence of PTI, Effurun graduates at our establishment as excellent. They possess a solid understanding of the latest technologies and demonstrate great proficiency in applying them to practical scenarios (Chevron 2).

I would rate the level of technological competence of PTI, Effurun graduates at a solid 8. They generally come with a strong foundation in technical skills and have a good understanding of the practical aspects of their field. However, there is still room for improvement in terms of adapting to new technologies and keeping up with industry trends (Agip 1).

PTI, Effurun graduates exhibit a high level of technological competence. Their training equips them with the necessary skills to tackle complex technical challenges, and their ability to adapt to new technologies is commendable (NPDC 1).

Not all the participants rated the technological competence of PTI graduates so high. Some of them were of the opinion that:

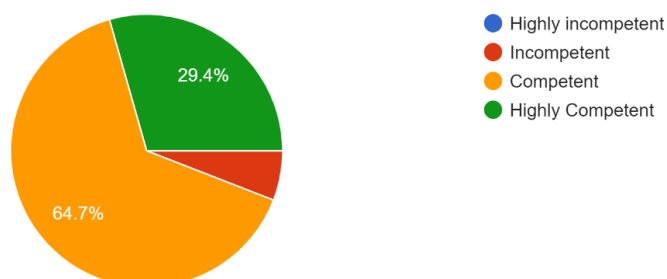
I think the technological competence of PTI, Effurun graduates is above average. While they have a good grasp of the fundamentals, there is still room for improvement in terms of adapting to rapidly evolving technologies and exploring innovative solutions (Chevron 3).

I have mixed feelings about the level of technological competence among PTI, Effurun graduates. Some of them showcase exceptional skills and have a deep understanding of specific domains, while others may need further training to keep up with industry demands (Mobil 2).

I would rate the level of technological competence of PTI, Effurun graduates as below expectations. While they may have theoretical knowledge, there seems to be a gap when it comes to applying that knowledge in real-world scenarios. More emphasis should be placed on practical training (SPDC 2).

5. How would you rate the level of technological competence of PTI graduates at your establishment?

17 responses



6.5 *In your opinion, how well are PTI graduates prepared for the future technological requirements of the oil and gas industry?*

Majority of participants 88.2% of participants viewed PTI graduates as well prepared for the future technological requirements of the oil and gas industry and just 11.8% saw the PTI graduates as not well prepared for the future technological requirements of the oil and gas industry. One of the ways of assessing how well PTI graduates are prepared for the future technological requirements of the oil and gas industry is to evaluate whether the curriculum of PTI covers the latest technological advancements and industry trends in the oil and gas sector. Given this, one of the discussants said:

I believe PTI graduates are adequately prepared for the future technological requirements of the oil and gas industry. The institute has a strong curriculum that covers both traditional and emerging technologies. However, considering the rapidly evolving nature of the industry, there is always room for improvement and continuous learning (Mobil 4).

Another discussant agreed with the above statement when he said:

PTI graduates receive comprehensive training that equips them with a solid foundation in the industry. However, to stay ahead of the rapidly changing technological landscape, graduates should engage in continuous professional development and keep up with the latest advancements (Agip 4).

Practical training is essential for students to develop the skills required for the oil and gas industry. Some of the discussants also noted that PTI offers hands-on training opportunities, access to modern laboratories, and industry-standard software and equipment. According to one of them:

I believe PTI graduates are well-prepared for the future technological requirements of the oil and gas industry. The institute places a strong emphasis on practical training and hands-on experience. They are taught to adapt to changing technologies and are prepared to learn and apply new advancements in their careers (Agip 2).

I have found PTI graduates to be well-prepared for the future technological requirements of the oil and gas industry. They have a strong foundation in both traditional and emerging technologies. Their ability to quickly adapt and learn new technologies is impressive, and it shows that the institute is providing them with the necessary tools to succeed in the future (Agip 5).

However, one of them asserted:

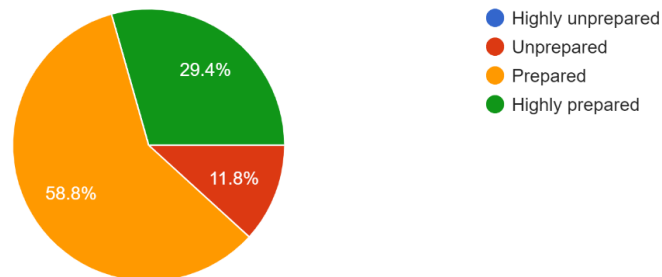
While PTI graduates have a good understanding of the current technological requirements of the oil and gas industry, I believe there is room for improvement in their preparedness for the future. The industry is evolving rapidly, and graduates need to be exposed to the latest advancements to stay ahead.



Continuous learning and professional development would enhance their future readiness (Chevron 1).

6. In your opinion, how well are PTI graduates prepared for the future technological requirements of the oil and gas industry?

17 responses



6.6 Have you observed any specific area or specialization on emerging technologies relevant to the oil and gas industry where PTI graduates are deficient?

It's important to note that the oil and gas industry is vast and continuously evolving, so there are emerging technologies that are relevant to the oil and gas industry. As a result of emerging technological trends in the oil and gas industry globally, 52.9% of participants agreed that PTI graduates are deficient or lack knowledge in certain areas or emerging technologies relevant to the oil and gas industry. However, 47.1% were of the opinion that PTI graduates are not deficient. The specific areas of deficiency may vary based on individual skill sets and the curriculum of the PTI, Effurun program.

There seems to be lack of expertise in data analytics and interpretation: Some participants feel that PTI graduates are not adequately equipped with the necessary skills to analyze and interpret large datasets generated by emerging technologies, such as sensor networks and IoT devices. Some of the participants believe that graduates should receive more training in data analytics to optimize production and decision-making processes. According to one of the discussants, he noted:

I think there is an inadequate familiarity with renewable energy technologies: Given the growing emphasis on sustainability and the transition towards renewable energy sources, I perceived a lack of exposure to renewable energy technologies among PTI graduates. This might suggest that the curriculum should include more content related to renewable energy sources, such as solar, wind, or geothermal, and their integration with traditional oil and gas operations (Mobil 5).

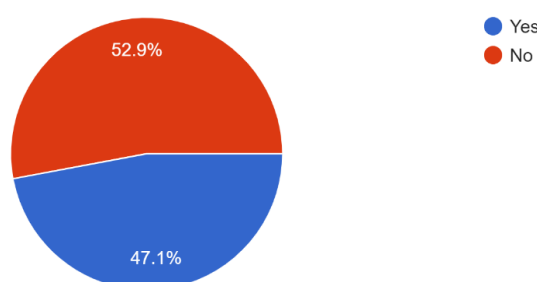
For me, I see a limited understanding of cybersecurity: With the increasing connectivity and digitization of oil and gas infrastructure, there are concerns about PTI graduates' insufficient knowledge of cybersecurity practices and threats. I think incorporating cybersecurity courses or modules into the curriculum to ensure graduates possess the necessary

skills to protect critical systems and data from cyber threats is important (Mobil 1).

The focus on emerging environmental regulations is inadequate: I feel PTI graduates need more exposure to emerging environmental regulations and sustainability practices within the oil and gas industry. This can be done by incorporating courses or workshops that cover topics such as emissions reduction, carbon capture and storage, and environmental impact assessments to ensure graduates are well-prepared for the evolving regulatory landscape (Mobil 2).

7. Have you observed any specific area or specialization on emerging technologies relevant to the oil and gas industry where PTI graduates are deficient?

17 responses



6.7 If your answer to 6 is Yes, please specify

There are a few areas where emerging technologies are increasingly important in the oil and gas industry as the following quotes from FGD participants indicate:

In the context of the oil and gas industry, emerging technologies related to Electronic Data processing, like data acquisition and monitoring, digital twin technology, cloud computing and big data, etc, can have significant relevance and impact (Agip 4).

Most of the equipment required for effective training are not functional, some new ones are not even put to use due to lack of manpower to operate the machines. PTI graduates were better several years ago when the equipment they had was very much in vogue and there were skilled personnel to train the students (NPDC 1).

Artificial intelligence (AI) and data science (DS) are indeed valuable skills that can significantly enhance the capabilities of graduates in the oil and gas industry. These fields offer numerous opportunities for improving efficiency, reducing costs, and making informed decisions based on data-driven insights. While there may be insufficient knowledge of AI and DS at PTI, Effurun, it is essential to recognize the potential benefits and consider ways to incorporate these skills into the curriculum (Mobil 2).

Information technology (IT) plays a vital role in the oil and gas industry by enabling efficient operations, improving safety, and optimizing exploration and production activities (SPDC 3).

Improving industry attachment programs can indeed enhance the quality of graduates from the Petroleum Training Institute (PTI), Effurun, in the oil



and gas industry. Industry attachment, also known as internships or practical training, provides students with real-world work experience and exposure to industry practices (Mobil 1).

Compressed Natural Gas (CNG) is a type of fuel that is increasingly being used as an alternative to traditional gasoline and diesel fuels. It is a cleaner and more environmentally friendly fuel option, as it emits fewer pollutants and greenhouse gases when burned. Enhancing the knowledge of graduates from the Petroleum Training Institute (PTI) in Effurun with respect to CNG can be beneficial for their future technological skills (Chevron 1).

The use of ICT-driven training at PTI, Effurun can play a crucial role in preparing graduates for future technological skills in the oil and gas industry. By leveraging the benefits of ICT, such as enhanced learning experiences, realistic simulations, access to industry knowledge, and technical skill development, PTI can equip its graduates with the competencies required to thrive in a technology-driven industry (Agip 4).

Artificial intelligence (AI) and machine learning (ML) are indeed valuable tools for enhancing the skills and knowledge of graduates in the oil and gas industry. These technologies have the potential to revolutionize various aspects of the industry, including exploration, production optimization, safety management, predictive maintenance, and more. By leveraging AI and ML, graduates of PTI (Petroleum Training Institute), Effurun, can gain a competitive edge and prepare themselves for future technological skills in the oil and gas sector (Mobil 4).

6.8 How effective has PTI products be in incorporating practical hands on training with modern technologies in your establishment?

A key finding here is that incorporating practical hands-on training with modern technologies have had tremendous benefits for PTI products. This is exhibited in the data below as majority of the participants, 88.2% have agreed that PTI products have effectively incorporated practical hands on training with modern technologies in places they work. Among those participants, a small fraction, about 11.8% said that PTI products have not incorporated practical hands on training with modern technologies in places they work effectively. Some of the discussants expressed their feelings in the following quotes:

I have had the opportunity to work with PTI graduates, and overall, I have been impressed with their technical skills and knowledge. They seem well-prepared for the challenges of the industry. However, I think PTI could focus more on providing training in emerging technologies and industry trends to keep up with the rapidly evolving oil and gas sector (NPDC 4).

While I haven't directly experienced PTI's training, I have heard positive feedback from industry professionals about the institute. It is known for its practical approach and industry-oriented programs. However, it would be helpful if PTI could collaborate more with industry partners to ensure that the training aligns with the latest industry needs (NPDC 1)

I have attended a few training programs at PTI, and I found them to be highly valuable. The instructors were knowledgeable and experienced, and the course content was up to date. The training programs also

emphasized hands-on learning, which helped me develop practical skills that I could apply in my work (Agip 3).

I'm a recent graduate of PTI. While the institute does provide extensive technical training, I believe there is room for improvement in terms of incorporating emerging technologies, such as artificial intelligence and renewable energy, into the curriculum (Agip 2).

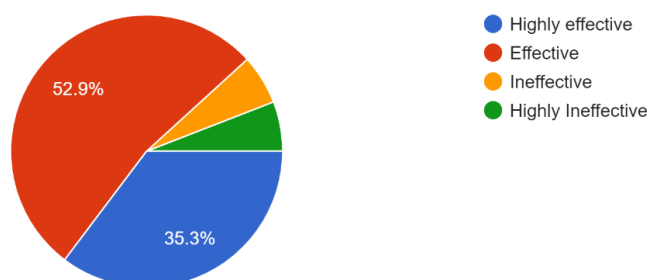
Some participants equally agreed that while the institution has a good rating, more still needed to be done as some discussants remarked below:

I agree with John that PTI has a solid reputation for producing competent graduates. However, I've noticed that the training infrastructure needs upgrades to keep up with industry standards. Additionally, there could be more collaboration between PTI and industry experts to enhance the curriculum (SPDC 1).

While the technical training was top-notch, I felt that there were limited opportunities for hands-on experience with real-world equipment. More industry internships or apprenticeships could greatly enhance students' practical skills (NPDC 2).

9. How effective have PTI products being in incorporating practical hands-on training with modern technologies in your establishment?

17 responses



6.9 *In your opinion, what areas or technologies should PTI focus on to better prepare its graduates for the future technological demands of the oil and gas industry?*

Some of the suggestions as revealed by the discussants on technologies that PTI could focus on to better prepare its graduate include:

- PTI needs to focus more on digital skills so that it can better prepare its graduates for the future technological demands of the oil and gas industry (Mobil 1).
- PTI should be more practical, by ensuring all their laboratory equipment are functional and being used during experiment (Mobil 3).
- PTI Effurun, like any educational institution, must adapt and focus on new global technological innovations to better prepare its graduates for the future technological demands of the oil and gas industry (Mobil 4)



- PTI, Effurun, must focus on Higher Education and Skills (HES) and competence development to better prepare its graduates for the future technological demands of the oil and gas industry (Mobil 2).
- In any area in oil and gas Technology where their expertise can cover (Chevron 2).
- They should focus more on practical works (training) in their various workshops and laboratories (Chevron 1).
- Another area to focus on is gas exploration which is very important in today's highly competitive oil and gas industry (Agip 1).
- The oil and gas industry is undergoing a significant transformation due to advancements in technology. Data science, sustainability, and AI are playing crucial roles in optimizing operations, improving efficiency, and reducing environmental impact. By focusing on these areas, PTI Effurun can equip its graduates with (Agip 4).
- Information technology can better prepare PTI graduates for the future technological demands of the oil and gas industry. It enables them to adapt to technological advancements, improve operational efficiency, handle digital systems, work with emerging technologies, and develop versatile skills that are in high demand (Mobil 2).
- By focusing on the refining of crude oil at PTI can better prepare its graduates for the future technological demands of the oil and gas industry. It equips them with specialized skills, knowledge of advanced refining technologies, and an understanding of downstream operations, making them valuable assets to refineries and related industries (NPDC 4).
- Emphasis on the use of more digital application and AI will enable PTI to produce industry-ready professionals who can drive innovation, enhance efficiency, improve safety, and meet the evolving needs of the sector (Mobil 1).
- Renewables and Decarbonization need to be prioritized at PTI, as this shift will equip graduates with the necessary skills to adapt to a changing energy landscape, diversify their skill set, align with industry transitions, leverage technological advancements, and contribute to sustainable development goals (Agip 2).
- Petrochemicals play a crucial role in the oil and gas industry, as they are the building blocks for various products and processes. Focusing on petrochemical education and training at PTI (Petroleum Training Institute) in Effurun would indeed better prepare its graduates for the future technological demands of the industry (NPDC 4).
- The oil and gas industry is becoming increasingly digitized and technologically advanced. The integration of IT systems, automation, and data analytics is

transforming various aspects of the industry, including exploration, production, refining, and distribution. So, emphasis must be placed on information technology (Mobil 4).

- PTI Effurun should prioritize courses on AI to equip graduates with skills to analyze and interpret large volumes of data. The ability to leverage AI technologies for predictive maintenance, reservoir modeling, and process optimization will be crucial for the future of the oil and gas industry (Chevron 1).
- The Institute has evolve to training with the help of ICT driven trainings. The application of IT solutions can significantly enhance operational efficiency and reduce costs in the oil and gas industry. For example, advanced data analytics can optimize production processes, predictive maintenance can minimize downtime, and remote monitoring can improve safety and efficiency (Chevron 5).
- Technologies for Oil & Gas Assets surveillance and protection must also be top priority. The oil and gas industry operates in a hazardous environment, and safety is a top priority. Technological solutions can help in early detection of equipment failures, leaks, and potential hazards, thus minimizing the risk to personnel and the environment. By focusing on technologies for surveillance and protection, PTI Effurun can contribute to the industry's efforts in ensuring safety and minimizing the environmental impact (Agip 5).

6.10 Please, is there any additional comments or suggestions you would like to make regarding the effectiveness of PTI in providing high technological training and skills for the oil and gas industry going into the future?

FGD participants provided additional comments on what steps to take by PTI to ensure its effectiveness in providing high technological training and skills for the oil and gas industry. The suggestions are mentioned below:

- PTI need to train specifically for the Oil and Gas Industry (Mobil 1).
- The classes used for conducting semesters' exams are filled with writings on the wall (EXPO), which pose as a threat to the PRODUCT being produced from PTI. My point is that all blocks of class room should be painted and a plan show be put in place to expel such culture and mindset from PTI school (Mobil 4).
- None (Mobil 2)
- PTI should be legally back as the go to institution for all oil and gas training. All oil and gas company must fund PTI and be internationally recognized as a hub for oil and gas in west Africa and Africa in general. Short and long-term courses should be well advertise and encouraged. Both in HES training and competence development to be



used not only in oil and gas, but in aviation industry, transportation and highly risky industries (SPDC 1).

- PTI requires a general over haul in the area of equipment maintenance and renewal, so as to bring it to the current level of expectations in the oil and gas industry (SPDC 2).
- Discipline (Chevron 4).
- Ensure that qualified hands are employed to meet with future required standard for the oil and gas industry (Chevron 5)
- No (Agip 1).
- Just Improve on IT (Agip 2).
- No comment (Agip 3).
- Expose the teaching staff with the applicable best work practices in the oil and gas industry (NPDC 1).
- Let there be practical aligned with theory (NPDC 4).

The key findings that emerged from the Focus Group Discussions are:

- (1) PTI's curriculum and training programs are designed to cater specifically to the needs of the oil and gas industry. The institute offers a wide range of courses and programs that cover various aspects of the sector, including drilling operations, production engineering, petroleum geosciences, refining, and petrochemical engineering. By focusing on industry-relevant subjects, PTI ensures that its graduates are equipped with the necessary skills and knowledge to meet the demands of the oil and gas industry.
- (2) PTI placed a strong emphasis on practical training and hands-on experience. The institute provides state-of-the-art facilities and equipment that allow students to gain practical experience in various industry-related tasks. This practical exposure is vital in preparing individuals for the real-world challenges they will face in their careers. PTI's commitment to practical training helps bridge the gap between theoretical knowledge and practical application, enhancing the employability of its graduates.
- (3) The oil and gas industry is continually evolving due to technological advancements, environmental concerns, and changing market dynamics. PTI recognizes the need for ongoing learning and adaptation. The institute emphasizes research and innovation, encouraging students to stay updated with the latest industry trends and emerging technologies. This focus on continuous learning equips graduates.
- (4) PTI's effectiveness is informed by its focus on research and innovation. The institute encourages students and faculty members to engage in research projects that address industry challenges and explore new technologies. By promoting research and

innovation, PTI contributes to the development and advancement of the oil and gas industry.

7. Conclusion

The oil and gas industry plays a critical role in the global economy, and the need for a skilled workforce is paramount to its success. As the industry evolves, the demand for high technological training and skills becomes increasingly important. This Focus Group Discussion thus provided an assessment of the contribution of Petroleum Training Institute, Effurun, in developing the required skills and manpower for the oil and gas industry in Nigeria. By analyzing factors such as curriculum relevance, industry collaboration, faculty qualifications, facilities, alumni success, and continuous learning, we have gained insights into the effectiveness of PTI Effurun and identified areas for improvement.

The effectiveness of PTI Effurun in delivering high technological training and skills for the oil and gas industry is of significant importance for producing a competent workforce. Through this literature review, incorporating insights from focus group discussions, a comprehensive evaluation was provided, highlighting the institute's strengths and weaknesses. Ultimately, the findings of this discussion will help inform strategies to enhance the institute's contribution to the growth and development of the oil and gas industry in Nigeria.

However, it is essential to acknowledge potential areas for improvement. There is a consensus on PTI's strong practical training and reputation for producing competent professionals. But the study make the following recommendations:

1. A comprehensive digital deployment in teaching and learning in order to measure up to global competition in technological skill and knowledge production. The Institute should incorporate emerging technologies, sustainable practices, and global perspectives into its curriculum to ensure its effectiveness going into the future, the institute should focus on incorporating emerging technologies. Thus the institute should focus on training students in areas like renewable energy, carbon capture, and digitalization to ensure their graduates remain competitive in the evolving global market of jobs in oil and gas industries.
2. Whereas this study's findings highlighted the effectiveness of PTI in providing high technological training and skills for the oil and gas industry, it is essential to note that the industry is constantly evolving. To maintain its effectiveness, PTI should continue to adapt its curriculum and training methodologies to align with emerging technologies, sustainability practices, and industry demands. By staying responsive to the evolving industry landscape, PTI can ensure its graduates remain well-prepared for the future challenges and opportunities in the oil and gas sector.



A Technical Review of Local Content in the Oil and Gas Sector of Nigeria: Challenges, Prospects and the Way Forward

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Abstract: International oil companies (IOCs) previously dominated the Nigerian oil and gas sector. However, a lack of skilled workers existed, particularly in terms of the participation of indigenous people in the sector. The result is, the IOCs heavily relied on expatriates to handle projects that could have been carried out by Nigerians. To promote local participation in oil and gas projects and increase local residents' employment chances, the Nigerian federal government enacted the Local Content Act in 2010, acknowledging the need for the growth of locally sourced human capital in the sector. Despite the government's efforts, the carrying out of the Local Content development policy has encountered some challenges, hindering its effectiveness and efficiency. This study aims to explore the growth of local content in Nigeria's oil and gas sector, identify the issues and opportunities associated with Nigerian local content, and propose solutions to effectively address these challenges.

Keywords: Local Content, Oil and gas industry, Policy, Problems and Prospects.

1. Introduction

Local content is described by the Nigerian Oil and Gas Development Law 2010 as "the quantum of composite value added to or created in Nigeria through utilization of Nigerian resources and services in the petroleum industry leading to the development of indigenous capability without compromising quality, health, safety, and environmental standards" (NCDMB, 2016). It is established within the framework of Nigerian entrepreneurial development as well as the domestication of resources to completely accomplish strategic objectives for development in Nigeria. The journey of implementing the local content policy in Nigeria's petroleum sector dates back to 1971 with the creation of the NOC, or Nigerian National Oil Corporation as part of the indigenization policy. In 1977, NOC combined with the petroleum ministry to form the Nigerian National Petroleum Corporation (NNPC), which played a pivotal role in initiating the local content initiative. NNPC took significant strides in promoting local content by acquiring interest in how international oil companies (IOCs) conduct its business. These interests increased over time to roughly 70%, granting NNPC control over various acreages and activities. Despite some prior efforts through Regulation 26 of the 1969 Petroleum Act to enforce local content policies, real progress in achieving sustainable economic transformation through local content remained limited, often limited to paperwork and lacking in substantial impact. The Nigerian government finds the current situation unacceptable, especially considering that 80% of government revenue and 95% of foreign exchange gains come from the oil and gas sector. Nigeria's rapid and steady growth in oil and gas production has propelled the country to take on a vital role within OPEC. The exploration of oil, which began onshore, has led to a substantial increase in Nigeria's capability for daily production, reaching 2.3 million barrels per day, increasing the amount of proved reserves to around 37 billion barrels. However, despite the country's increasing riches and notoriety, it continues to be among the poorest and least advanced technological countries globally. The expected prosperity from this wealth has not resulted in an increase in peoples' welfare. One of the contributing factors to this situation is that capital flight accounts for over 90% of industry spending annually, hindering the country's ability to fully harness the benefits of its oil and gas resources.

2. Legal Framework for Nigerian Content Development

Among the laws governing the creation of Nigerian Content are:

(a) The Immigration Act

According to the Act, individuals who are not Nigerian citizens cannot accept employment, except when working with the Federal or State governments, and without the Director of Immigration's prior written authorization. Additionally, they are not allowed to engage in a career, start or take over any business or trade, or register or take over any corporation without the Minister's written permission. The Minister may specify conditions regarding the locality of operation and the employment of individuals on behalf of such person (NCDMB, 2018).

(b) The Petroleum Act

Section 1 of the Act grants complete ownership and management of all Nigerian petroleum resources to the State. The Act further authorizes the Minister will grant permits and leases to petroleum operators. All such permits and leases are subject to the provisions of the First Schedule to the Act. Under the First Schedule, the Minister has the authority to place unique restrictions on, not conflicting with the Act, on any license or lease if he deems it to be in the public interest. These unique restrictions may include provisions relating to the Federal Government's participation in the venture linked with the license or lease, subject to discussion between the Minister and the applicant (Olusola, 2023).

(c) Petroleum (Drilling and Production) Regulations

The Regulations require a thorough program for the recruitment and training of Nigerians to be submitted for the Minister's approval by both the licensee of an OPL within a year after receiving his license and the lessee of an OML after receiving his lease. No matter whether the lessee



handles the phases whether directly or via contractors and agencies, the program must offer training to Nigerians in all facets of the petroleum industry (PDPAR, 2020).

(d) Petroleum Development Fund Act

The Act was passed with the primary aim of creating the Petroleum Technology Development Fund, which is intended for Nigerians' education and training in the petroleum sector. The Act's Section 1 outlines the establishment of the Petroleum Technology Development Fund and mandates that specific funds be deposited into the Fund. These funds include: (a) the remaining financial resources held in the Gulf Oil Company Training Fund accounts at the time the Act was passed; (b) all additional funds due to or received by the Minister in charge of petroleum development under agreements between the Government and any company regarding petroleum oil prospecting or mining concessions; and (c) any additional funds that are voluntarily donated to the Government or the Fund and intended for the education and training of individuals (Orji, 2018).

(e) National Office for Technology Acquisition and Promotion Act

In Nigeria, the Transfer of Technology Act controls this. The National Office for Technology Acquisition and Promotion (NOTAP) was created by the Act. According to the Act's requirements, NOTAP's main duty is to keep an eye on the importation of foreign technology into Nigeria. The Act mandates that each and every contract and agreement for the transfer of foreign technology made by anyone in Nigeria be registered with NOTAP within 60 days of the contract's execution or conclusion. It also mandates that all contracts and agreements that a Nigerian person has with a foreigner be registered with NOTAP (NOTAP, 2023).

(f) The Nigerian Oil and Gas Industry Content Development Act

The National Assembly passed the Act with the primary objective of revitalizing and reinforcing efforts to promote local content development in the Nigerian petroleum industry. The Act's main purpose, as expressed in its lengthy title, is to facilitate the advancement of Nigerian content in the country's oil and gas sector. It aims to establish the Nigerian content plan, oversee, coordinate, monitor, and implement Nigerian content initiatives, and address related matters. The Act's core focus is to enhance Nigerians' involvement in the oil and gas sector and ensure that the industry's benefits create value for Nigerians. Section 1 of the Act ensures that its provisions apply to all aspects related to the oil and gas sector's use of Nigerian content, overriding any conflicting provisions in other statutes. Section 2 of the Act mandates that all stakeholders, including licensees and subcontractors, create a plan for local content and integrate it into their policies (NCDMB, 2018).

(g) The Petroleum Industry Bill

By allowing for the granting of licenses or leases to the National Oil business or any indigenous oil business. The Local Content Policy is supported by the Petroleum Industry Bill (PIB), which is described as one having at least 60% ownership by Nigerians. (PIB, 2020).

3. Problems of Local Content Development in the Oil and Gas Industry in Nigeria

(a) Lack of Infrastructural Base:

Nigeria's infrastructure is now unable to support industrial expansion; some of the issues include a lack of public utilities, a weak transportation and telecommunications system, and inadequate power supply (Nwankwo and Iyeke, 2022).

(b) Lack of Funds for Indigenous Companies

The oil and gas industry requires a lot of finance. Local businesses are unable to participate in this area of the economy effectively or efficiently due to a lack of funding. Nigerian banks do not have the financial resources necessary to significantly influence the creation of localized content. The biggest banks in Nigeria when it comes to financing for electricity are tiny banks. Due to the lack of suitable business structures among most domestic contractors, the majority of Nigerian banks operate in uncertain environments (Adenikinju, 2018).

(c) Political Instability and Bad Investment Climate

In the past few years, attacks against foreigners and foreign investments have occurred often in the Niger Delta. Certainly, this is bad for the development of industry. In addition to lacking infrastructural development, Nigeria has a very unfavorable investment climate. If these issues are not resolved, they would discourage foreign direct investment and make it difficult to afford decent housing, receive adequate healthcare, and form a motivated crime fighting force. In contrast, Canada has a conducive investment climate with a clean, secure environment and a calm mood (Akpanika, 2012).

(d) Discrimination and Under-Utilisation of Nigerian Services

Ihua claims that even though the local content policy has given small and medium-sized businesses (SMEs) in the industry more opportunities, which has led to more contracts being awarded in both instances, this cannot yet be viewed as a higher SMEs involvement given that there remains many obstacles standing in the way of the awarding of such contracts, such as drawn-out prequalification and tendering procedures. Although local oil service firms are now more numerous than ever before, their yearly gross profits still account for less than 5% of the sector's overall annual contractual budget (Nwaguru and Waribugo, 2022). In the oil and gas sector, there have also been charges of discrimination and underutilization of local talent.

(e) Issues of Non-Compliance

One of the main accusations made is that some multinational corporations have persisted in breaking the Nigerian Content Act's rules by employing foreigners to carry out work duties that Nigerians are qualified to do. The Nigerian Content Development and Monitoring Board (NCDMB), according to another claim, rather than using the Act's sanctions to compel compliance, employs a more cooperative or persuading strategy to persuade the offending multinationals to adhere to its requirements (Oguagha, 2017).

(f) Corruption

Nigeria's history is marked by pervasive corruption and a lack of transparency. Economic growth is not supported by corruption, which instead undermines trust in both public and private organizations. Accountability can be stifled by corruption, which can also undermine the economy. The widespread corruption in the nation is a significant barrier to the Act's enforcement. If not properly managed, the demand that a Nigerian content plan be submitted every six months as a prerequisite for contract awarding could lead to the formation of further inefficiencies in the contract award process lead to increased corruption (Onwuka, 2021).

(g) Inadequacy of Educational System

The educational system in Nigeria lacks proper training facilities. This is due to inadequate government funding, which it is believed has jeopardized efforts in technical education and research. Due to the Nigerian content policy's omission of research and advancement, universities and other research institutes do not collaborate on R&D with global oil firms operating in Nigeria, even though this is required by the Canadian-Newfoundland & Nova Scotia local content requirements (Onwuka, 2021).

(h) Uncertainties and imprecision of most key provisions of the Act

The Nigerian Oil and Gas Industry Content Development Act has several important concepts and laws, however there are many questions about their meaning and actual use. For instance, neither the definition of the word "first consideration" nor the rules for the Board to assess the validity of the operator's methods for determining first considerations in accordance with the Act's requirements are specified in the Act. Additionally, according to Omenikolo and Amadi (2010), the Act did not define the phrases "management position" and "intermediate cadre."

(i) Insufficient sanctions for breach

The Act's penalties with regard to non-compliance are small (5% of the project's budget), making them ineffective as a breach deterrent. In addition, under the Nigerian JV framework, around 60% of this insignificant penalty (5%) will ultimately be covered by Nigeria through NNPC. Many nations provide tax advantages and incentives to these IOC companies for following domestic laws, regulations, and guidelines on local content. In contrast to the



significant financial benefits, they receive for abiding by the local content regulations and policies of their home country, which are detrimental to Nigeria, this means that just 2% of the IOC's actual penalties will be paid for violations (Udok et al., 2020).

(j) Weak Institutions

The NCDMB lacks the capacity and willingness to hold negligent MNOCs that violate the Act's rules accountable. In light of the Act's provisions, it has been said that the Act is excessively ambitious. These clauses completely alter how indigenous people participate in oil and gas developments in all respects (Olawuyi and Olaniyan, 2021). Due to these regulations' bold nature, the NCDMB lacks the necessary political determination to execute them.

(k) Limited Scope of Local Content Act

The Local Content Act only applies to the oil and gas sector in Nigeria. Other crucial areas of the Nigerian economy are not covered by it. Road, rail, construction of dams and bridges, and also aviation and maritime development, are some of the other crucial industries. Due to the lack of implementation of a local content strategy in several critical sectors of the Nigerian economy, foreign workers are now filling many of the positions that were previously mostly held by Nigerians.

4. Prospects of Local Content Development in the Oil and Gas Industry in Nigeria

Since the Nigerian Oil and Gas Industry Content Development Act was enacted into law in 2010, the industry has seen a methodical yet modest improvement in local content as a result of the innovative and creative leadership of the Nigerian Content Development and Monitoring Board (NCDMB). Since Nigerians have become proficient in occupations that were formerly the sole domain of foreigners, the majority of professions that were once conducted outside of Nigeria are now done by Nigerians and in Nigeria. Due to this, a significant portion of industrial spending has been kept domestically, which has had a favorable impact on the creation of jobs and GDP development.

5. The Way Forward

Domestic oil service firms struggle to compete successfully with well-funded competitors from Europe and the United States due to the high cost of funding. Due to this unfortunate situation, Nigerian banks have been reduced to little more than "cash centers," because they desire quick transactions and profits, and are not yet equipped for long-term projects. The creation of a robust energy bank that would support regional contractors and investors is a concept that policymakers in Nigeria's oil and gas sector ought to take very seriously. As a result, they would participate more actively and get the essential skills to facilitate the transfer of technology. Technology transfer needs to be well planned and aggressively pursued if gains in the areas of business, defense, and politics are to be guaranteed. It has been noted that there are more Nigerians currently holding managerial and professional roles in businesses participating in both upstream and downstream activities. Evidence of technology transfer, however, has not yet been observed. Therefore, Nigeria needs to develop its own special plan for technical advancement, which must be seriously pursued if Nigerians are to soon have any real influence. The nation's incapacity to build its infrastructure was another factor that rendered ineffective earlier attempts to improve local content (and continues to provide a hurdle to present efforts). Additionally, there is a dearth of efficient foundries, a strong iron and steel industrial foundation, and machine tool manufacture. All of them are components of the overarching problem that the government must solve through its privatization plan. In particular, when awarding contracts for deepwater and other oil industry projects, the government must rigorously adhere to the local content blueprint, eliminate any discrepancies in the local content legislation, and show real respect for it. Such a policy ought to make sure that the downstream sub-sector as a whole, including the refining industry, is commercialized and made more accessible to private sector involvement. Additionally, it should guarantee the effective operation of the nation's current refineries. The ideal way to do this is to follow Indorama/EPCL, Nigeria, and bring in core investors to buy the bulk of the stock in the plants and take over management. The goal of the program should be to actively include private sector involvement in refining should be promoted,

and efforts should be made to attract investors to construct refineries that are primarily geared toward export markets.

Getting exploration and production businesses working in Nigeria's upstream as well as newcomers to the country to commit to downstream business, a crucial strategic objective of the local content policy should be to foster the development of energy infrastructure and assets. Networking, also known as the extended enterprise concept (virtual integration, outsourcing, collaborative R & D), must also be highlighted.

6. Conclusion

The broad goals of local content development include luring in foreign direct investments, creating jobs, building human capacity, transferring technology, and enhancing the economy. Nigeria's lackluster infrastructure, which makes it unable to support industrial expansion, and the unfavorable investment climate, which includes a lack of security and a clean environment, are only a few of the disparities between the success of the nations described in this article and Nigeria. It is crucial to pinpoint the reasons why the Nigerian local content policy has been poorly implemented, and the difficulties will be considered while creating a structure for improved application. Furthermore, the Local Content Act is crucial to the development of Nigeria's oil and gas sector and prioritizes Nigerian independent operators.

7. Recommendations

For the local content development initiative to be implemented efficiently the following suggestions are made:

- (i) To support the development of the industrial sector, adequate power must be generated. In order to build investor trust, it will be necessary to enhance the infrastructure for telecommunications and all other public utilities. A favorable openness in the public and private sectors should be sought after, and an investment atmosphere should be promoted.
- (ii) By implementing a sort of revolving financing program for indigenous businesses, the Nigerian Content Development Fund, established by the Act and overseen by the Board, shall assist indigenous peoples in participating meaningfully in the Nigerian economy's oil and gas sector. enterprises in need.
- (iii) The Nigerian Oil and Gas Industry Content Act needs to be modified in order to specify some phrases that are currently being used by various operators in violation of the Act's intentions to engage in a variety of contract staffing and casualization practices. To prevent abuse, the vast discretionary authority of the petroleum minister must also be limited.
- (iv) In order to strengthen the implementation of the Niger Delta's human capital development plan, private-public partnerships are also necessary. The establishment of cutting-edge technology infrastructure and a comprehensive social welfare program should be prioritized in order to turn the area from its current state into an industrial hub.
- (v) The Act has to be amended in order to expand the extent of its application to other significant Nigerian economic sectors, such as the construction of roads, dams, and railroad lines.

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Market Segmentation, Customer Value Proposition and Organizational Performance of Indigenous Oil And Gas Companies In Rivers State, As Moderated By Market Innovation And Local Content Policy.

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ABSTRACT

The study investigated the effect of market segmentation components (behavioural, geographic, demographic) and customer value proposition components (functional, monetary, psychological) as moderated by market innovation and local content policy on organizational performance of indigenous oil and gas companies in Rivers State, Nigeria.

The performance of many indigenous oil and gas companies has been a major concern for managers due to the dwindling level of productivity, low profitability, sales volume, market share and inability to compete favourably in the market place. However, market segmentation strategies and customer value proposition as moderated by market innovation and local content policy are critical elements that organisations can adopt to enhance performance within the oil and gas sector. Several studies have been carried out in many industries on the effect of market segmentation on organisational performance, however, there has not been enough research work to explain the effect of market segmentation, customer value proposition as moderated by market innovation and local content policy on the performance of indigenous oil and gas companies in Rivers state, Nigeria.. The study adopted survey research design. The population of study consist of 322 management staff of 11 indigenous oil and gas companies in Rivers State, Nigeria. Total enumeration method was adopted for the study. A validated and structured questionnaire was adopted to collect data with 95.3% response rate. Cronbach's alpha reliability coefficients for the construct's ranged from 0.60 to 0.78. Data were analysed using descriptive and inferential (Multiple and Hierarchical regression) statistics.

Findings revealed that the effect of market segmentation and customer value proposition on organisational performance of indigenous oil and gas companies in Rivers state is significantly moderated by the combination of market innovation and local content policy(= -2.322E-6, $R^2 = 0.006$; $F = 10.055$, $p < 0.05$).

The study concluded that market segmentation, customer value propositions as moderated by market innovation and local content policy have significant effect on organisational performance of the selected indigenous oil and gas companies in Rivers State, Nigeria. The study recommended that management of indigenous oil and gas companies should segment their market and tailor their products and services to meet customers' needs and develop a strong value proposition with adequate adherence to local content policy that can attract and retain customers with local contractors' participation, which ultimately leading to improved organizational performance

Keywords: Customer value proposition, Indigenous oil and gas companies, Local content policy, Market innovation, Market segmentation, Organisational performance.

BACKGROUND TO THE STUDY

The downstream sector of the oil and gas represents the primary engine of economic activity and prosperity in the countries of the world. The sector is considered as an important input of economic development within primary energy resources that cut across the value chain of petroleum trading, marketing, distribution, and retail segments. The need for petroleum is increasing everyday due to the use of fuel for increased needs such as transportation, electricity consumption, heating and so on. In spite the overall importance of this sector, relatively little quantitative information is available concerning the Marketing strategies and performance of the various components measured in terms of productivity, profitability, sales volume with particular focus on innovation and local content policy moderating their behaviour. The oil and gas sector has been challenged by the dramatic fluctuations in oil price as a result of poor segmentation practices (faulty behavioural, geographic and demographic segmentation practices) and their inability to meet up with satisfying the expectations of the global market via the introduction of strong value propositions (functional, monetary and psychological value propositions) that give them edge over their competitors. Globally, the oil and gas industry has witnessed price fluctuations and huge uncertainty in recent time due to Russia's invasion of Ukraine which has squeezed supplies, leading to fossil fuel shortages and skyrocketing prices thereby making energy majors to enjoy bumper profits, and as a source of attractive target for government tax raids. As oil and gas prices soared, energy super majors have reported bumper profits. For the first three quarters of 2022, the seven companies that are often lumped together as Big Oil – ExxonMobil, Chevron, Shell, BP, Eni, Total Energies and ConocoPhillips – saw profits reaching more above \$170bn, (Nash, 2022).

The United States of America (USA) faces prospect of oil supply shortages particularly after the President, Biden, drained the Strategic Petroleum Reserve (SPR) to its lowest level in 38 years. Despite pushing for green growth, Biden was forced to back US fossil fuels. And also the administration has already eased sanctions on Venezuela, with the US once again importing crude from the South American country and approving the development of oil and gas in Alaska (Currency direct 2023). In Africa, oil and gas is faced with several risks and challenges including, Political Risk, Geological Risk, Price Risk, Supply and Demand Risks, Cost Risks (Beattie, 2020). With an ever-growing market in the oil and gas, continuing price volatility and inflationary costs of production and raw materials, oil companies face serious challenges in production outputs remaining competitively priced and while diversifying their services to keep going in fluctuating market in Kenya in 1994 led to an increase in the number of petroleum dealers in the country (Institute of Economic-Affairs, 2020). There are over thirty oil marketing companies in Kenya and their market shares are based on the volumes of fuel traded within a given period (Energy Regulatory Commission, 2017). African countries can be categorized into four archetypes based on the resilience of their crude oil reserves and the extent of their economic reliance on oil and gas revenues. Countries with more than 50 percent of projected oil production at risk in the event of a more rapid energy transition (achieved commitments scenario) can be considered vulnerable, while those with less than 50 percent of production at risk are likely to be more resilient to global shifts. This analysis focuses primarily on the competitiveness of African crude supply, given the global nature of oil demand and supply dynamics (Acha, et al, 2022).

The Nigeria's petroleum sector performance has increasingly become unstable over the years (Oyakhire & Akpan, 2021). The oil and gas industry in Nigeria is currently facing significant challenges because of the volatility of oil prices, pipeline vandalism and illegal refining as a result of which the performance in the industry has been dropping. Aside from activities of oil theft, the drop in the productivity, profitability, sales volume, market share, competitive advantage and return on investment in the industry have been traced to the inability of the industry captains to harness the advantages inherent in segmenting the market along with adopting competitive customer value proposition dimensions over time. After years of letting the Integrated Oil Companies (IOCs) run the show in the petroleum industry, the Nigerian Federal Government passed a bill in 2010 that opened up the sector to indigenous participation and value retention. (Ameh, 2021) pointed out that even with the existence of regulatory agencies, indigenous oil and gas companies have experienced marketing problems. The indigenous oil and gas industries faced huge challenges. In every aspect, the Nigerian oil and gas industry proved to be unsatisfactory, failing to meet expectations (Addeh, 2022).



Several studies have examined market segmentation from different industries in several countries (Manyando & Haq, 2018; Plomaritou, & Menelaou, 2020; Zhang, & He, 2019). All these studies looked at market segmentation from behavioural perspective in different industries. However, the linkage between the combined moderating roles of market innovation and local content policy with respect to the relationship between market segmentation, customer value proposition and organisational performance of indigenous oil and gas companies in River State, Nigeria has not been properly established. This, therefore, leaves a gap that needs to be attended to in the case of market segmentation, customer value proposition and organisational performance of indigenous oil and gas companies in River State, Nigeria as moderated by combined effects of market innovation and local content policy. This area needs more studies in order to address the contradictions among scholars' findings as far as the level of significance or otherwise that have been reported across the different constructs.

Objective of the study: This study examined the effect of market segmentation components, customer value proposition on organizational performance of indigenous oil and gas companies in Rivers State, Nigeria as moderated by market innovation and local content policy.

REVIEW OF LITERATURE

Market segmentation

Market segmentation is a way of dividing an aggregate market into smaller (and more distinct) groups of consumers who have similar needs and wants (Desarbo & Blank, 2020). Relatedly, market segmentation characterizes the division of the market into distinct groups of buyers who have distinct needs, characteristics, or behavior and who might require separate products or marketing mixes (Kotler & Keller, 2017). Choi et al., (2020) described market segmentation as a way of subdividing a market that is composed of heterogeneous customers into multiple sub-markets composed of homogeneous customers, grouping the buyers based on their preferences. Thomas (2019) defines market segmentation at its most basic level, he called the term "market segmentation" as a way of subdividing a market along some commonality, similarity, or kinship. Wilki and Cohen (2006) cited in Thomas (2019) viewed market segmentation as the process by which the total heterogeneous market for a product is divided into several sub-markets or segments and each segment is homogeneous in all major aspects and is different from the other. According to Tyopine (2019), as cited in Oyakhire & Akpan (2021), market segmentation involves the grouping of customers with similar needs and buying behavior into segments, each of which can be reached by a distinct marketing programme. The purpose of segmentation is the concentration of marketing energy and force on the subdivision (or the market segment) to gain a competitive advantage within the segment. It's analogous to the military principle of "concentration of force" to overwhelm an enemy. The researcher defines market segmentation as the process of identifying and breaking down of both consumer and business markets into sizeable levels, based on their unique characteristics, that can be targeted or not targeted for customized benefits. Boateng et al., (2022) stresses that market segmentation is one of the fundamental principles of marketing. Put differently, market segmentation involves the grouping of customers together with the aim of better satisfying their needs whilst maintaining economies of scale. The basic tenet of market segmentation is that a heterogeneous group of customers can be grouped into homogeneous groups or segments exhibiting similar wants, preferences, and buying behaviour. The process of market segmentation enables companies to target specific market segment(s), design more suitable marketing mix variables (price, product, promotion, and distribution), and develop more effective positioning strategy.

Xu et al., (2022) argue that there are several benefits of market segmentation. It helps in designing products that match with the market demand. The concept attempts to bridge the gap between diverse customer needs and limited company resources, by encouraging distinct product and marketing offerings to be developed to suit the requirements of different customer segments (Srivastava, 2015).

Customer Value Proposition

Stakeholder theory proposes that value creation is a collaborative effort in relationships, ideally benefitting the focal business and all its stakeholders (Freeman 2010) as cited in Freudenreich et al, 2019). This corresponds to the idea of multi-directional value flows and supports an in-depth analysis of what types of value a stakeholder relationship creates, with whom and for whom. The view that value creation is a process resulting in different outcomes for different stakeholders is particularly prevalent in the sustainability-oriented business model literature (Dembek et al., 2017). Wellington (2022) posited that product value proposition is a statement that articulates the product's features, uses and differentiators while taking into account the customers' problems, wants and needs. It is the north star for product messaging. A product value proposition describes what the customer will get out of the product. It serves as the foundation for additional ancillary marketing copy that will branch off from it. The core idea of value creation for customers is expressed in a variety of terms, including value proposition, value object, offering, and customer benefit (Al-Debei & Avison 2010; Chesbrough 2007; Gordijn & Akkermans 2001; Teece (2010) as cited in Freudenreich et al., (2019). The term value proposition is predominantly used to refer to the bundle of products and services offered to customers to satisfy their needs and to create value for them. The customer value proposition reflects an offering's ability to create superior value for target customers relative to competitors' offerings. Because value depends on customer needs, the same offerings can have a different value for different customer segments. Depending on the type of customer needs, an offering can create value in three different domains: functional, monetary, and psychological (Chernev, 2012).. This process has also led to pervasive and overarching changes in organizations and their business models—a phenomenon known as digital transformation (Verhoef et al., 2021). While value propositions have been traditionally seen as provider-initiated statements of benefits targeting passive customers (Eggert et al., 2018), an emerging body of research increasingly recognizes the customer's active role in the value proposition development process (e. g., Ballantyne et al., 2011; Kowalkowski et al., 2012; Baumann et al., 2017).

In conclusion, customer value proposition is the recognition of a value chain in the distribution of oil and gas that is given to potential customer for a better organisational performance

Market innovation

Innovation agencies around the world are increasingly considering socioeconomic-technological challenges that can be tackled through innovation policies (EC Innovation Union; OECD Innovation Strategy). The idea is that, through such challenges, which can relate to such issues as climate change, cancer, or the demographic-aging crisis, innovation policy can produce solutions for societal problems. Societal challenges require technological, behavioral, and systemic changes and have much to learn from those mission-oriented feats that led to putting humans on the moon and to the emergence of new general-purpose technologies ranging from the Internet to biotechnology and nanotechnology (Foray et al., 2012). Systematic strategic thinking about market-creating value propositions establishes a basis for conceiving and developing products or services innovations that will appeal to both current and new customers. The aim is to create products or services that enjoy little competition, precisely because they meet a need that is not being met (Denning, 2017). Market innovations tend to create new markets i.e., new market that serve people for whom either no products existed or existing products were not accessible for a variety of



reasons, including cost or a lack of the expertise required to use them. Market-creating innovations transform complicated and expensive products into ones that are much more affordable and accessible to an increased number of consumers. In some cases, such innovation can create entirely new product categories (Christensen et al., 2018). In view of the above, market innovation is the discovery of a novel idea based on which a product or service can be added to an existing customer in the market place.

Local content policy (local content Act)

Local content policy is essentially concerned with how best to govern the relationship between humans and the economic activities for their mutual benefit. This broad definition successfully captures the unbounded character of the government as a distinct public policy field, touching as it does upon virtually every aspect of modern life. However, it lacks specificity: most areas of local content policy affect the world 'out there.' Local content policy is a declaration of government political activities, plans and intentions relating to a particular cause (Anjaneyulu & Kubendiran, 2022). Khan (2016) discussed it as the legal and regulatory framework within which an enterprise operates influences its survival and growth potentials. Regulations by governments produce an atmosphere for businesses to grow or crumble (Mwasiaji, 2019). This became necessary as public service delivery is key in ensuring that governments meet commitment to citizens as well as realization of development. The public service delivery in terms of quality, ease of access, speed of provision and cost is key in poverty alleviation and achievement of developmental goals such as Sustainable Development Goals (SDGs) (Opio et al., 2022). Nigerian oil and gas industry content development act 2010 (act no. 2) for instance is an act to provide for the development of Nigerian content in the Nigerian oil and gas industry, Nigerian content plan, supervision, coordination, monitoring and implementation of Nigerian content; and for related matters. This regulatory act was enacted by the national assembly of the federal republic of Nigeria in April 22nd day of April, 2010.

The Nigerian content development in oil and gas industry stipulates that: Notwithstanding anything to the contrary contained in the Petroleum Act or in any other enactment or law, the provisions of this Act shall apply to all matters pertaining to Nigerian content in respect of all operations or transactions carried out in or connected with the Nigerian oil and gas industry: All regulatory authorities, operators, contractors, subcontractors, alliance partners and other entities involved in any project, operation, activity or transaction in the Nigerian oil and gas industry shall consider Nigerian content as an important element of their overall project development and management philosophy for project execution

The Nigerian Oil and Gas Industry Content Development (Local Content Act 2010) was enacted to promote the indigenous participation in the Nigeria's oil and gas industry for the purpose of improving the economic and social wellbeing of those engaged in operating in the oil and gas industry. The Act provides for the development of Nigerian content in the Nigerian oil and gas industry, Nigerian content plan, supervision, coordination, monitoring, and implementation of the Nigerian content. The Act stated in section 2 of the local content act the requirement of any company or organization that intends to operate in the Oil and Gas Industry. In section 2 of the local content act, which states that "all regulatory authorities, operators, contractors, subcontractors, alliance partners and other entities involved in any project, operation, activity or transaction in the Nigerian oil and gas industry shall consider Nigerian content as an important element of their overall project development and management philosophy for project execution". The Act defines Nigerian content in Section 106 as "the quantum of composite value added to or created in Nigeria through the utilization of Nigerian resources and services in the petroleum industry resulting in the development of indigenous capability without compromising quality, health, safety, and environmental standards". The sole purpose of the Local Content is to increase Nigerian participation in the oil and gas industry by prescribing minimum thresholds for the use of local services and materials for the promotion of technology innovation, capacity building and skillful manpower to serve the Nigerian labour in the oil and gas industry (Jegede 2020).

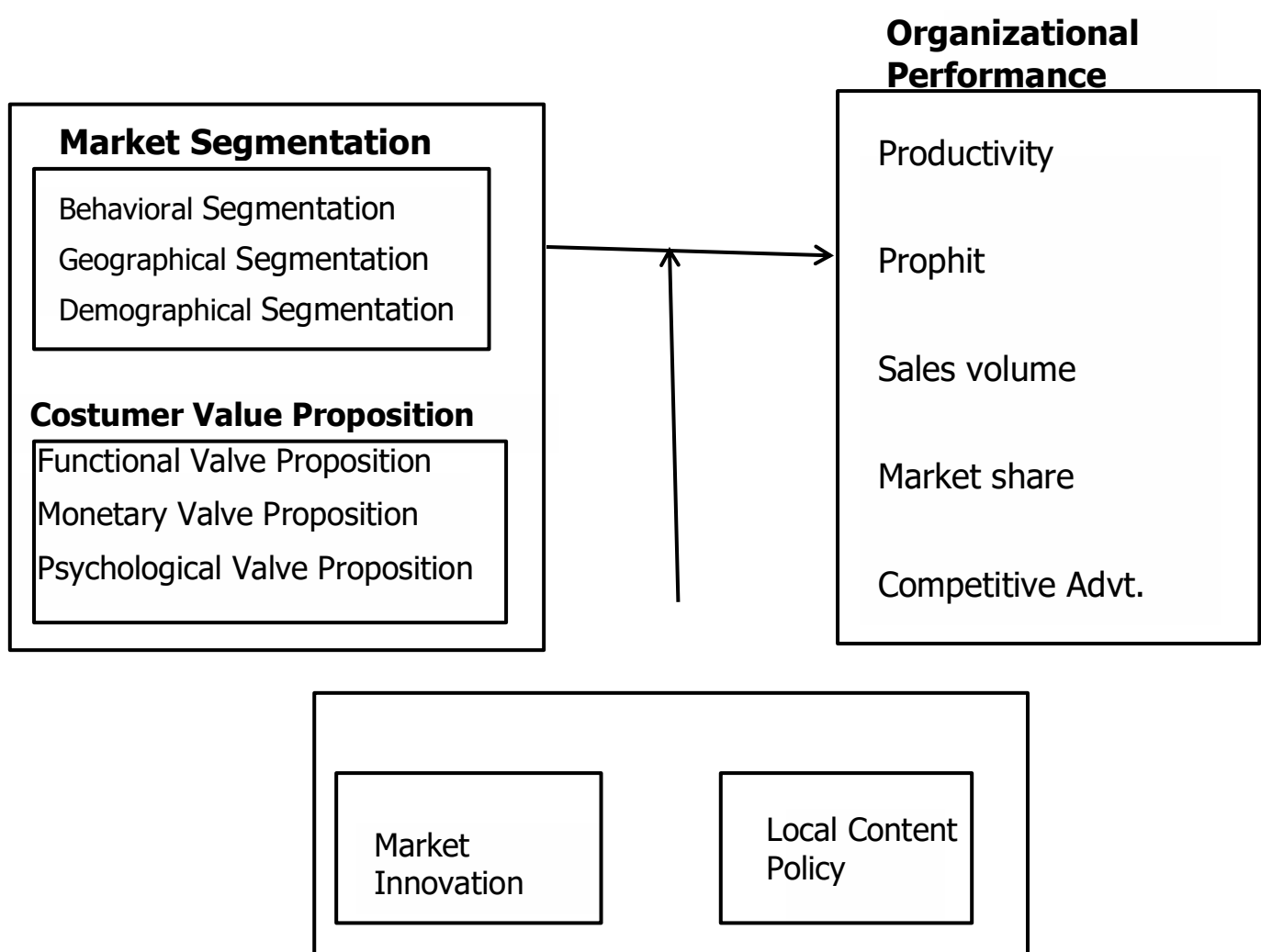
This policy has greatly affected the indigeneous oil and gas companies operating in Rivers state by providing employment opportunities to the local contractors and promoting local material usage for the production, supply and distribution of oil and gas projects and products.

Organisational Performance

The definition of organisational performance has been described as the development of value in a corporation that may refer to a positive change in a company's financial state as a result of better performance. Better performance can be due to better use of raw materials, labor, money, and efficient resource management (Alchian & Demsetz, 2017). Venkatraman and Ramanujam (2018) agree that there is no consensus about how to define firm performance. Organizational performance can also be used to view how an enterprise is doing in terms of level of profit, market share and product quality in relation to other enterprises in the same industry. Dim and Nzube (2020) corroborated this view by claiming that organizational performance is a reflection of productivity of members of an enterprise measured in terms of revenue, profit, growth, development and expansion of the organization. Firm performance, on the other hand, is described by Armstrong (2015) as both behavior and results, which are the primary concerns of any business manager. Studies by Richard et al., (2019) proposed that firm performance encompasses three specific firm outcome areas: financial performance that is profits, volume of sales, return on assets, market share return on investment and shareholder return. Adjei & Adjei, (2017) indicated the ability of an organization to attain large market shares and good financial outcomes constitute organisation success. Ochieng et al., (2015) on the other hand defined organizational performance as a measure of how well a business is managed in delivering value to customers and stakeholders. Contu (2020) defined organizational performance as the degree to which the organization, with some informational, financial, and human resources, positions itself effectively on the business market. Ahmed (2018) viewed firm performance as the output of an organization with regards to its interaction with the external and internal environment.

Moreover, organizational performance is the ultimate measure of the success of any local contractor. It involves the analysis of a firm's performance measured against its goals and objectives (Simon et al., 2021). According to Rolstadas (1998), cited in Simon et al., (2021) the organizational performance of any system is a complex relationship that involves seven critical performance criteria: efficiency, effectiveness, productivity, quality, quality of work, profitability and innovation. USAID (2021) posited that the organizational performance index (OPI) is an indicator that measures the overall performance of organizations in four main domains; in terms of short-term aspects by analyzing elements such as (i) Effectiveness and (ii)Efficiency, and in the long term by reviewing those of (iii)Relevance and (iv)Sustainability. There are, however, challenges in using these measures; for starters most managers are reluctant to share their financial records, leading to researchers relying on perceived results and not actual results.

Conceptual Framework



SOURCE: Researcher Model 2023

Empirical Review

Few related studies have contributed empirically to the moderating effects of market innovation on the relationship between market segmentation, Customer value proposition and firm performance (Capittelo & Todirica, 2021; Mallik, 2019; Müller et al., 2018). Kiveu et al., (2019) defines market innovation as the process of introducing new product with higher quality than existing products, new methods of production, the opening of new markets, the use of new sources of supply and new forms of competition to existing and prospective customers, that lead to the restructuring of an industry. Mallik (2019) in his study of effect of VALS (values and lifestyle) psychographic market segmentation on Areca tea revealed that consumers want reduced price of areca tea and also using psychographic segmentation, areca health tea products are most liked by the females. Ullah and Arshad (2021). carried out a study on impact of internal innovation on firm performance with moderating role of collaborative innovation. The study primarily aim to examine the effect of internal innovation in determining the firm performance for the textile industry of Pakistan by testing the research framework of innovation diffusion theory. The study found that product innovation and process innovation both serve as critical drivers for the firm performance in the textile industry of Pakistan, and that, collaborative innovation plays a complementary role as the significant moderator between the internal innovation process and the firm's performance in the target population. Rajapathirana and Hui (2018) carried out a study on relationship between innovation capability, innovation type, and firm performance and confirmed that relationship between innovation capabilities; innovation efforts and firm performance are significant and strong.

However, Latifi et al., (2021) reported that direct link between BMI and firm performance is not significant, and that, the path is fully mediated through efficiency growth, organisational capabilities and revenue growth. In the same vein, Alao et al., (2020) investigated the effect of strategic marketing on competitive advantage and how resource capabilities moderated the effect. A cross-sectional survey research design was adopted. Based on population of 1568 managerial staff of selected petroleum products marketing firms in Lagos State, Nigeria, Cochran's formula was applied to determine a sample size of 564. Response rate of 497 respondents was achieved in the study. Data collected were analysed using multiple regression analysis. The findings established that strategic marketing has a positive and significant effect on competitive advantage and that resource capability has no significant moderation on the effect of strategic marketing on competitive advantage. However, to the best of researcher's knowledge, literature on the combined moderating effects of market innovation and government local content policy on the relationship between market segmentation, customer value proposition and performance of indigenous oil and gas companies is scarce and need to be researched on. Ladipo et al., (2023) investigated market segmentation and competitive advantage in Nigerian Telecommunications. The study results indicate that measurability, accessibility, uniqueness, and size of a market segment are the most significant market segmentation variables that affect business market share in the Nigerian telecommunication industry.

Theoretical framework

The theory underpinning this study is the resource-advantage (R-A) theory of competition (Hunt & Morgan, 1995): R-A theory maintains that firms that are successful in developing market offerings that provide more value to consumers in specific market segments and/or provide market offerings at a lower cost (relative to their competitors) will occupy marketplace positions of competitive advantage, and in turn, positions of

competitive advantage led to superior financial performance. Therefore, R-A theory shows how market segmentation strategies impact firm performance. R-A theory permits the success of market segmentation strategy. The theory's foundational premises view intra-industry demand as inherently heterogeneous. The theory, by means of its treatment of resources, accounts for the behaviors of firms that often choose to produce and market a variety of market offerings in the same industry. The theory, by means of the marketplace position matrix, explicates the mechanism by which a market segment strategy can lead to superior financial performance.

R-A theory suggests that, to be successful, firms must examine the nature of competition in the segments that they target (or are considering targeting). The nature of segments differs; thus, organizations do attain different competitive level as far as each segment is concerned, therefore, there is need for organizations to familiarize themselves with market offerings of their rivals on a segment basis. Main considerations to determine an organisation's marketplace standing include: Consumers' perceptions regarding the value of market offerings compared to those of rivals and; the cost (relative to competitors) of the resources used to produce the market offerings. Firms will have competitive advantages over rivals when they produce more valuable market offerings and/or produce market offerings more efficiently than rivals. The fit between a firm and its strategy is one of the key factors influencing successful strategy implementation (Vorhies & Morgan 2003). That is, firms whose resource sets match more closely those required by a specific segmentation strategy are more likely to be successful.

METHODOLOGY

The study adopted survey research design. The population of study consist of 322 management staff of 11 indogenous oil and gas companies in Rivers State, Nigeria. Total enumeration method was adopted for the study. A validated and structured questionnaire was adopted to collect data with 95. 3% response rate. statistically, Cronbach's alpha reliability coefficients for the construct's ranged from 0.60 to 0.78. Data are analysed using descriptive and inferential (Multiple and Hierachical regression) statistics.

Data Analysis

Hypothesis Testing

Restatement of Research Hypothesis

Ho₁: The effect of market segmentation and customer value proposition on organisational performance is not significantly moderated by the combination of market innovation and local content policy.

Table 3.1 Summary of multiple regression analysis for effect of Market Segmentation Components on Performance of indigenous Oil and Gas Companies in Rivers State, Nigeria.

Oil Performance of Indigenous Oil and Gas Companies in Rivers State, Nigeria								
N	Model	B	T	Sig.	ANOVA (Sig.)	R	Adjusted R ²	F (3, 303)
307	(Constant)	6.599	6.867	.000	0.000 ^b	0.794 ^a	0.627	172.594
	Behavioural segmentation	.332	6.107	.000				
	Geographical segmentation	.244	4.618	.000				
	Demographic segmentation	.197	3.663	.000				
a. Dependent Variable: organisational performance								
b. Predictors: (Constant), Demographic, Behavioural, Geographical Segmentation								

Source: Researchers' Field Survey, 2023

Table 3.1 shows the multiple regression analysis results for the market segmentation components on performance of oil and gas companies in Rivers State, Nigeria as a case study. The results showed that

behavioural segmentation ($\beta = 0.332$, $t = 6.107$, $p < 0.05$), geographical segmentation ($\beta = 0.244$, $t = 4.618$, $p < 0.05$) and demographic segmentation ($\beta = 0.197$, $t = 3.663$, $p < 0.05$) all have positive and significant effect on performance of indigenous oil and gas companies in Rivers State, Nigeria. The results of the analysis revealed that all three components of market segmentation (behavioural segmentation, geographical segmentation and demographic segmentation) have significant effect on the performance of oil and gas companies in Rivers State, Nigeria. This implies that, behavioural segmentation, geographical segmentation and demographic segmentation are important predictors of performance among the surveyed oil and gas companies in Rivers state.

The R value of 0.794 supports this result and it indicates that market segmentation components have a fairly strong and positive relationship with the performance of indigenous oil and gas companies in Rivers State, Nigeria. The coefficient of multiple determination Adj. $R^2 = 0.627$ indicates that about 62.7% variation that occurs in the performance of oil and gas companies in Nigeria can be accounted for by the components of market segmentation components while the remaining 37.3% changes that occurs is accounted for by other variables not captured in the model.

The effect of market segmentation and customer value proposition on organisational performance is not significantly moderated by the combination of market innovation and local content policy.

A hierarchical multiple regression was performed to test the null hypothesis where the analysis was carried out in the order of hierarchy. An interaction term was obtained by multiplying the composite score for market segmentation and customer value proposition by local content policy. The hypothesis would be supported if the effect of the interaction of Market segmentation and customer value proposition and local content policy on organizational performance of the selected Oil and Gas companies in Rivers State, Nigeria is statistically significant. The results of the analysis are presented in Table 3.2

Table 3.2: Goodness of Fit for Summary of multiple regression analysis for effect of Market segmentation and customer value proposition on Organizational performance as moderated by Market innovation and local content policy

Model Summary									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.897 ^a	.805	.804	5.24090	.805	625.047	2	303	.000
2	.903 ^b	.815	.814	5.10585	.011	17.241	1	302	.000
3	.906 ^c	.821	.819	5.03099	.006	10.055	1	301	.002
a. Predictors: (Constant), Customer Value Proposition, Market Segmentation									
b. Predictors: (Constant), Customer Value Proposition, Market Segmentation, Market Innovation									
c. Predictors: (Constant), Customer Value Proposition, Market Segmentation, Market Innovation, local content, Interaction term (MSEG*CVP*MI*LCP)									

Source: Researcher's Field Survey, 2023

Table 3.2 presents the summary of hierarchical regression analysis which was used to test how market innovation and local content policy moderates the effect of market segmentation, customer value proposition and organizational performance of oil and gas companies in River State, Nigeria. The predictors are market segmentation and customer value proposition, market innovation, local content policy and interaction of market segmentation, customer value proposition, market innovation and local content policy aggregated (MS*CVP*MI*LCP) while the dependent variable is oil and gas companies' organizational performance aggregated. The results in Table 3.2 shows that $R^2 = 0.805$ and adjusted $R^2 = 0.804$ for Model I. This indicates that market segmentation and customer value proposition explained 80.4% variation in the organizational performance of oil and gas companies in River State, Nigeria. With the inclusion of market innovation and local content policy in Model II as an independent variable, there was an increase (R^2 change) of 0.11 from 0.805 to 0.815. Therefore, market segmentation, customer value proposition, market

Corrosion Behavior Of Cast Iron In 1.5M Of Hcl Using Water Hyacinth Extract As Inhibitor

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ABSTRACT

The use of plants extract as inhibitors has been identified by several researchers as an alternative way to prevent metal and alloys from corrosion. Numerous failures and eventual losses in the chemical industry have been traced to corrosion processes. This research work focused on the investigation of corrosion inhibition of cast iron in 1.5M of HCL with water hyacinth extract as inhibitor. The extract from water hyacinth was obtained from Ilaje in Ondo state and analyzed for chemical composition. Samples of cast iron were subjected to acidic solution with and without the inhibitor at different concentrations for an exposure period of 168hrs. Corrosion rate was examined using weight loss and inhibition efficiency calculated, The phytochemical analysis has shown that they contain compounds such as Alkaloids, Phenols, Flavonoid, Saponins, Tannins that may help protect cellular systems which is within the range of inhibitor concentration studied and proved that its usage will extremely reduce corrosion rate. The result obtained show that as the concentration of inhibitor increases, the harshness of corrosion reduces on the cast iron. Inhibition efficiency of the extract increased with an increase in concentration of the extract.

Keywords: mild steel; inhibitors; corrosion resistance; water hyacinth; acid



1.0 INTRODUCTION

Corrosion is the deterioration of materials by chemical interaction with their environment. Degradation implies deterioration of physical properties of the material (Singh and Quraishi, 2011 ; Louis *et al* , 2017). The environment could be of any type such as atmosphere, water; sea water, acids, alkaline, steam, gases, soils, liquid metals, etc. Corrosion is one of the most critical problems ravaging storage and transportation facilities among others; and in most cases, the traditional corrosion protection systems (cathodic protection and coatings) are either not efficient enough, have short service life, or cannot be applied (Mathina and Rajalashmi, 2016). Researchers have therefore created tremendous effort work in the field of corrosion inhibition of metals using natural products and corrosion inhibiting ability of plant phytochemicals such as tannins, alkaloids, organic amino acids and organic dyes, etc. This has brought about sustained interest on the corrosion inhibiting property of natural products of plant origin (Satapathy *et al.*, 2009). The most practical method of protecting metal is the use of inhibitors especially in acid solutions to prevent unexpected metal dissolution and acid consumption (Znini *et al.*, 2011). A number of organic and inorganic compounds have been studied as inhibitors to prevent metals from corrosive attacks. Usually, organic compounds exert a significant influence on the extent of adsorption on the metal surface and therefore can be used as effective corrosion inhibitor. The efficiency of these organic corrosion inhibitors is associated with the presence of polar functions with Sulphur (S), Oxygen (O) or Nitrogen (N) atoms in the molecule, heterocyclic compounds and electrons (Solomon *et al.*, 2010). Due to the hazardous effects of most synthetic inorganic inhibitors and restrictive environmental regulations, researchers have focused on the need for the design of cheap, non-toxic and environmentally benign natural products as corrosion inhibitors. Natural organic compounds which are either synthesized or extracted from aromatic herbs, spices and medicinal plants have played major roles in this regard. Plants extracts are seen as rich sources of naturally synthesized chemical compounds that can be produced by simple processes at low cost and are biodegradable in nature. The application of naturally extracted compounds from leaves or seeds as corrosion inhibitors have been widely reported by several authors (Orubite and Oforka, 2004; Roslliza and Wan Nik, 2010). Thus, in this research work, investigation was carried out on water hyacinth as a potential inhibitor on cast iron.

Water hyacinth (*Eichhornia crassipes*) is an aquatic plant which can live and reproduce floating freely on the surface of fresh waters or can be anchored in mud. The plant originated in the Amazon basin and was introduced into many parts of the world as an ornamental garden pond plant due to its beauty. It has proliferated in many areas and can now be found on all continents apart from Europe. The mature plant consists of long pendant roots, rhizomes, stolons, leaves, inflorescence and fruit clusters. Water hyacinth has recently been reported as a good substance for the production of paper, biogas, fiber board, yarn and rope, basket work and charcoal briquetting. Water hyacinth entered Nigeria waters through pot-novo creek in Benin Republic in September 1984. Since that time, it has been spreading and has gotten to the confluence of Rivers Niger and Benue, and spread to other rivers in Nigeria. This aquatic plant is problematic in Nigeria coastal waters as it blocks the navigation way, blocking irrigation channels and rivers, recreational activities, hydropower generation, restricting livestock access to water, destroying natural wetland, eliminating native aquatic plants, reducing infiltration of sunlight, changing the temperature, pH and oxygen levels of water and reducing gas exchange at the water surface. Others include, increasing water loss through transpiration (greater than evaporation from an open water body), altering the habitats of aquatic organisms, reducing aesthetic values of waterways, reducing water quality from decomposing plants, destroying fences, roads and other infrastructure when large floating rafts become mobile during flood events, and destroying pastures and crops when large floating rafts settle over paddocks after flood events.

Corrosion has a vast environmental and economic impact on all the surfaces of national infrastructure like highways, bridges, buildings, chemical processing units, waste water treatment and virtually on all metallic objects in our day to day life use. In addition to our everyday encounters with this form of degradation, corrosion causes plant shutdowns, waste of valuable resources, loss or contamination of product, reduction in efficiency, costly maintenance, and expensive overdesign. It can also jeopardize safety and inhibit technological progress.

Cast iron is a generic term that identifies a large family of ferrous alloys. Cast irons are primarily alloys of iron that contain more than 2% carbon and 1% or more silicon with a wide variety of properties. Cast irons

are complex materials with stable and meta-stable phases and have elements in the solution which influence the extent and stability of the desirable properties not obtained by other alloys. Cast irons are intended to be cast to shape rather than formed in the solid state. Cast irons have low melting temperature, high fluidity when molten, don't form undesirable surface film when poured due to less reactivity with molten materials and have slight to moderate shrinkage during solidification and cooling.

However, cast irons have relatively low impact resistance and ductility which may limit their use (Henkel and Pense, 2002)

Mechanical properties of cast irons like strength, ductility, and modulus of elasticity depend strongly on structure and distribution of micro structural constituents. Physical properties such as thermal conductivity and damping capacity are strongly influenced by microstructure. Cast irons popularity stems from an ability to cast complex shapes at relatively low cost and the wide range of properties that can be achieved by careful control over composition and cooling rate without radical changes in production methods (Milekhine, 2003).

Alloying elements can play a dominant role in the susceptibility of cast irons to corrosion attack. Silicon is the most important alloying element used to improve the corrosion resistance of cast irons. Silicon is generally not considered an alloying element in cast irons until levels exceed 3%. Silicon levels between 3 and 14% offer some increase in corrosion resistance to the alloy, but above about 14% Si, the corrosion resistance of the cast iron increases dramatically (Rana *et al.*, 2001). Carbon in cast iron can exist in two forms, namely as free graphite or combined with some of the iron to form iron carbide (cementite). Due to existing form of carbon, cast irons can be defined as; a family of alloys which contain carbon at a level which will cause precipitation of graphite or iron carbide (Fe_3C) during the final phase of solidification (Loper, 2002). Since high carbon contents tend to make cast irons very brittle, other metallic and non-metallic alloying elements are, therefore added to control and vary the microstructure and mechanical properties (Sinha, 2003). Cast Iron has, for hundreds of years, been the preferred piping material throughout the world for drain, waste, and vent plumbing applications and water distribution. Gray iron can be cast in the form of pipe at low cost and has excellent strength properties. Unique corrosion resistance characteristics make cast iron soil pipe ideally suited for plumbing applications (Zhang *et al.*, 2001). Cast iron and steel corrode; however, because of the free graphite content of cast iron (3% - 4% by weight or about 10% by volume), an insoluble graphitic layer of corrosion products is left behind in the process of corrosion. These corrosion products are very dense, adherent, have considerable strength, and form a barrier against further corrosion. Because of the absence of free graphite in steel, the corrosion products have little or no strength or adherence and flake off as they are formed, thus presenting fresh surfaces for further corrosion. In tests of severely corroded cast iron pipe, the graphitic corrosion products have withstood pressures of several hundred pounds per square inch although corrosion had actually penetrated the pipe wall (Fredric, 2004).

2.0 MATERIAL AND METHOD

2.1 MATERIAL

The materials utilized for the research include, Water hyacinth, Cast iron, Laboratory oven, Electronic weighing balance, Pulverizer, Ethanol, Measuring cylinder, Filter paper, Conical flask, Zinc oxide, Distilled water, Hydrochloric acid

2.2 METHODS

2.2.1 Electrolytic Solution

The corrosive solution, 200ml of 1.5M HCL was prepared with distilled water. The concentration range of the extract used was 10ml, 20ml, 40ml and 60ml and volume of the electrolyte used was 30ml in each experiment

2.2.2 Preparation of Inhibitor

Water hyacinth samples were sourced from Ilaje Ondo State. The leaves samples was washed, oven dried at 70° C for 10hours and pulverized using mechanical pulverizer. About 10g of the pulverized leaves was weighed and soaked in 450ml of ethanol for 72 hours. The mixture was then filtered to obtain the extract. Water hyacinth was prepared in volumetric concentration of 10, 20, 40, 60ml in 30ml of the acid solution respectively.

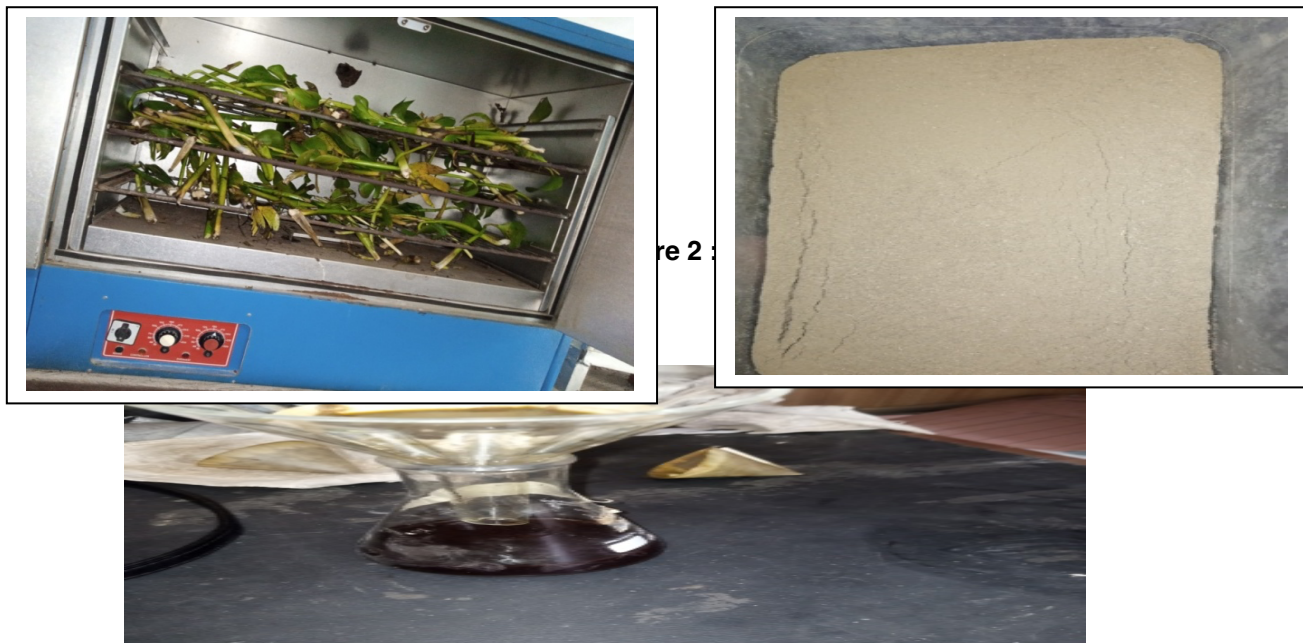


Figure 3 : Extraction process of water hyacinth

2.2.3 Elemental Analysis Preparation

To determine metal concentrations in the plant sample, 0.1g of the sample was wet-digested with 15ml of concentrated HNO_3 and HClO_4 (3:1) acid mixture in 25ml pyrex tube. About 2ml of the HF acid was added, the mixture were heated to a clear solution, and then near dry. The cooled residue was dissolved in 5ml 10% HNO_3 and the solution was diluted to 25ml with deionized water for AAS measurement. The result is as shown in table 3.

2.2.4 Phyto-chemical screening

The phyto-chemical constitution of the water hyacinth was evaluated following standard procedures (in accordance to Trease and Evans, 1991). Specifically, the tannin, saponnin, terpenoids, phenols, alkaloids, and flavonoids were determined as follows:

2.2.5 Cast iron coupon preparation

Cast iron was sought out from Niger- match Eleyele, Ibadan. Chemical compositions was carried out with the using XRF composition (wt. %) of cast iron samples used for all the experiment was as follows: Carbon (3.01), Manganese (0.60), silicon (1.07), phosphorus (0.03), chromium (12.9), Nickel (0.07), lithium (0.05), tungsten (0.03). The specimens were cut into separate dimensions of 1.0 cm x 1.0 cm and thickness 5 mm. The alloy specimen were polished mechanically using emery papers and then, washed thoroughly with distilled water and degreased with ethanol and acetone, air dried before being immersed in the acid solution.

2.2.6 Weight Loss Method

The weight loss measurements were carried out by weighing the specimens in triplicate before use and after immersion in 30ml for 24hrs in 1.5M HCL solution in the absence and presence of 10ml, 20ml, 40ml, and 60ml respectively. Each of the test specimens was taken out every 24hrs, washed thoroughly with liquid soap (morning fresh), rinsed with distilled water, cleaned with acetone, dried and re-weighed. The test was also carried out progressively for 168hrs (7days). The corrosion rate (C_R), inhibition efficiency (%IE) and surface coverage (θ) were calculated from the following equations in accordance with ASTM G31 standard.

$$CR (mmpy) = \frac{87.6W}{DAT} \quad (1)$$

$$\% IE = 1 - \frac{CR_{inh}}{CR_{blank}} \times 100 \quad (2)$$

$$\theta = 1 - \frac{CR_{inh}}{CR_{blank}} \quad (3)$$

Where W is the weight loss in grammes, D is the density in g/cm³, A is the surface area in cm², T is the immersion time in hrs, CR_{inh} and CR_{blank} are corrosion rates of mild steel with and without inhibitor respectively. The procedure was utilized for all the 168hrs (7days), in order to study the inhibition efficiency of the extract inhibitor. The study gives details about the inhibition mechanism prevalent. Where W is the weight loss in grammes, D is the density in g/cm³, A is the surface area in cm², T is the immersion time in hrs, CR_{inh} and CR_{blank} are corrosion rates of mild steel with and without inhibitor respectively. The procedure was utilized for all the 168hrs (7days), in order to study the inhibition efficiency of the extract inhibitor. The study gives details about the inhibition mechanism prevalent.

2.2.7 Mass Loss Measurement

Pre-weighed cast iron samples were immersed in 100ml of the blank/inhibitor solutions for 7days (168 hours). The weight loss of each coupon was determined at every 24 hours intervals by retrieving the samples from the solution, cleaned with acetone and then reweighed. The mass loss for each sample was evaluated by dividing its weight loss by the surface area of the coupon.

2.2.8 Atomic Adsorption Spectrometric Analysis

Atomic adsorption analysis was performed using atomic adsorption spectrometer model bulk 200. This was required to determine the concentrate of iron(II) ions in the acidic solutions after gravimetric measurements. The calibration curve of iron (II) ions was drawn before analyzing the electrolyte solution. All sample containing irons were diluted with distilled water to ensure that the concentrations of metal ions are within the range of the calibration curve.

2.3 Inhibitive Medium

The inhibitor used is water hyacinth extract (**Eichhomia crassipes**). This water hyacinth inhibitor used in corrosion medium of sample B,C,D and E

2.4 Experimental Set Up

- A – 30ml HCL + sample without inhibitor
- B – 30 ml HCL + 10ml water hyacinth + sample
- C – 30ml HCL + 20ml water hyacinth + sample
- D – 30ml HCL + 40ml water hyacinth + sample
- E – 30ml HCL + 60ml water hyacinth + sample

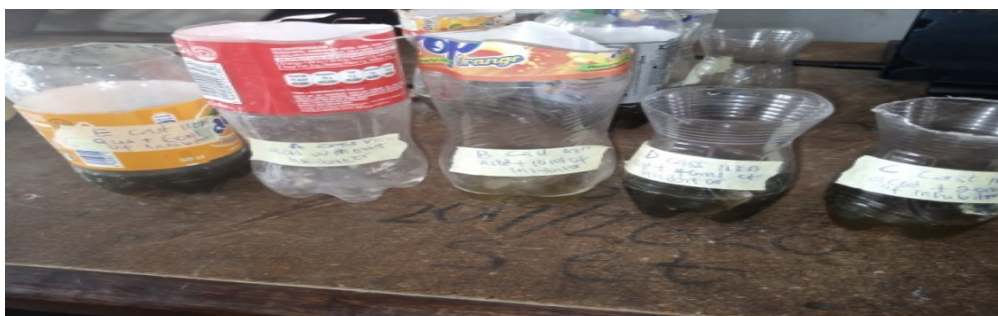


Figure 4: Samples in solution

2.5 Experimental Method

The sample prepared were immersed in the corrosion medial and the weight losses were taken on daily basis. The initial weight of the cast iron, which have cut were measured before immersion. Each of this rod



was totally immersed in each of the beaker(contain acid sample). This steel rod were removed sample (where they immersed) everyday and returned after cleaning with tooth brush and distilled water and weighing. The final weight was subtracted from the initial weight to get the weight loss during the test. However, the weight loss was calculated in grams as difference between the initial and final weight prior to immersion and the weight after removal of corrosion product.

3.0 RESULTS AND DISCUSSION

3.1 Phytochemical and Elemental Analysis

The water hyacinth extracts were preliminarily screened for their phytochemical constituents and the results are presented in Table 1&2 below. It was found that the extract samples contain many organic compounds such as Tannins, Terpenoids, Saponins, Phenols, Flavonoids, Alkaloids, oxalate, and Qurnone. Most of these constituents are known to have good inhibitive action. The presence of the compounds has been reported to promote the inhibition of cast iron in aggressive acidic media (Alaneme *et al.*, 2016). These compounds are also bioactive which contains functional groups and possess molecular structures of inhibitory potentials of plant extracts(Nnanna *et al.*, 2016). This showsthe variation of inhibition efficiency with the extract concentration on cast iron in 1.5 M HCL solutions as observed in the plots. This indicates that the plant extracts mitigate the corrosion of cast iron in 1.5 M HCL solutions and that the extent of corrosion inhibition depends on the amount of the inhibitor constituents present in the extract (Xiaoyuan and Singh, 2010). The works reveals water hyacinth hashigh of potassium (132. 400ppm), and appreciable amount of calcium (98.100ppm), sodium (46.200ppm), and others. All these are nutritionally valuable minerals. However these values maybe lower than those found in other plants.

Phytochemical Screening of Water Hyacinth

Table 1: Quanlitative screening

Secondary metabolite	Test	Colour changes	Inference
Alkaloid	Mayers test	Greenish yellow	++ve
Saponin	Frothing test	Nil	-ve
Phenol	Swallowsh	Reddish brown	+++ve
Flavonoid	Ferric chloride test	Yellowish green	+++ve
Tannin	FecI3 test	Brown	+ve
Terpernoid		Light yellow	+ve
Oxalate		Nil	-ve
Quinone			-ve

Key:

- +++ve= Abundant
- ++ve =moderate
- +ve = low

Table 2: Quantitative screening

Secondary metabolite	Constituent
Alkaloid	1.9
Tannin	1.4
Flavonoid	3.9
Terpernoid	3.25

Phenon	5.0
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Table 3: Elemental composition of water hyacinth

Parameters	Composition (ppm)
Na	46.200
Ca	98.100
K	132.400
Mg	16.916
As	0.011
Fe	5.275

Table 4: Elemental Composition of Cast Iron(%)

Element	Composition
C	3.01
Si	1.07
Mn	0.60
P	0.03
S	0.05
Cr	12.9
Ni	0.07
Li	0.05
W	0.031
Ti	0.005
Cu	0.039
Fe	Balance

3.2 Gravimetric measurements

The gravimetric monitoring of corrosion rate (C_R) and inhibitors efficiency (I.E %) is a versatile tool because of the simplicity of the method and this has been employed by several researchers (Anyiamet *al.*, 2020;

Ogunleye *et al.*, 2020; Nnanna *et al.*, 2016). The values of C_R and I.E % obtained from the weight loss experiment at varying concentration of *Eichhornia crassipes* extract in 1.5M HCL solution are also shown below. The results showed that the mass loss increased with increase in exposure time in the blank solution. However, the addition of the extracts in 1.5M HCL solution resulted in a significant reduction in the mass loss of cast iron substrate in comparison with the blank solution. The reduction can be attributed to the adsorption of the extract active constituents on the cast as revealed by the phytochemical screening. The adsorption of such compounds on the metal surfaces result in the formation of a surface layer barrier for charge and mass transfer between the base metals and the corrosive environments (Alaneme *et al.*, 2015). Moreover, the C_R as indicated in below decreases noticeably with increase in inhibitor concentration, that is, the inhibition efficiency (IE) increases with increase in the inhibitor concentration. The progressive increase of the extract shows greater reduction of corrosion rate, given maximum inhibitive efficiency at 60ml in a 24hour immersion. The inhibitive effect of the inhibitor is attributed mainly to the presence of the bioactive compounds in their constituents which was in tandem with the phytochemical screening results (Nnanna *et al.*, 2016).

Table 5: Experiment A (cast iron immersed in HCL without inhibitor)

Time (hrs)	Concentrate	Sample area (cm ²)	Initial weight (g)	Final weight (g)	Weight loss	Corrosion rate (mm/hrs)	IE%	Θ	Mass loss
24	Blank	15.5	24.2	23.4	800	27.50	0	0	51.61
48	Blank	15.5	24.2	23.0	1200	20.62	0	0	77.42
72	Blank	15.5	24.2	22.8	1400	16.04	0	0	90.32
96	Blank	15.5	24.2	22.5	1700	14.61	0	0	109.68
120	Blank	15.5	24.2	22.3	1900	11.69	0	0	122.55
144	Blank	15.5	24.2	22.2	2000	11.46	0	0	129.03
168	Blank	15.5	24.2	21.9	2300	11.30	0	0	148.39

Table 6: Experiment B (cast iron immersed in HCL with 10ml of inhibitor)

Time (hrs)	Concentration	Sample area (cm ²)	Initial weight (g)	Final weight (g)	Weight loss	Corrosion rate (mm/hrs)	IE%	Θ	Mass loss
24	10ml	15.5	20.6	19.6	1000	34.37	-25	-0.25	64.52
48	10ml	15.5	20.6	19.6	1000	17.18	16.64	0.1664	64.52
72	10ml	15.5	20.6	19.6	1000	11.45	28.56	0.2856	64.52
96	10ml	15.5	20.6	19.4	1200	10.31	29.44	0.2944	77.42
120	10ml	15.5	20.6	19.4	1200	8.26	29.44	0.2944	77.41
144	10ml	15.5	20.6	1.4	1400	8.03	30.02	0.3002	90.32
168	10ml	15.5	20.6	1.4	1400	6.89	60.88	0.6088	90.32

Table 7: Experiment C (cast iron immersed in HCL with 20ml inhibitor)

Time (hrs)	Concentration	Sample area (cm ²)	Initial weight (g)	Final weight (g)	Weight loss	Corrosion rate (mm/hrs)	IE%	Θ	Mass loss
24	20ml	15.5	18.6	18.0	600	20.63	24.99	0.2499	38.71
48	20ml	15.5	18.6	18.0	600	10.31	50.0	0.50	38.71
72	20ml	15.5	18.6	17.8	800	9.17	42.84	0.4284	51.61
96	20ml	15.5	18.6	17.5	1100	9.15	35.32	0.3532	70.97
120	20ml	15.5	18.6	17.3	1300	8.94	23.53	0.2353	83.87
144	20ml	15.5	18.6	17.3	1300	7.45	35.0	0.35	83.87
168	20ml	15.5	18.6	17.0	1600	7.86	30.45	0.304	103.2

Table 8: Experiment D(cast iron immersed in HCL with 40ml of inhibitor)

Time (hrs)	Conc	Sample area (cm ²)	Initial weight (g)	Final weight (g)	Weight loss	Corrosion rate (mm/hrs)	IE%	Θ	Mass loss
24	40ml	15.5	29.2	0.4	400	13.75	50.0	0.50	25.81
48	40ml	15.5	29.2	0.4	400	6.88	66.64	0.66	25.81
72	40ml	15.5	29.2	0.6	600	6.88	57.11	0.571	38.71
96	40ml	15.5	29.2	0.6	600	5.16	64.70	0.647	38.71
120	40ml	15.5	29.2	0.8	800	5.50	52.96	0.529	51.61
144	40ml	15.5	29.2	0.9	900	5.16	54.97	0.5497	58.06
168	40ml	15.5	29.2	1.0	1000	4.91	56.55	0.5655	64.51

Table 9: Experiment E (cast iron immersed in HCL with 60ml of inhibitor)

Time (hrs)	Conc	Sample area(cm ²)	Initial weight(g)	Final weight (g)	Weight loss	Corrosion rate (mm/hrs)	IE%	Θ	Mass loss
24	60ml	15.5	22.8	22.8	0	0	0	0	0
48	60ml	15.5	22.8	22.7	100	1.72	91.60	0.9160	6.45
72	60ml	15.5	22.8	22.6	200	2.30	85.67	0.8567	12.90
96	60ml	15.5	22.8	22.5	300	2.58	85.91	0.8591	19.35
120	60ml	15.5	22.8	22.5	300	2.06	82.38	0.8238	19.35
144	60ml	15.5	22.8	22.5	300	1.72	85.0	0.850	19.35
168	60ml	15.5	22.8	22.4	400	1.96	82.66	0.8266	21.81

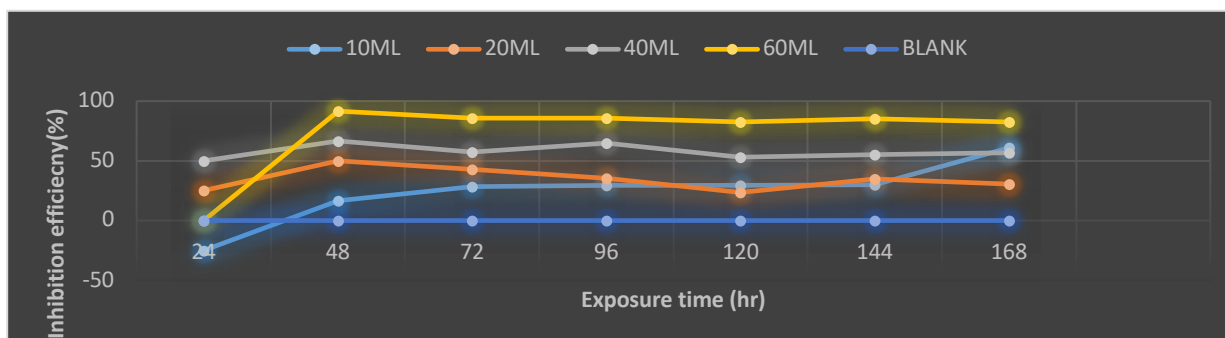


Figure 5: inhibition efficiency against period of immersion

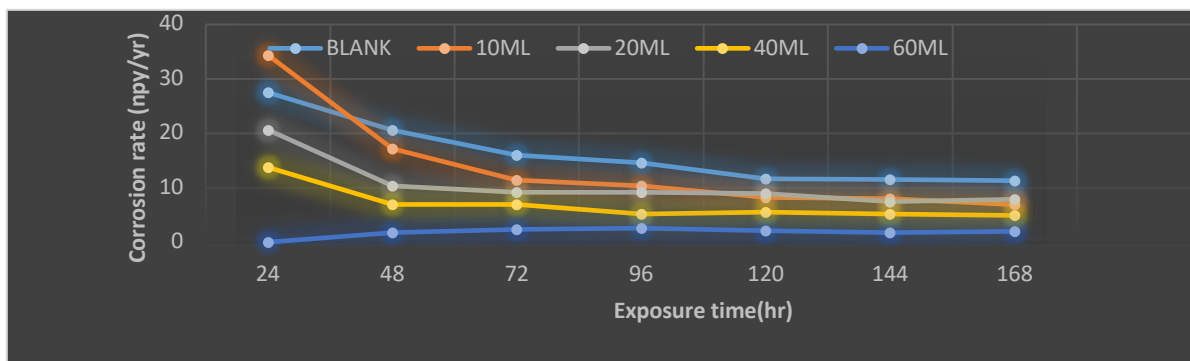


Figure 6 :corrosion rate against Exposure time.

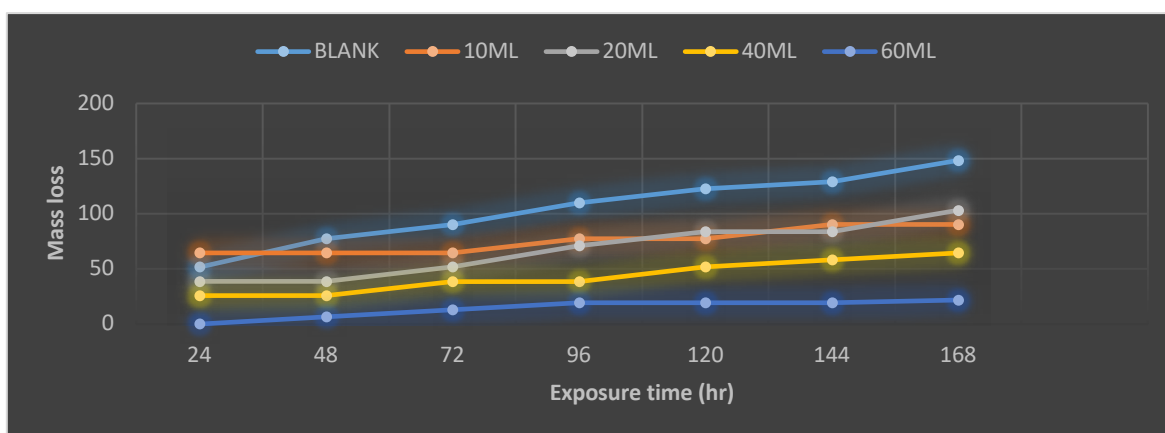


Figure 7:Mass loss against Exposure time

The corrosion rate of the uninhibited samples and the inhibited samples are shown in table 4 -table4.5. Figure16.1 shows the graphical representation of the corrosion rate as function of exposure time for the inhibited and uninhibited samples. It was observed from the graph that the corrosion rate for the uninhibited cast iron in 1.5m of HCL experienced a stable increase from the start of the experiment to the end. The mass loss also increased steadily as the exposure time increases (figure 16.2). This could be attributed to the presences of dissolved air in the system. The corrosion of the inhibited cast iron sample shown a nominal corrosion rate as the exposure time increases. As presented in figure 16.1 the corrosion rate of the inhibited sample decreases with exposure time. The decrease in corrosion rate of the inhibited sample was as a result of the alteration (passivation) in the mechanism of corrosion due to the presence of inhibitor and also because the exposure time under investigation fall between the active region for corrosion. Figure 16 shows the graphical representation of the inhibition efficiency against exposure time for cast iron sample. From the graph, it was observed that the inhibition efficiency increased with increase with exposure time.

Table 10: Result of Atomic Absorption Spectrometric Analysis

Sample	Description	Concentration of Fe (ppm)
A	cast iron immersed in acid without inhibitor	65296.15
B	cast iron immersed in acid with 10ml of inhibitor	44201.25

C	cast iron immersed in acid with 20ml of inhibitor	35620.70
D	cast iron immersed in acid with 40ml of inhibitor	25789.75
E	cast iron immersed in acid with 60ml of inhibitor	29719.30

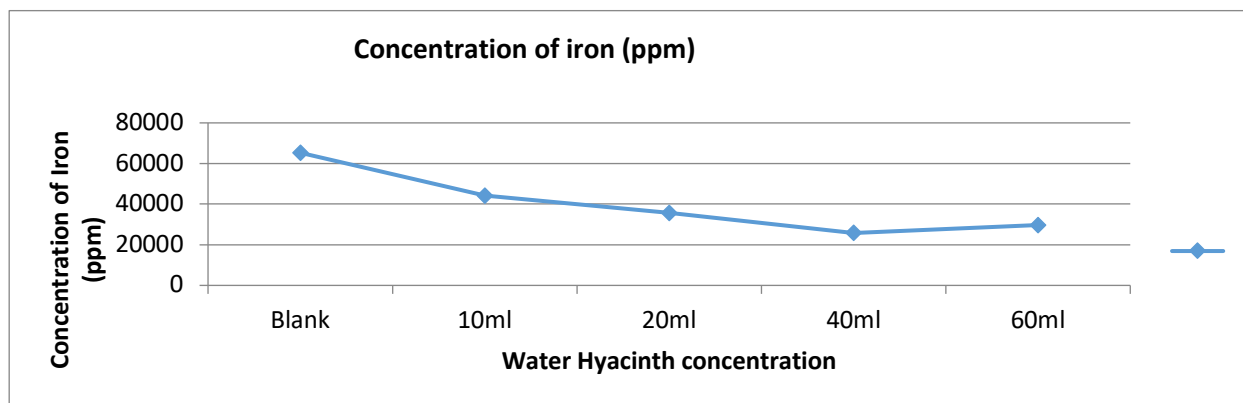


Figure 8: Graph representation of concentration of iron against concentration of water hyacinth extract.

The result shown from table shows the sample without inhibitor contains more iron deposit of about (65296.15)ppm as compared to the sample with varying concentration of inhibitor, as the inhibitor concentration increases the deposition of iron reduces. This impulses that the water hyacinth extract is a good inhibitor.

4.0 CONCLUSION

Having carried out a research work on the investigation and evaluation of the corrosion inhibition properties of water hyacinth extract on cast iron, the following deductions were drawn:

- Water Hyacinth extract possess remarkable anti-corrosion properties on cast iron in acidic medium.
- The mechanism of action of the Water Hyacinth extract was predicted to be a formation of protective film on the surface of the cast iron in acidic medium.
- The corrosion rate of cast iron is a combined function of the concentration of Water Hyacinth extract and exposure time.
- Significantly high Inhibitor Efficiency (IE) was found at highest concentration (60ml) of Water Hyacinth. Thus, cast iron efficiency increases with the increase of inhibitor concentration and exposure period.

4.1 RECOMMENDATIONS

- Government should tailor their research towards development of green inhibitors in such a way that it will be readily available in the market.
- Government should focus on local source to reduce the importation of chemical inhibitor.
- Government should enforce the use of eco friendly inhibitors to guide against the use of toxic chemical inhibitors.

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